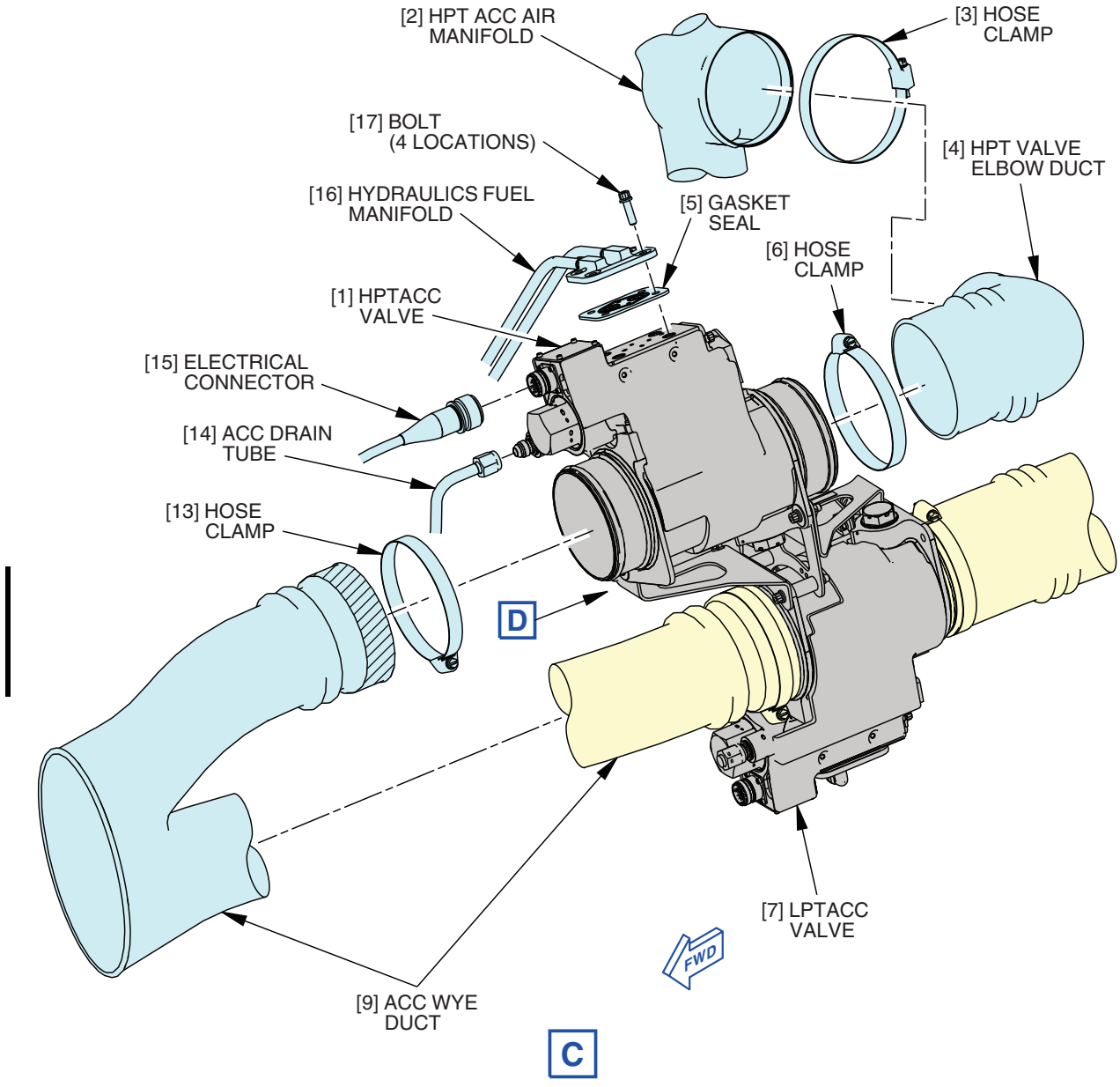
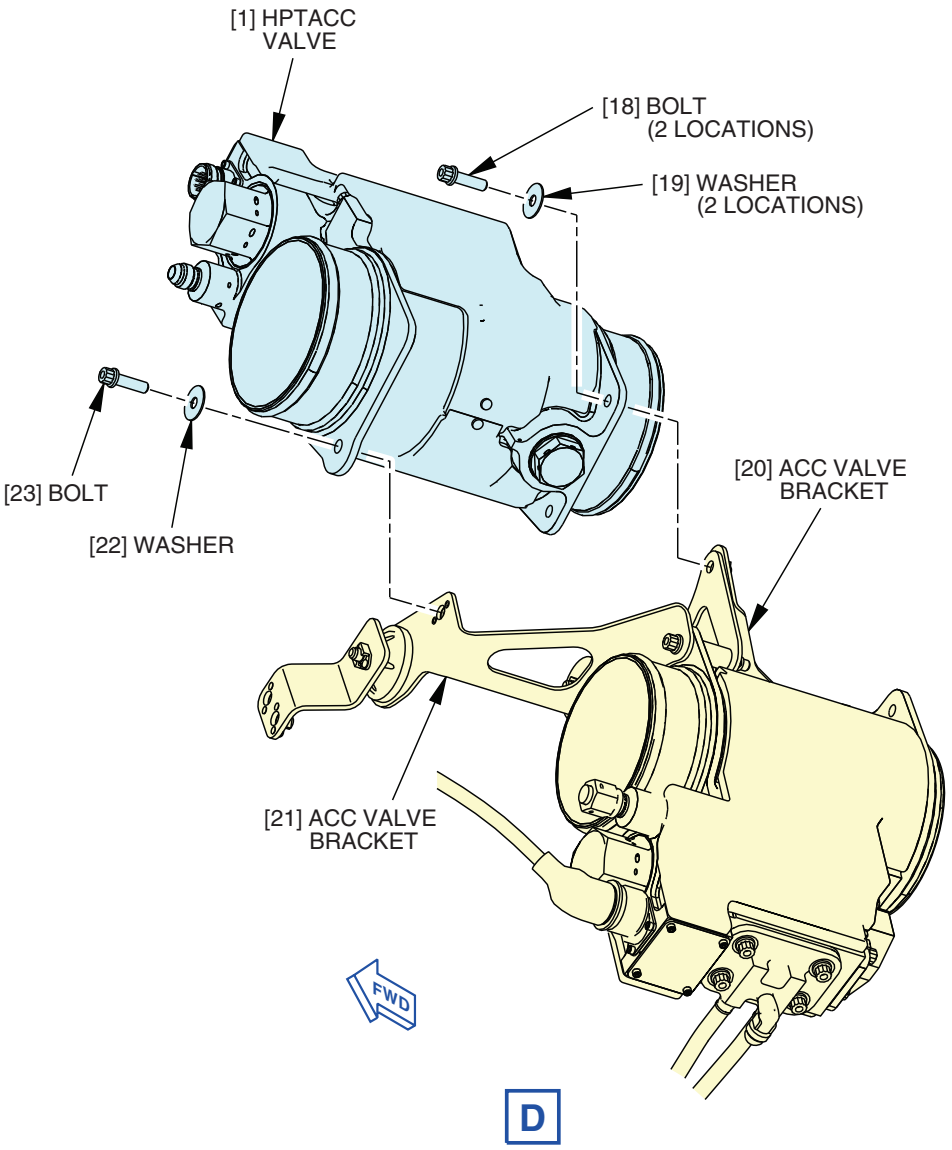


HPTACC Valve Installation
Figure 401/75-24-04-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5035977-00
2391973 S0000550085_V3

HPTACC Valve Installation
Figure 401/75-24-04-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-24-04-400-801-G00

3. High Pressure Turbine Active Clearance Control (HPTACC) Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the High Pressure Turbine Active Clearance Control (HPTACC) valve.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G51669 [CP2975]	Tape - Silicone Rubber, Glass Cloth Fabric	MIL-I-22444

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	HPTACC valve	75-24-04-01-030	PTF ALL
5	Gasket seal	73-11-00-20-270	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. High Pressure Turbine Active Clearance Control Valve Installation

SUBTASK 75-24-04-420-001-G00

(1) Install the HPTACC valve [1] as follows:

(a) Remove the protective covers from these components:

1) The HPT valve elbow duct [4]

2) The ACC wye duct [9]

3) The HPTACC valve [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Put the HPT valve elbow duct [4] in position on the HPTACC valve [1] and the HPT ACC air manifold [2].

(c) Move the hose clamp [3] on the aft end of HPT valve elbow duct [4] and put it in position so that there is access to the head of the screw.

(d) Use a back and forth motion to twist and push the HPT valve elbow duct [4] aft end point on the HPT ACC air manifold [2] past the HPT ACC air manifold bead a minimum distance of the hose clamp width.

(e) Put the hose clamp [3] in position on the rubber joint between the raised HPT ACC air manifold [2] bead and the HPT valve elbow duct [4].

(f) Move the hose clamp [6] on the forward end of HPT valve elbow duct [4] and put it in position so that there is access to the head of the screw.

(g) Use a back and forth motion to twist and push the HPT valve elbow duct [4] forward end point on the HPTACC valve [1] past the HPTACC valve bead a minimum distance of the hose clamp width.

(h) Put the hose clamp [6] in position on the rubber joint between the raised HPTACC valve [1] bead and the HPT valve elbow duct [4].

(i) Lubricate the threads of the two bolts [18] and bolt [23] with engine oil, D00599 [CP2442].

(j) Install the two bolts [18] and two washers [19] that attach the HPTACC valve [1] to the ACC valve bracket [20].

(k) Install the bolt [23] and washer [22] that attach the HPTACC valve [1] to the ACC valve bracket [21].

(l) Tighten the hose clamp [3] and hose clamp [6] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

SUBTASK 75-24-04-430-001-G00

CAUTION MAKE SURE TO LIFT THE HYDRAULICS FUELING MANIFOLD TO GIVE SPACE TO INSTALL THE GASKET SEAL. DO NOT RUB OR SLIDE THE GASKET TOO MUCH. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(2) Connect the hydraulics fuel manifold [16] to the HPTACC valve [1] as follows:

(a) Remove the protective covers from the hydraulics fuel manifold [16] and the HPTACC valve [1].

(b) Install a new gasket seal [5] between the hydraulics fuel manifold [16] and the HPTACC valve [1].

NOTE: Install the gasket seal [5] dry with no lubrication.

(c) Lubricate the threads of the four bolts [17] with engine oil, D00599 [CP2442].

(d) Install the four bolts [17] that attach the hydraulics fuel manifold [16] to the HPTACC valve [1].

SUBTASK 75-24-04-860-003-G00

(3) Remove the protective covers from the ACC drain tube [14] and the HPTACC valve [1] fitting.

SUBTASK 75-24-04-430-002-G00

(4) Connect the ACC drain tube [14] to the HPTACC valve [1].

(a) Tighten the ACC drain tube [14] B-nut to 262.0 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

(b) Put the ACC wye duct [9] in position between the HPT ACC duct [11] and the HPTACC valve [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Move the hose clamp [13] on the aft end of ACC wye duct [9] and put it in position so that there is access to the head of the screw.

(d) Use a back and forth motion to twist and push the ACC wye duct [9] aft end point on the HPTACC valve [1] past the HPTACC valve bead a minimum distance of the hose clamp width.

(e) Make sure that the ACC wye duct [9] has silicone rubber tape, G51669 [CP2975], in the aft side clamping area.

1) If the ACC wye duct [9] does not have silicone rubber tape, G51669 [CP2975], in the aft side clamping area, do this step:

a) Apply one row of silicone rubber tape, G51669 [CP2975], 0.020 in. (0.51 mm) thick and 1.00 in. (25.4 mm) wide to the ACC wye duct [9] end directly forward of the aft edge bead of the ACC wye duct [9].

(f) Put the hose clamp [13] in position on the rubber joint between the raised HPTACC valve [1] bead and the ACC wye duct [9].

(g) Tighten the two bolts [18] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(h) Tighten the bolt [23] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(i) Tighten the four bolts [17] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(j) Tighten the hose clamp [13] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

SUBTASK 75-24-04-430-003-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(5) Use a teflon-jawed pliers, STD-664, to connect the electrical connector [15] to the HPTACC valve [1] as follows:

(a) Remove the protective covers from the electrical connector [15] and the electrical receptacles on the HPTACC valve [1].

(b) Connect the electrical connector [15] to the HPTACC valve [1] (TASK 70-00-01-910-803-G00).

1) Tighten the electrical connector [15].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-24-04-410-001-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-24-04-860-004-G00

(2) For the left engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-04-860-006-G00

(3) For the right engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-04-865-002-G00

(4) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

J. High Pressure Turbine Active Clearance Control Valve Installation Test

SUBTASK 75-24-04-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COMPRESSOR CONTROL - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Compressor control detailed inspection.

TASK 75-30-00-211-801-G00

2. Compressor Control Detailed Inspection

A. General

(1) This task provides the instructions on how to do the detailed inspection of the compressor control system.

(a) You do an inspection of these components:

1) Left and right Variable Stator Vane (VSV) actuators
2) VSV system hardware
3) Left and right Variable Bypass Valve (VBV) actuators
4) Transient Bleed Valve (TBV)
5) Start Bleed/Booster Anti-Ice (SB/BAI) valve.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-30-00-200-801-G00	High Pressure Compressor (HPC) Section Inspection (P/B 601)
73-22-01-000-801-G00	Electronic Engine Control (EEC) Electrical Harness (HJ3A) Removal (P/B 401)
73-22-01-400-801-G00	Electronic Engine Control (EEC) Electrical Harness (HJ3A) Installation (P/B 401)
75-11-01-000-801-G00	Start Bleed/Booster Anti-Ice Valve Removal (P/B 401)
75-11-01-400-801-G00	Start Bleed/Booster Anti-Ice Valve Installation (P/B 401)
75-31-02 P/B 401	VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION
75-32-01 P/B 401	VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION
75-33-01-000-801-G00	Transient Bleed Valve Removal (P/B 401)
75-33-01-400-801-G00	Transient Bleed Valve Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 75-30-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-30-00-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Compressor Control Inspection

SUBTASK 75-30-00-212-001-G00

(1) Examine the left and right VSV actuators as follows:

(a) Look for fuel leaks on the actuator body and valve shaft.

1) A fuel leak that makes drops of fuel is not permitted.

2) A fuel stain or wetting on the actuator body or actuator valve shaft is permitted if you do not find drops of fuel.

3) If you find a fuel leak, do these steps:

a) Replace the gasket seal between the fuel manifold and the VSV actuator (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

b) Examine the fuel fittings on the fuel manifold for damage or contamination.

<1> If you find damage or contamination on the fuel fittings, replace the fuel manifold (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

c) Examine the VSV actuator fuel tubes for damage or contamination.

<1> If you find a VSV actuator fuel tube with damage or contamination, replace the fuel tube (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

d) Make sure that the VSV fuel tubes to the fuel manifold connections are correct and tight.

<1> If the installation of the fuel tubes are not correct or the fuel tubes are loose, re-install and tighten the fuel tube B-nuts (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

e) If the fuel leak continues, replace the VSV actuator (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

(b) Make sure that the VSV actuator is not loose.

1) If the VSV actuator is loose, tighten the bolts and nuts (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

2) If the VSV actuator has bearings with cracks or damage, replace the VSV actuator (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Look for cracks and dents on the VSV actuator.

1) If you find cracks or dents, replace the VSV actuator (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

(d) Look for scores on the piston rod of the VSV actuator.

1) If you see scores on the piston rod, replace the VSV actuator (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

(e) Look for a bent VSV actuator piston rod.

1) If you see a bent piston rod, replace the VSV actuator (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

(f) Examine the VSV actuator electrical receptacle and the electrical connector.

1) If there is leakage through the electrical connector or signs of wetting around the electrical connector, replace the VSV actuator (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

2) If it is loose, tighten the electrical connector (TASK 70-00-01-910-803-G00).

a) If the electrical connector has damage, replace the wiring harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

3) If there is more than small damage to the VSV actuator electrical receptacle, replace the VSV actuator (VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-31-02/401).

NOTE: Small damage to the electrical receptacle threads that is repairable is permitted.

a) Repair small damage to the threads so that the connector can be tightened to hold torque.

SUBTASK 75-30-00-211-001-G00

(2) Examine the VSV system hardware as follows:

(a) Do this task: High Pressure Compressor (HPC) Section Inspection, TASK 72-30-00-200-801-G00.

SUBTASK 75-30-00-212-002-G00

(3) Examine the left and right VBV actuators as follows:

(a) Look for fuel leaks on the actuator body and valve shaft.

1) A fuel leak that makes drops of fuel is not permitted.

2) A fuel stain or wetting on the actuator body or actuator valve shaft is permitted if you do not find drops of fuel.

3) If you find signs of a fuel leak, do these steps:

a) Replace the gasket seal between the fuel manifold and the VBV actuator (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

b) Examine the fuel fittings on the fuel manifold for damage and contamination.

<1> If you find damage or contamination on the fuel fittings, replace the fuel manifold (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

c) Examine the VBV actuator fuel manifold for damage and contamination.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> If you find a VBV actuator fuel manifold with damage or contamination, replace the fuel manifold (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

d) Make sure that the connections between the VBV fuel tubes to the fuel manifold connections are correct and are tight.

<1> If the installation of fuel tubes are not correct or the fuel tubes are loose, re-install and tighten the fuel manifold B-nuts (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

e) If the fuel leak continues, replace the VBV actuator (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

(b) Make sure that the VBV actuator is not loose.

1) If the VBV actuator is loose, tighten the mounting bolts (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

2) If the VBV actuator has cracks or damaged bearings, replace the VBV actuator (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

(c) Look for cracks and dents on the VBV actuator.

1) If you find cracks or dents, replace the VBV actuator (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

(d) Look for cracks in the VBV actuator mounting flange.

1) If you see cracks, replace the VBV actuator (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

(e) Examine the VBV actuator electrical receptacle and electrical connector.

1) If there is leakage through the electrical connector or signs of wetting around the electrical connector, replace the VBV actuator (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

2) If it is necessary, tighten the electrical connector (TASK 70-00-01-910-803-G00).

a) If the electrical connector has damage, replace the wiring harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

3) If there is more than small damage to the VBV actuator electrical receptacle, replace the VBV actuator (VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION, PAGEBLOCK 75-32-01/401).

NOTE: Small damage to the electrical receptacle threads that is repairable is permitted.

a) Repair small damage to the threads so that the connector can be tightened to hold torque.

SUBTASK 75-30-00-212-003-G00

(4) Examine the Transient Bleed Valve (TBV) as follows:

(a) Look for signs of fuel leaks.

1) Fuel stain on the TBV is permitted if you do not find drops of fuel.

2) A fuel leak that makes drops of fuel is not permitted.

3) If you find signs of a fuel leak, do these steps:

a) Make sure that the installation of the TBV fuel tubes are correct and tight.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> If the installation of the fuel tubes are not correct or the fuel tubes are loose, re-install and tighten the fuel tube B-nuts (TASK 75-33-01-000-801-G00 and TASK 75-33-01-400-801-G00).

b) Examine the fuel fittings on the TBV actuator for damage or contamination.

<1> If you find damage or contamination on the TBV fuel fittings, replace the TBV (TASK 75-33-01-000-801-G00 and TASK 75-33-01-400-801-G00).

c) If the fuel leak continues, replace the TBV (TASK 75-33-01-000-801-G00 and TASK 75-33-01-400-801-G00).

(b) Make sure that the TBV is not loose.

1) If the TBV is loose, tighten the mount locations and mount brackets (TASK 75-33-01-400-801-G00).

(c) Look for cracks and dents on the TBV.

1) If you find cracks or dents, replace the TBV (TASK 75-33-01-000-801-G00 and TASK 75-33-01-400-801-G00).

(d) Look for signs of air leaks from the E-seals on the TBV.

1) If you see signs of air leaks, replace the ring-seals (TASK 75-33-01-000-801-G00 and TASK 75-33-01-400-801-G00).

(e) Examine the TBV electrical receptacles and electrical connectors.

1) If it is loose, tighten the electrical connector (TASK 70-00-01-910-803-G00).

a) If an electrical connector has damage, replace the wiring harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

2) If there is more than small damage to a TBV receptacle, replace the TBV (TASK 75-33-01-000-801-G00 and TASK 75-33-01-400-801-G00).

NOTE: Small damage to the electrical receptacle threads that is repairable is permitted.

a) Repair small damage to the threads so that the connector can be tightened to hold torque.

SUBTASK 75-30-00-210-001-G00

(5) Examine the Start Bleed/Booster Anti-ice (SB/BAI) valve.

(a) Look for signs of fuel leaks as follows:

1) It is permitted to have an SB/BAI valve that has a fuel stain if you do not find drops of fuel.

2) A fuel leak that makes drops of fuel is not permitted.

3) If you find signs of a fuel leak, do these steps:

a) Make sure that the installation of the SB/BAI fuel tubes are correct and tight.

<1> If the installation of the fuel tubes are not correct or the tubes are loose, re-install and tighten the fuel tubes (TASK 75-11-01-000-801-G00 and TASK 75-11-01-400-801-G00).

b) Examine the fuel fittings on the SB/BAI valve for damage or contamination.

<1> If you find damage or contamination on the SB/BAI valve fuel fittings, replace the valve (TASK 75-11-01-000-801-G00 and TASK 75-11-01-400-801-G00).

c) Examine the SB/BAI fuel tubes for damage or contamination.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> If you find a SB/BAI fuel tube with damage or contamination, replace the fuel tube (TASK 75-11-01-000-801-G00 and TASK 75-11-01-400-801-G00).

d) Make sure that the installation of the SB/BAI fuel tubes are correct and tight.

<1> If the installation of the fuel tubes are not correct or the fuel tubes are loose, re-install and tighten the fuel tube B-nuts (TASK 75-11-01-000-801-G00 and TASK 75-11-01-400-801-G00).

e) If the fuel leak continues, replace the SB/BAI valve (TASK 75-11-01-000-801-G00 and TASK 75-11-01-400-801-G00).

(b) Make sure that the SB/BAI valve is not loose.

1) If the SB/BAI valve is loose, tighten the mount locations and mount brackets (TASK 75-11-01-400-801-G00).

(c) Look for cracks and dents on the SB/BAI valve.

1) If you find cracks or dents, replace the SB/BAI valve(TASK 75-11-01-000-801-G00 and TASK 75-11-01-400-801-G00).

(d) Look for signs of air leaks from the E-seals on the SB/BAI valve.

1) If you see signs of air leaks, replace the ring-seals (TASK 75-11-01-000-801-G00 and TASK 75-11-01-400-801-G00).

(e) Examine the SB/BAI electrical receptacles and electrical connector.

1) If it is loose, tighten the electrical connector (TASK 70-00-01-910-803-G00).

2) If an electrical connector has damage, repair or replace the wiring harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

3) If there is more than small damage to the SB/BAI valve receptacle, replace the valve (TASK 75-11-01-000-801-G00 and TASK 75-11-01-400-801-G00).

NOTE: Small damage to the electrical receptacle threads that is repairable is permitted.

a) Repair small damage to the threads so that the connector can be tightened to hold torque.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-30-00-860-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-30-00-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VARIABLE STATOR VANE SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has one task:

(1) Variable Stator Vane Actuator Manual Operation.

TASK 75-31-00-910-801-G00

2. Variable Stator Vane Actuator Manual Operation

(Figure 201)

A. General

(1) This task gives the instructions to manually operate the variable stator valve system to the open or close position.

(2) The variable stator valve is referred to as the VSV.

B. References

Reference	Title
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14774	Adapter - VSV Pneumatic Part #: 956A1607G02 Supplier: 58828
SPL-14802	Cart- Fuel Manifold Pressure Check Part #: 956A1256G01 Supplier: 58828
SPL-14835	Actuator - Portable Hydraulic, VSV Part #: 856A1084G04 Supplier: 58828 Part #: 9465M11P01 Supplier: 58828 Part #: 9471M46G01 Supplier: 58828
SPL-15596	Adapter - VSV Hydraulic Part #: 956A1607G01 Supplier: 58828
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A
G02272	Fuel - Turbine, Aviation (Grades JP-4, JP-5, JP-5/JP-8ST)	MIL-DTL-5624
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Operation

SUBTASK 75-31-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-31-00-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Variable Stator Vane Actuator Manual Operation

SUBTASK 75-31-00-020-001-G00

(1) Remove the segment shroud 4 and segment shroud 5 (TASK 72-23-04-000-801-G00).

SUBTASK 75-31-00-800-001-G00

WARNING DO NOT LET THE FUEL GET ON YOUR SKIN OR CLOTHES. REMOVE CLOTHES THAT HAVE FUEL ON THEM. FUEL CAN BURN YOUR SKIN. FULLY REMOVE FUEL FROM YOUR SKIN WITH WATER.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

(2) Do these steps to manually operate the Variable Stator Vane (VSV) system:

(a) Disconnect the VSV fuel manifold [1] from the left VSV actuator [4] as follows:

1) Cut and remove the safety wire or safety cable from the B-nuts and fittings that connect the VSV fuel manifold [1] to the fuel hydraulics manifolds [6].
2) Put a fuel resistant container (5 gal), STD-201, below the VSV fuel manifold [1].
3) Disconnect the fuel hydraulics manifold [6] (2 locations) from the VSV fuel manifold [1].
4) Install the protective caps on the fuel hydraulics manifold [6] open ports.
5) Disconnect the VSV-MTC drain manifold [5] from the VSV fuel manifold [1].
6) Remove the four bolts [2] that attach the VSV fuel manifold [1] to the left VSV actuator [4].
7) Let the fuel drain into the fuel resistant container (5 gal), STD-201.
8) Remove the VSV fuel manifold [1] from the engine.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

9) Remove and examine the gasket seal [3] for these types of damage:
   a) Nicks
   b) Cuts
   c) Abrasions
   d) Missing material
   e) Flattened areas.
10) If there is no damage, keep the gasket seal [3] for the installation task.
11) If there is damage, replace the gasket seal [3].

(b) Remove the segment shroud 2 (TASK 72-23-04-000-801-G00).

(c) Disconnect the VSV fuel manifold [9] from the right VSV actuator [12] as follows:

1) Cut and remove the safety wire or safety cable from the B-nuts and fittings that connect the VSV fuel manifold [9] to the fuel hydraulics manifolds [6].
2) Put a fuel resistant container (5 gal), STD-201, below the VSV fuel manifold [9].
3) Disconnect the fuel hydraulics manifold [6] (2 locations) from the VSV fuel manifold [9].
4) Install the protective caps on the fuel hydraulics manifold [6] open ports.
5) Disconnect the VSV drain tube [11] from the VSV fuel manifold [9].
6) Disconnect the drain manifold [13] from the VSV fuel manifold [9].
7) Install the protective caps on the drain manifold [13] open ports.
8) Remove the four bolts [10] that attach the VSV fuel manifold [9] to the right VSV actuator [12].

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

9) Let the fuel drain into the fuel resistant container (5 gal), STD-201.
10) Remove the VSV fuel manifold [9] from the engine.
11) Remove and examine the gasket seal [8] for these types of damage:
   a) Nicks
   b) Cuts
   c) Abrasions
   d) Missing material
   e) Flattened areas.
12) If there is no damage on the gasket seal [8], keep the seal for the installation task.
13) If there is damage on the gasket seal [8], replace the seal.

(d) Connect the VSV hydraulic adapter, SPL-15596, to the left VSV actuator [4] as follows:

NOTE: You can use the VSV pneumatic adapter, SPL-14774, with cart, SPL-14802, as an alternative.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put the gasket seal (included as part the VSV hydraulic adapter, SPL-15596) on the left VSV actuator [4].
2) Put the VSV hydraulic adapter, SPL-15596, into position on the left VSV actuator [4] flange.
3) Install the four bolts (included as part of the VSV hydraulic adapter, SPL-15596) that attach the VSV hydraulic adapter, SPL-15596, and the gasket to the left VSV actuator [4].
4) Tighten the four bolts to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

   NOTE: The bolt and gasket are included as part of the VSV hydraulic adapter, SPL-15596.

5) Connect the VSV hydraulic adapter, SPL-15596, to the VSV-MTC drain manifold [5].
   a) Tighten the VSV-MTC drain manifold [5] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).
6) Connect the VSV hydraulic adapter, SPL-15596, to right VSV actuator [12] as follows:
   a) Put the gasket seal (included as part the VSV hydraulic adapter, SPL-15596, on the right VSV actuator [12].
   b) Put the VSV hydraulic adapter, SPL-15596, into position on the right VSV actuator [12] flange.
   c) Install the four bolts (included as part the VSV hydraulic adapter, SPL-15596) that attach the VSV hydraulic adapter, SPL-15596, and the gasket to the right VSV actuator [12].
   d) Tighten the four bolts to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

      NOTE: The bolts and gaskets are included as part of the VSV hydraulic adapter, SPL-15596.

   e) Connect the VSV drain tube [11] B-nut to the VSV hydraulic adapter, SPL-15596.

      <1> Tighten the VSV drain tube [11] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).
7) Connect the hydraulic actuator, SPL-14835, and return hoses to the fittings on the VSV hydraulic adapter, SPL-15596, with quick disconnect couplings.
8) Operate the right and left VSV actuators manually as follows:

   NOTE: This task describes the use of the hydraulic actuator, SPL-14835, to manually operate the VSV. If the hydraulic actuator, SPL-14835, tool is not available, you can also use nitrogen, G00018, as an alternative pressure source. The operating pressure for the nitrogen, G00018, is the same as the fluid pressures provided below.

   a) Make sure that you use the clean oil, D00599 [CP2442], or the fuel, G02272, as the hydraulic fluid for the hydraulic actuator, SPL-14835.
   b) Operate the VSV system to the fully open position as follows:

      NOTE: The VSV actuator shafts will be retracted when the VSV system is in the fully opened position.

      <1> Operate the hydraulic actuator, SPL-14835, at 50.0 psig (344.7 kPa) - 100.0 psig (689.5 kPa) to retract the shafts of the VSV actuators.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<2> Make sure that the actuators move smoothly to the fully opened position.

NOTE: A bind in the system will cause the pressure reading to increase during actuation.

9) Operate the VSV system to the fully closed position.

NOTE: The VSV actuator rods will be extended when the VSV system is in the fully closed position.

a) Operate the hydraulic actuator, SPL-14835, at 50.0 psig (344.7 kPa) - 100.0 psig (689.5 kPa) to extend the shafts of the VSV actuators.

b) Make sure that the actuators move smoothly to the full close position.

NOTE: A bind in the system will cause the pressure reading to increase during actuation.

10) If it is not necessary, release the hydraulic pressure from the hydraulic actuator, SPL-14835.

SUBTASK 75-31-00-000-001-G00

(3) Do these steps to disconnect the VSV hydraulic adapter, SPL-15596, and install the left and right VSV actuators.

(a) Disconnect the quick-disconnect couplings on the VSV hydraulic adapter, SPL-15596, fittings.

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

1) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(b) Remove the VSV hydraulic adapter, SPL-15596, from the right VSV actuator [12] as follows:

1) Disconnect the VSV hydraulic adapter, SPL-15596, from the VSV drain tube [11].

2) Remove the four bolts that attach the VSV hydraulic adapter, SPL-15596, to the right VSV actuator [12].

3) Remove the VSV hydraulic adapter, SPL-15596, and gasket from the right VSV actuator [12].

NOTE: The bolts and gaskets are included as parts of the VSV hydraulic adapter, SPL-15596.

(c) Remove the VSV hydraulic adapter, SPL-15596, from the left VSV actuator [4] as follows:

1) Disconnect the VSV hydraulic adapter, SPL-15596, from the VSV-MTC drain manifold [5].

2) Remove the four bolts that attach the VSV hydraulic adapter, SPL-15596, to the left VSV actuator [4].

3) Remove the VSV hydraulic adapter, SPL-15596, and gasket from the left VSV actuator [4].

NOTE: The bolts and gasket are included as part of the VSV hydraulic adapter, SPL-15596.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Install the VSV fuel manifold [9] on the right VSV actuator [12] as follows:

1) Put the gasket seal [8] on the right VSV actuator [12].
2) Apply the oil, D00599 [CP2442], on the threads of the four bolts [10].
3) Install the four bolts [10] that attach the VSV fuel manifold [9] to the right VSV actuator [12].
4) Tighten the four bolts [10] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).
5) Remove the protective caps from the fuel hydraulics manifold [6] open ports.
6) Connect the fuel hydraulics manifold [6] (2 locations) to the VSV fuel manifold [9].
   a) Tighten the two B-nuts to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).
7) Connect the VSV drain tube [11] to the VSV fuel manifold [9].
   a) Tighten the VSV drain tube [11] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).
8) Remove the protective caps from the drain manifold [13] open ports.
9) Connect the drain manifold [13] to the VSV fuel manifold [9].
   a) Tighten the VSV fuel manifold [9] 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).
10) Install safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006], on the B-nuts and fittings that connect the VSV fuel manifold [9] to the fuel hydraulics manifolds [6] (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

(e) Install the VSV fuel manifold [1] on the left VSV actuator [4] as follows:

1) Put the gasket seal [3] on the left VSV actuator [4].
2) Apply the oil, D00599 [CP2442], on the threads of the four bolts [2].
3) Install the four bolts [2] that attach the VSV fuel manifold [1] to the left VSV actuator [4].
4) Tighten the bolts [2] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).
   a) Tighten the two B-nuts to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).
5) Remove the protective caps from the fuel hydraulics manifold [6] open ports.
6) Connect the VSV-MTC drain manifold [5] to the VSV fuel manifold [1].
   a) Tighten the VSV-MTC drain manifold [5] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).
7) Install safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006], on the B-nuts and fittings that connect the VSV fuel manifold [1] to the fuel hydraulics manifolds [6] (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

SUBTASK 75-31-00-420-001-G00

(4) Install the segment shroud 4 and the segment shroud 5 (TASK 72-23-04-400-801-G00).

SUBTASK 75-31-00-420-002-G00

(5) Install the segment shroud 2 (TASK 72-23-04-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-31-00-860-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-31-00-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

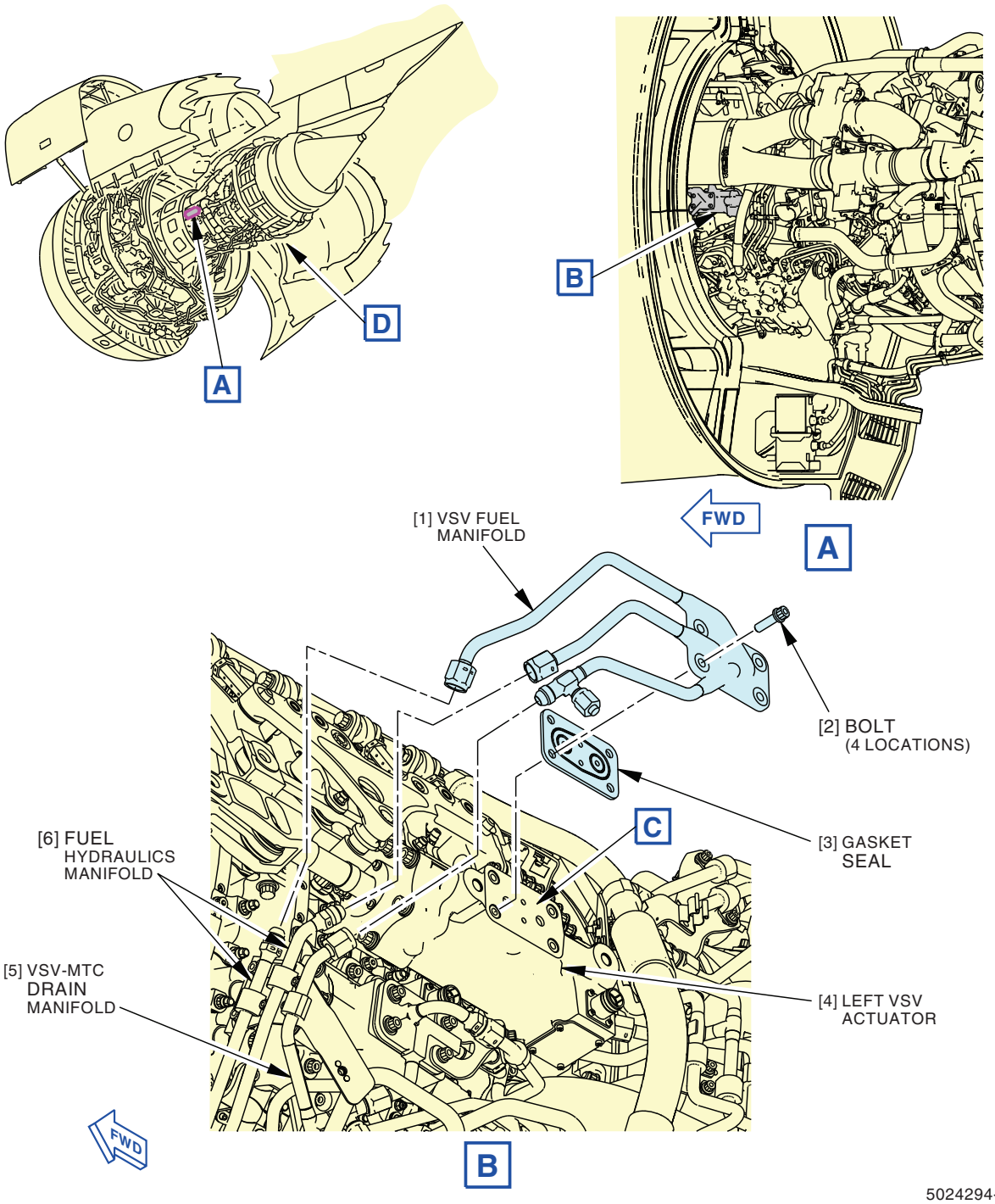
J. Variable Stator Vane Actuator Test

SUBTASK 75-31-00-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

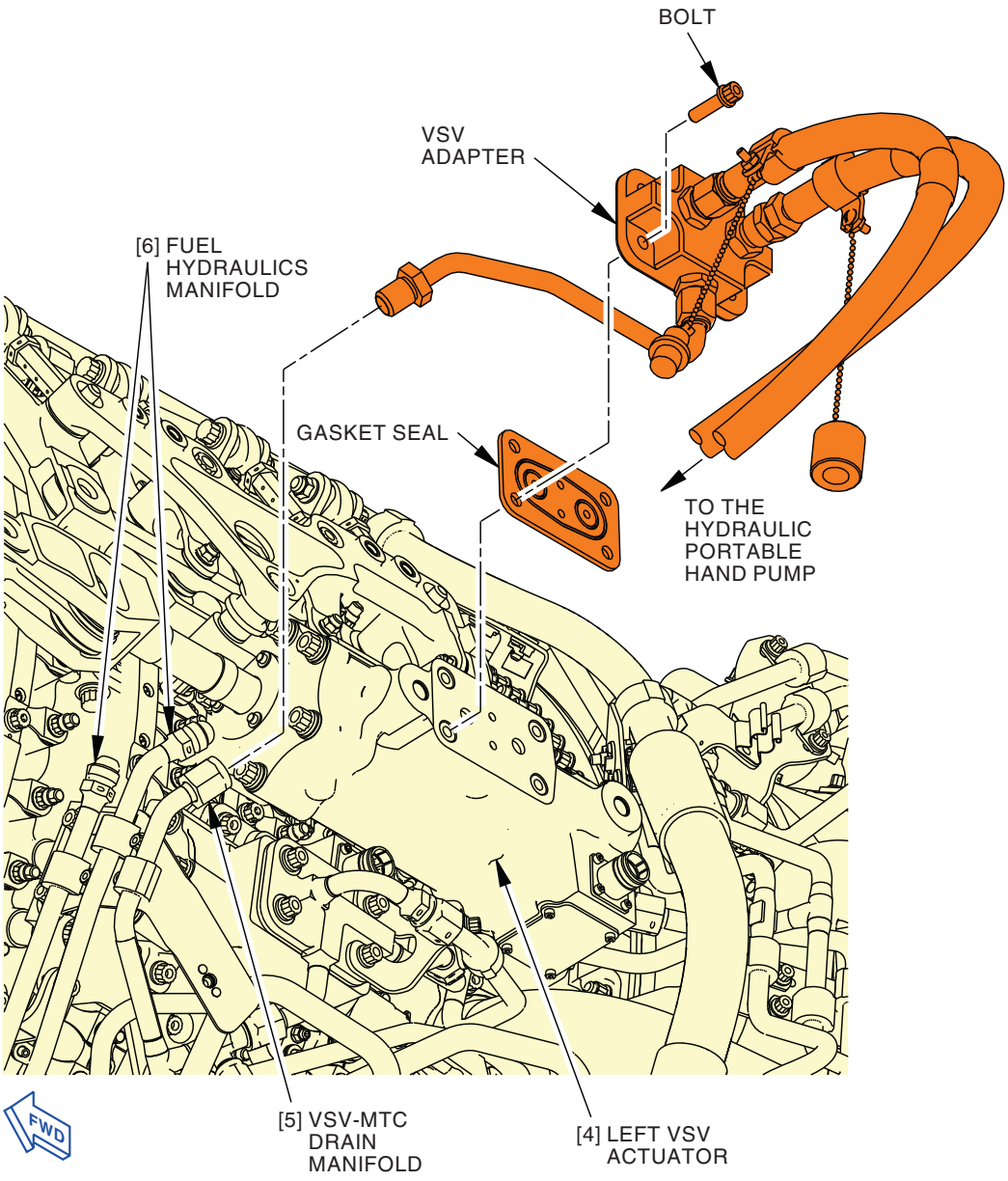
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Variable Stator Vane Actuator (Manually) Operation
Figure 201/75-31-00-990-801-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

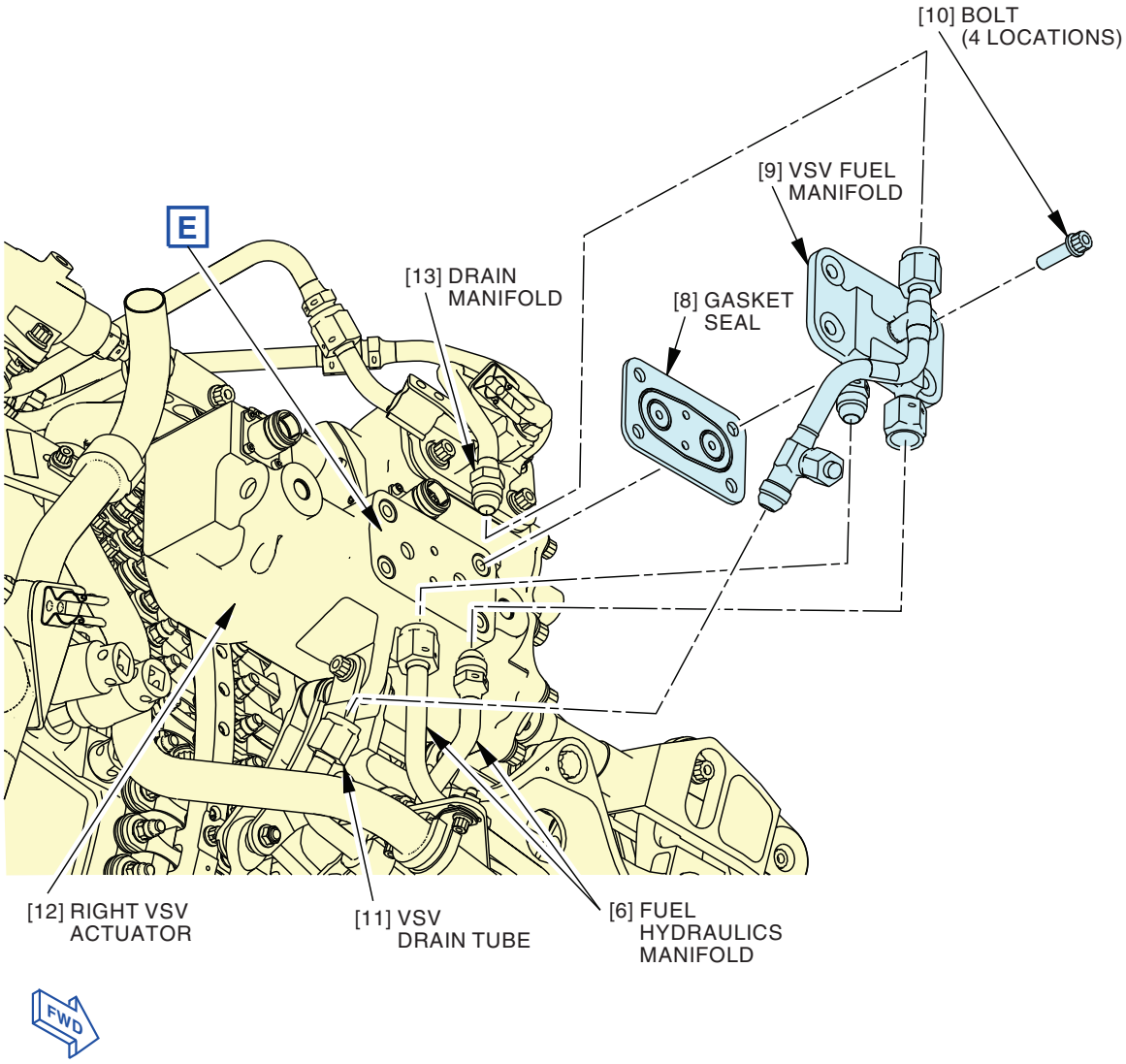


5024295-00

2354972 S00061516290_V1

Variable Stator Vane Actuator (Manually) Operation
Figure 201/75-31-00-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

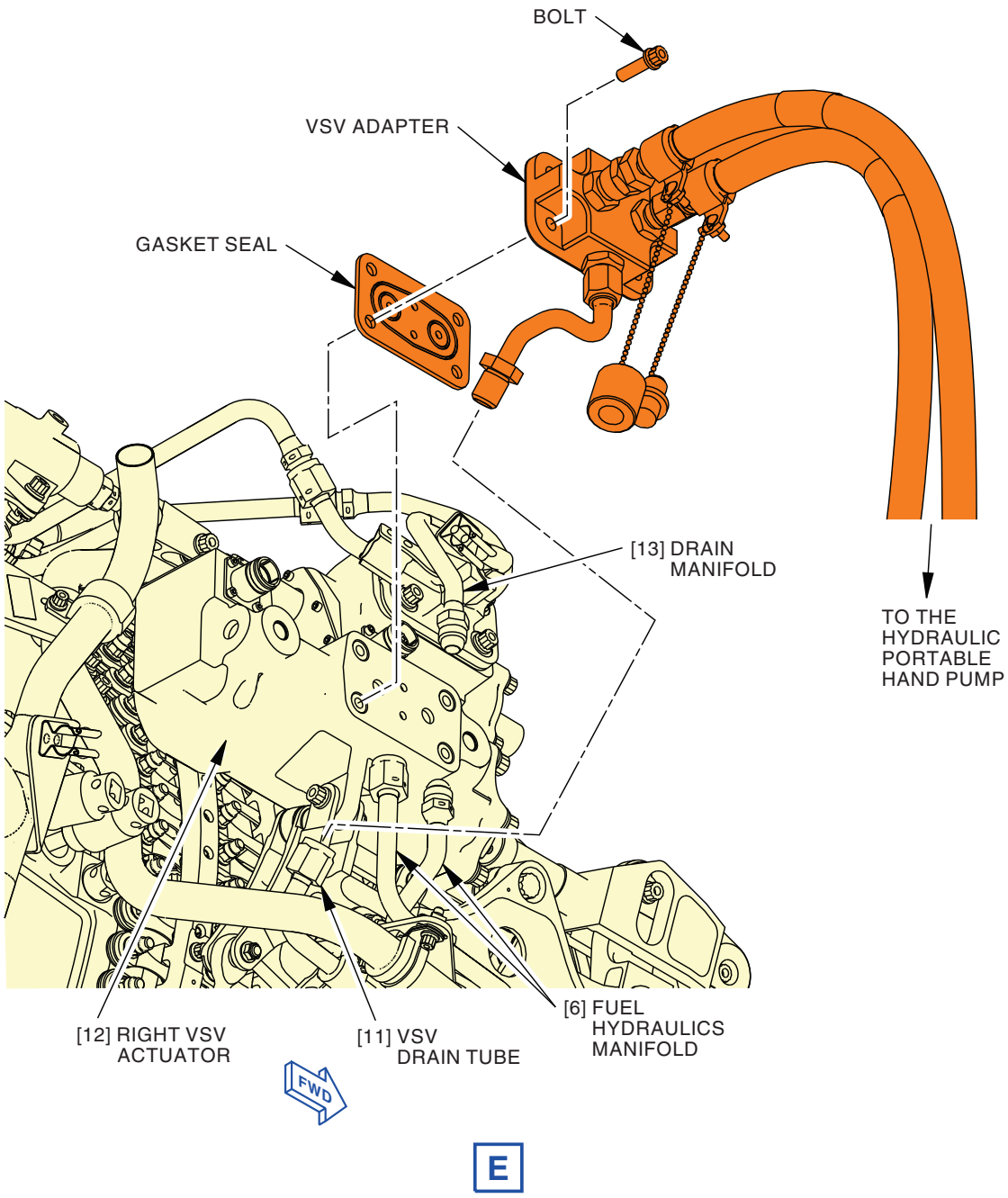


5024296-00

2458318 S0000572354_V1

Variable Stator Vane Actuator (Manually) Operation
Figure 201/75-31-00-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5024314-00
2458323 S0000572355_V1

Variable Stator Vane Actuator (Manually) Operation
Figure 201/75-31-00-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VARIABLE STATOR VANE ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the left VSV actuator

(2) An installation of the left VSV actuator

(3) A removal of the right VSV actuator

(4) An installation of the right VSV actuator.

TASK 75-31-02-000-801-G00

2. Left Variable Stator Vane (VSV) Actuator Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the left Variable Stator Vane (VSV) actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE
STD-4049	Container - Fuel Resistant, 1 Gallon (4 Liter)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-31-02-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-31-02-010-001-G00

(2) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. Left Variable Stator Vane (VSV) Actuator Removal

SUBTASK 75-31-02-010-003-G00

(1) Remove the segment shroud 5 (TASK 72-23-04-000-801-G00).

SUBTASK 75-31-02-010-004-G00

(2) Remove the segment shroud 4 (TASK 72-23-04-000-801-G00).

SUBTASK 75-31-02-420-001-G00

(3) Remove the VOSS oval clamp [4] that attaches the ACC wye duct [3] to the center support ACC bracket [2] and HPT ACC inlet duct [1].

SUBTASK 75-31-02-010-005-G00

(4) Remove the HPT ACC inlet duct [1].

SUBTASK 75-31-02-020-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(5) Disconnect the electrical connector [6] as follows (TASK 70-00-01-910-803-G00):

(a) Use a teflon-jawed pliers, STD-664, to disconnect the electrical connector [6] from the left VSV actuator [7].

(b) Install the protective covers on the electrical connector [6] and electrical receptacle on the left VSV actuator [7].

SUBTASK 75-31-02-020-002-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Disconnect the VSV-MTC drain manifold [16] from the fuel hydraulics manifold [13] as follows:

(a) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the VSV-MTC drain manifold [16].

(b) Disconnect the VSV-MTC drain manifold [16] B-nut from the fuel hydraulics manifold [13].

(c) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

SUBTASK 75-31-02-020-003-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Disconnect the fuel hydraulics manifold tubes [17] from the fuel hydraulics manifold [13] B-nuts as follows:

(a) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the fuel hydraulics manifold [13].

(b) Disconnect the fuel hydraulics manifold [13] B-nuts from the fuel hydraulics manifold tubes [17].

(c) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

SUBTASK 75-31-02-010-006-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Remove the fuel hydraulics manifold [13] from the left VSV actuator [7] as follows:

(a) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the fuel hydraulics manifold [13].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the four bolts [14] that attach the fuel hydraulics manifold [13] to the left VSV actuator [7].

(c) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

(d) Remove the fuel hydraulics manifold [13] from the engine.

SUBTASK 75-31-02-010-007-G00

(9) Remove and discard the gasket seal [15] from between the fuel hydraulics manifold [13] and left VSV actuator [7].

SUBTASK 75-31-02-020-004-G00

(10) Install protective covers on the fuel hydraulics manifold tube [17], VSV-MTC drain manifold [16], left VSV actuator [7], and fuel hydraulics manifold [13].

SUBTASK 75-31-02-010-008-G00

(11) Remove the bolt [20], bushing [21], washer [23], and nut [24] that attach the left VSV actuator [7] to the connecting link [22].

PTF ALL; PRE LEAP-1B SB 72-0343

SUBTASK 75-31-02-010-009-G00

(12) Remove the bolt [19], washer [26], bushing [18], and nut [25] that attach the left VSV actuator [7] to the connecting link [22] and bell crank assembly [5].

(a) Remove the connecting link [22].

PTF ALL; POST LEAP-1B SB 72-0343

SUBTASK 75-31-02-010-028-G00

(13) Remove the bolt [19], washer [26], and nut [25] that attach the left VSV actuator [7] to the connecting link [22] and bell crank assembly [5].

(a) Remove the connecting link [22].

PTF ALL

SUBTASK 75-31-02-010-010-G00

(14) Remove the bolt [8], spacer [9], and nut [10] that attach the bell crank assembly [5] to the left VSV actuator [7].

NOTE: Do not remove the safety wire that attaches the spacer to the bell crank assembly.

SUBTASK 75-31-02-980-001-G00

(15) Hold the left VSV actuator [7] in position.

SUBTASK 75-31-02-010-011-G00

(16) Remove the bolt [11] and nut [12] that attach the left VSV actuator [7] to the bell crank assembly [5].

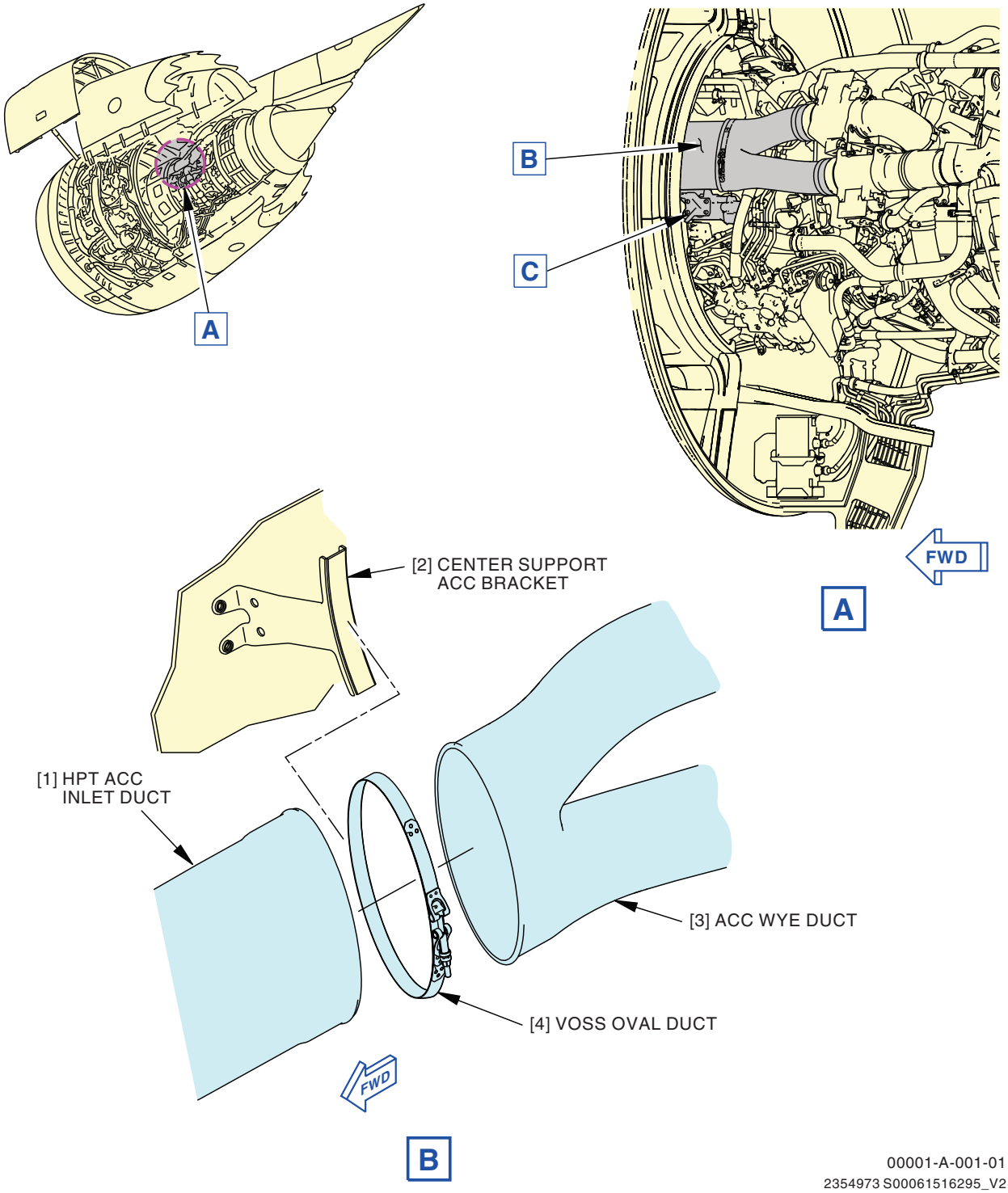
(a) Make sure that the bushing [64] stays in the left VSV actuator [7] housing.

SUBTASK 75-31-02-010-012-G00

(17) Remove the left VSV actuator [7] from the engine.

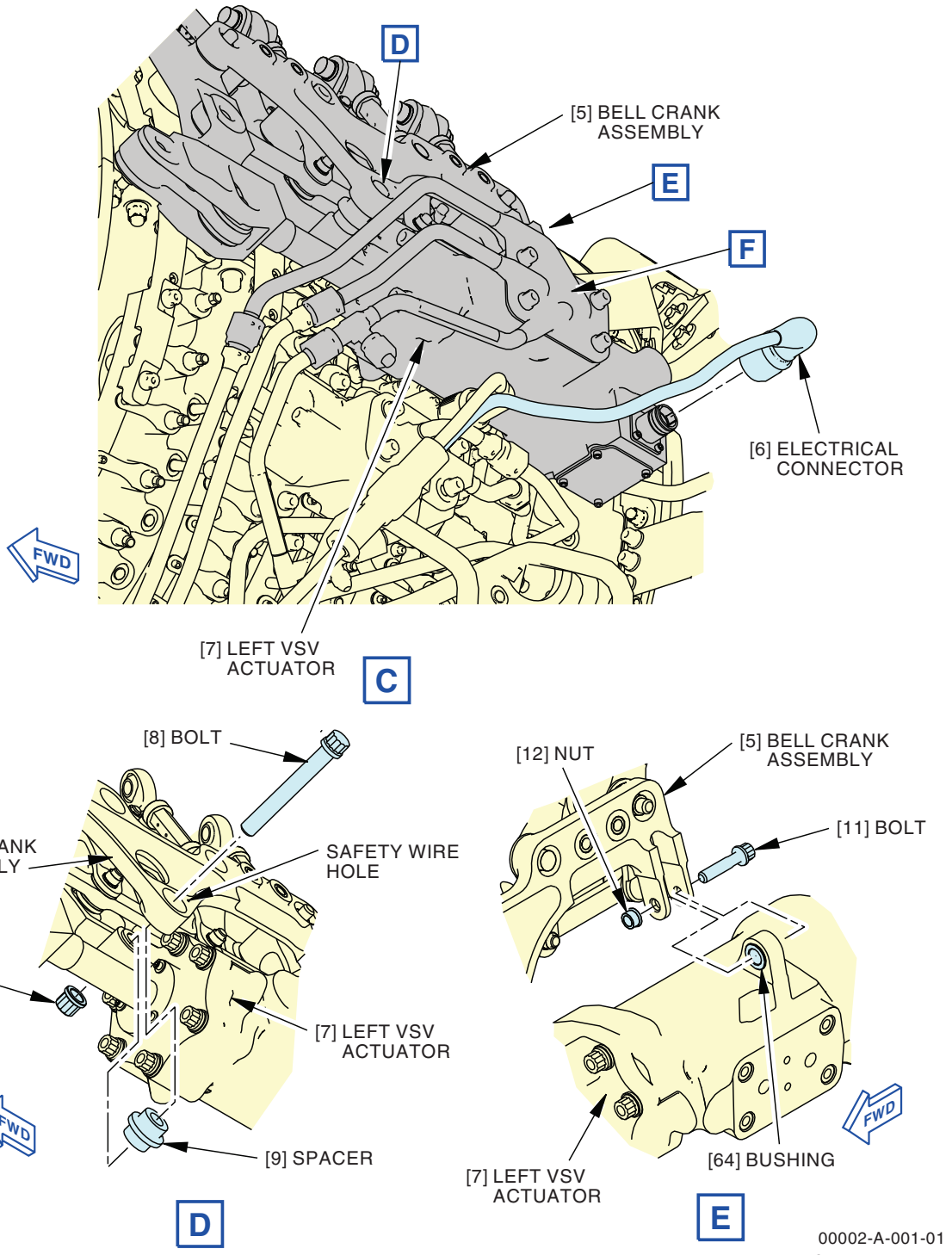
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



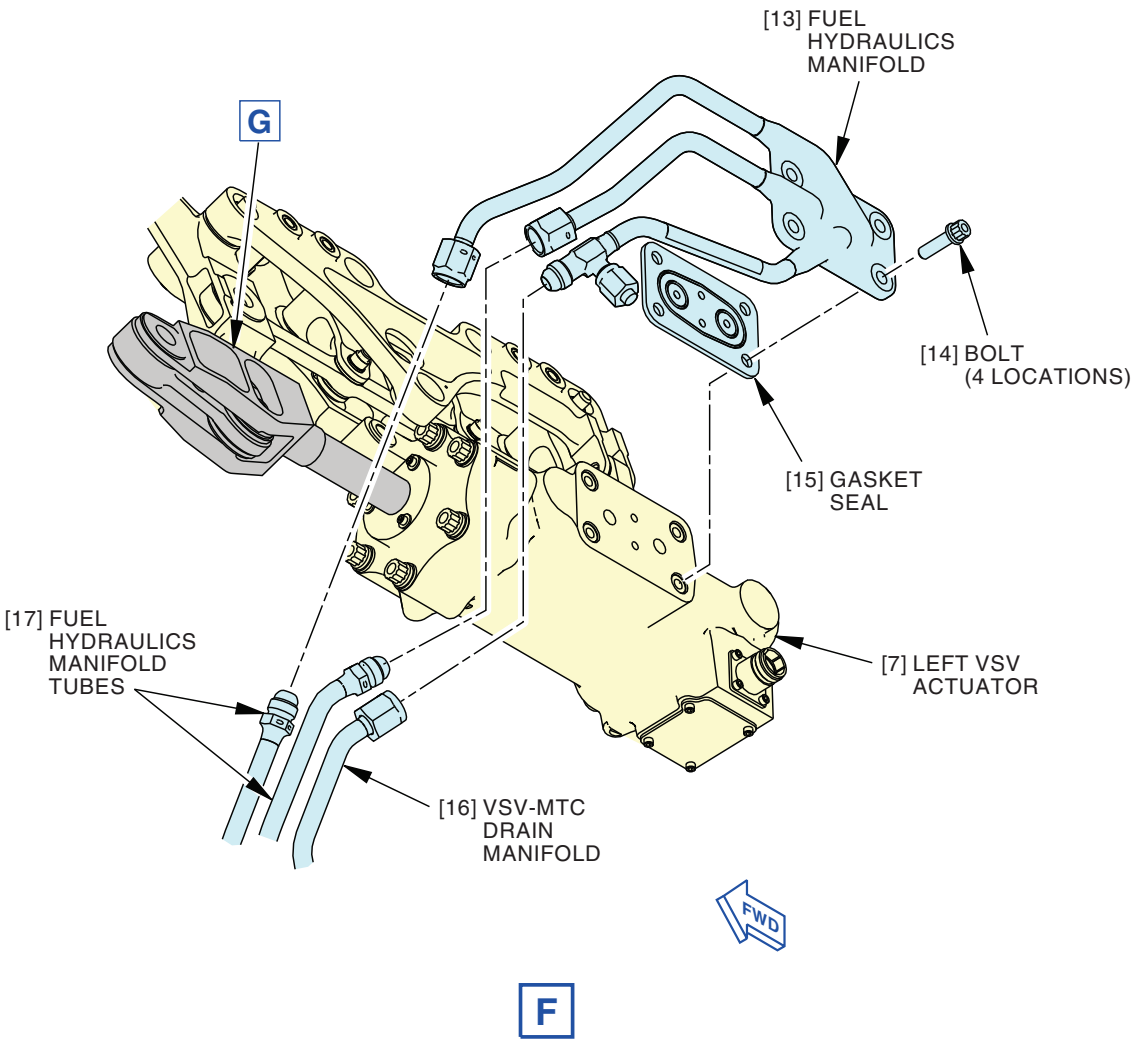
Left VSV Actuator Installation
Figure 401/75-31-02-990-801-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left VSV Actuator Installation
Figure 401/75-31-02-990-801-G00 (Sheet 2 of 5)

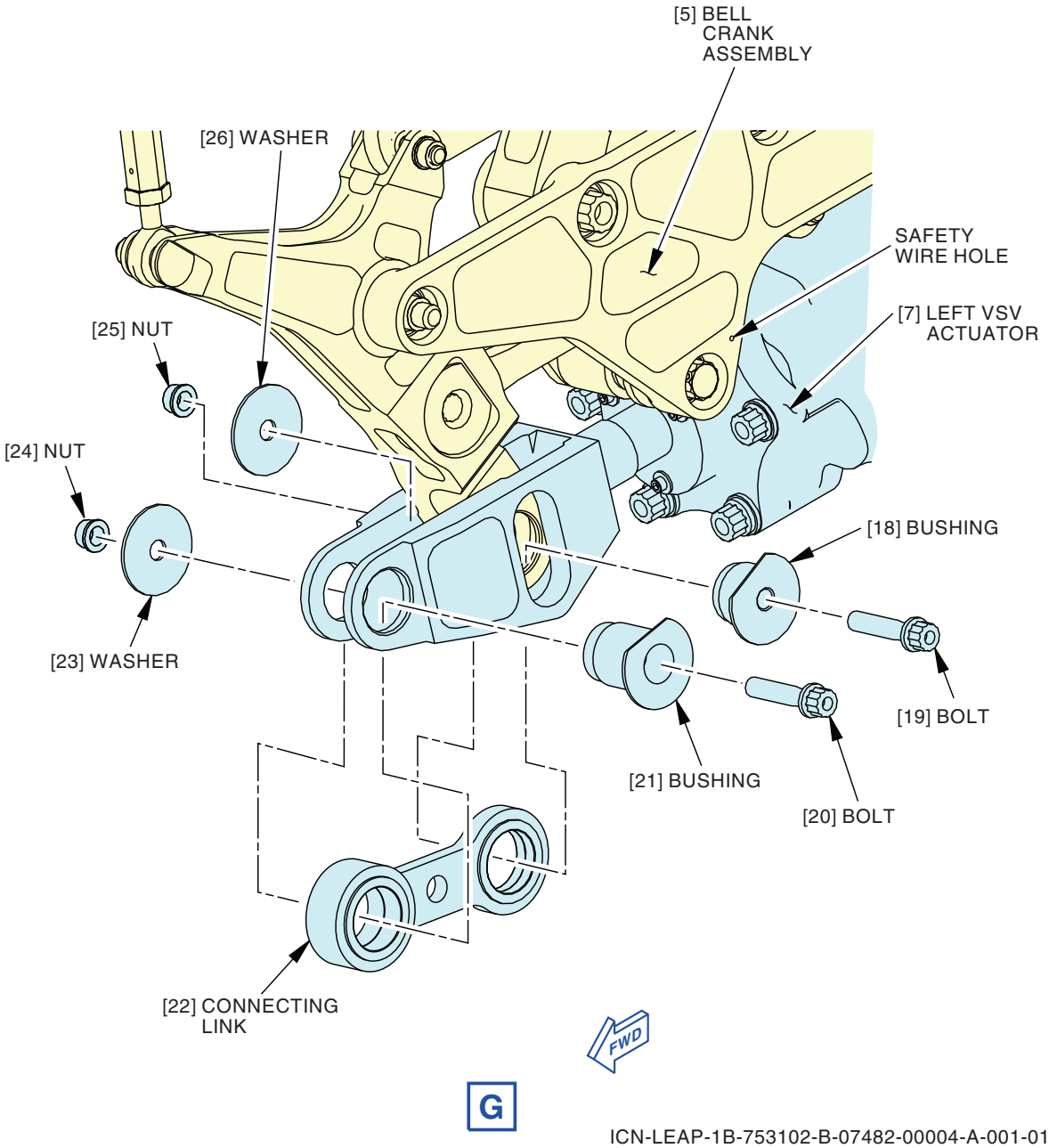
ECCN 9E991 BOEING PROPRIETARY - See title page for details



00003-A-001-01
2391351 S0000549970_V1

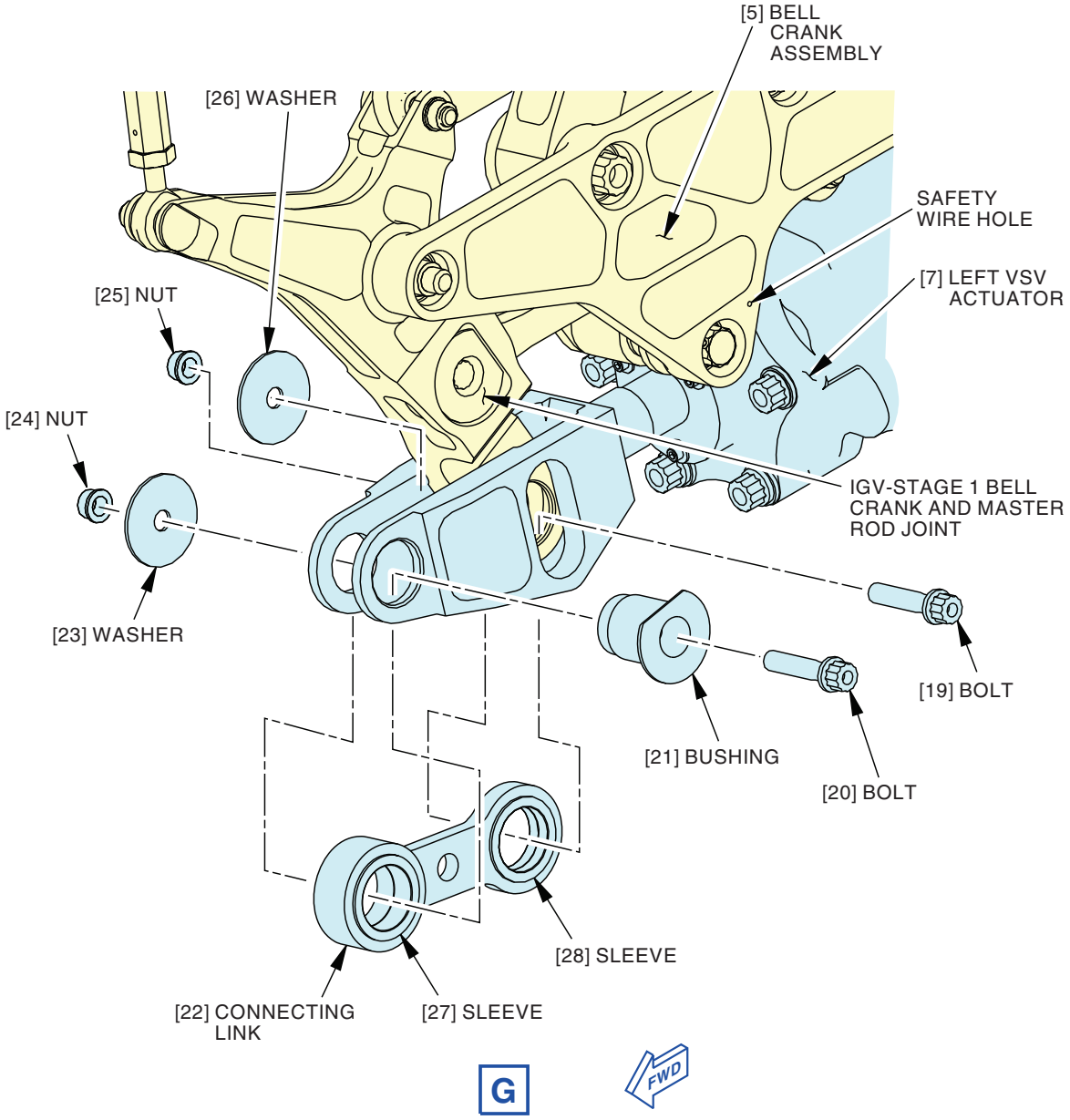
Left VSV Actuator Installation
Figure 401/75-31-02-990-801-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left VSV Actuator Installation
Figure 401/75-31-02-990-801-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



6035548-00
3010726 S0000784715_V1

Left VSV Actuator Installation
Figure 401/75-31-02-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-31-02-400-802-G00

3. Left Variable Stator Vane (VSV) Actuator Installation

(Figure 401)

A. General

(1) This task gives the instructions on how to install the left Variable Stator Vane (VSV) actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Left VSV actuator	75-31-02-01-010	PTF ALL
15	Gasket seal	73-11-00-20-370	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. Left Variable Stator Vane (VSV) Actuator Installation

SUBTASK 75-31-02-640-001-G00

(1) Apply engine oil, D00599 [CP2442], on the threads of the nut [25], nut [24], nut [12], and nut [10].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SB 72-0343

SUBTASK 75-31-02-420-002-G00

(2) Put the bushing [18], washer [26], and connecting link [22] in position on the left VSV actuator [7].

SUBTASK 75-31-02-420-003-G00

(3) Put the left VSV actuator [7] in position on the bell crank assembly [5].

SUBTASK 75-31-02-420-004-G00

(4) Loosely install the bolt [19] and nut [25] that attach the bushing [18], washer [26], and connecting link [22] to the bell crank assembly [5].

PTF ALL; POST LEAP-1B SB 72-0343

SUBTASK 75-31-02-420-017-G00

(5) Put the washer [26] and connecting link [22] in position on the left VSV actuator [7].

SUBTASK 75-31-02-420-018-G00

(6) Put the left VSV actuator [7] in position on the bell crank assembly [5].

SUBTASK 75-31-02-420-019-G00

(7) Loosely install the bolt [19] and nut [25] that attach the washer [26] and connecting link [22] to the bell crank assembly [5].

PTF ALL

SUBTASK 75-31-02-410-002-G00

(8) Loosely install the bolt [20], bushing [21], washer [23], and nut [24] that attach the left VSV actuator [7] to the connecting link [22].

PTF ALL; POST LEAP-1B SB 72-0343

SUBTASK 75-31-02-210-003-G00

(9) Check that the assembly fit of the connecting link [22] with the bushing [21] and bolt [11] has play on the direction perpendicular to the left VSV actuator [7] extending/retracting direction similar to the fit of the IGV-stage 1 bell-crank and master rod joint on the bell crank assembly [5].

(a) If the assembly fit feels noticeably looser, make sure that the sleeve [27] and sleeve [28] are properly installed.

PTF ALL

SUBTASK 75-31-02-210-001-G00

(10) Make sure that the bushing [64] is on the left VSV actuator [7].

SUBTASK 75-31-02-410-003-G00

(11) Loosely install the bolt [11] and nut [12] that attach the left VSV actuator [7] to the bell crank assembly [5].

SUBTASK 75-31-02-410-004-G00

(12) Loosely install the bolt [8], spacer [9], and nut [10] that attach the bell crank assembly [5] to the left VSV actuator [7].

SUBTASK 75-31-02-410-005-G00

(13) Tighten the nut [25], nut [24], nut [12], and nut [10] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-31-02-010-014-G00

(14) Remove the protective covers from the fuel hydraulics manifold tube [17], VSV-MTC drain manifold [16], left VSV actuator [7], and fuel hydraulics manifold [13].

SUBTASK 75-31-02-410-006-G00

(15) Connect the fuel hydraulics manifold [13] to the left VSV actuator [7] as follows:

(a) Install a new gasket seal [15] between the fuel hydraulics manifold [13] and left VSV actuator [7].

(b) Apply engine oil, D00599 [CP2442], on the threads of the four bolts [14].

(c) Install the four bolts [14] that attach the fuel hydraulics manifold [13] to the left VSV actuator [7].

(d) Tighten the bolts [14] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 75-31-02-410-007-G00

(16) Connect the fuel hydraulics manifold tubes [17] to the fuel hydraulics manifold [13] as follows:

(a) Connect the fuel hydraulics manifold [13] B-nuts to the fuel hydraulics manifold tubes [17].

(b) Tighten the two B-nuts to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

(c) Install a safety wire on the two B-nuts (TASK 70-41-00-400-801-G00).

SUBTASK 75-31-02-410-008-G00

(17) Connect the VSV-MTC drain manifold [16] to the fuel hydraulics manifold [13] as follows:

(a) Connect the VSV-MTC drain manifold [16] B-nut to the fuel hydraulics manifold [13].

(b) Tighten the VSV-MTC drain manifold [16] B-nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

SUBTASK 75-31-02-420-005-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(18) Connect the electrical connector [6] as follows (TASK 70-00-01-910-803-G00):

(a) Remove the protective covers from the electrical connector [6] and electrical receptacle on the left VSV actuator [7].

(b) Use a teflon-jawed pliers, STD-664, to connect the electrical connector [6] to the left VSV actuator [7].

1) Tighten the electrical connector [6].

SUBTASK 75-31-02-020-005-G00

(19) Put the HPT ACC inlet duct [1] in position on the ACC wye duct [3].

SUBTASK 75-31-02-410-009-G00

(20) Install the VOSS oval clamp [4] that attaches the ACC wye duct [3] to the center support ACC bracket [2] and HPT ACC inlet duct [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-31-02-420-010-G00

(21) Tighten the VOSS oval clamp [4] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

SUBTASK 75-31-02-410-011-G00

(22) Install the segment shroud 5 (TASK 72-23-04-400-801-G00).

SUBTASK 75-31-02-410-012-G00

(23) Install the segment shroud 4 (TASK 72-23-04-400-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-31-02-860-002-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-31-02-410-013-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

J. Left Variable Stator Vane Actuator (VSV) Installation Test

SUBTASK 75-31-02-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

TASK 75-31-02-000-802-G00

4. Right Variable Stator Vane (VSV) Actuator Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the right Variable Stator Vane (VSV) actuator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE
STD-4049	Container - Fuel Resistant, 1 Gallon (4 Liter)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-31-02-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-31-02-010-015-G00

(2) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

G. Right Variable Stator Vane (VSV) Actuator Removal

SUBTASK 75-31-02-010-016-G00

(1) Remove the segment shroud 2 (TASK 72-23-04-000-801-G00).

SUBTASK 75-31-02-020-006-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(2) Disconnect the electrical connector [44] as follows (TASK 70-00-01-910-803-G00):

(a) Use a teflon-jawed pliers, STD-664, to disconnect the electrical connector [44] from the right VSV actuator [43].

(b) Install protective covers on the electrical receptacle on the electrical connector [44] and receptacle on the right VSV actuator [43].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-31-02-020-007-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Disconnect the fuel hydraulic manifold [45] from the VSV fuel manifold [47] as follows:

(a) Cut and remove the safety wire or safety cable from the B-nuts and fittings that connect the VSV fuel manifold [47] to the fuel hydraulic manifolds [45].

(b) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the VSV fuel manifold [47].

(c) Disconnect the fuel hydraulic manifolds [45] from the VSV fuel manifold [47].

(d) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

SUBTASK 75-31-02-020-008-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Disconnect the VSV right drain tube [46] from the VSV fuel manifold [47] as follows:

(a) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the VSV fuel manifold [47].

(b) Disconnect the VSV right drain tube [46] B-nut from the VSV fuel manifold [47].

(c) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

SUBTASK 75-31-02-010-017-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Disconnect the right MTC drain manifold [41] from the VSV fuel manifold [47] as follows:

(a) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the VSV fuel manifold [47].

(b) Disconnect the VSV fuel manifold [47] B-nut from the right MTC drain manifold [41].

(c) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

SUBTASK 75-31-02-010-018-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Remove the VSV fuel manifold [47] from the right VSV actuator [43] as follows:

(a) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the VSV fuel manifold [47].

(b) Remove the four bolts [48] that attach the VSV fuel manifold [47] to the right VSV actuator [43].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

(d) Remove the VSV fuel manifold [47] from the right VSV actuator [43].

SUBTASK 75-31-02-010-019-G00

(7) Remove and discard the gasket seal [42] from between the VSV fuel manifold [47] and right VSV actuator [43].

SUBTASK 75-31-02-020-009-G00

(8) Install protective covers on the right VSV actuator [43], VSV fuel manifold [47], fuel hydraulic manifolds [45], VSV right drain tube [46], and right MTC drain manifold [41].

SUBTASK 75-31-02-010-020-G00

(9) Remove the bolt [62], bushing [63], washer [51], and nut [52] that attach the right VSV actuator [43] to the connecting link [53].

PTF ALL; PRE LEAP-1B SB 72-0343

SUBTASK 75-31-02-010-027-G00

(10) Remove the bolt [60], bushing [61], washer [49], and nut [50] that attach the right VSV actuator [43] to the connecting link [53] and bell crank assembly [55].

(a) Remove the connecting link [53].

PTF ALL; POST LEAP-1B SB 72-0343

SUBTASK 75-31-02-010-021-G00

(11) Remove the bolt [60], washer [49], and nut [50] that attach the right VSV actuator [43] to the connecting link [53] and bell crank assembly [55].

(a) Remove the connecting link [53].

PTF ALL

SUBTASK 75-31-02-010-022-G00

(12) Remove the bolt [58], spacer [59], and nut [54] that attach the bell crank assembly [55] to the right VSV actuator [43].

(a) Do not remove the safety wire that attaches the spacer [59] to the bell crank assembly [55].

SUBTASK 75-31-02-980-002-G00

(13) Hold the right VSV actuator [43].

SUBTASK 75-31-02-010-023-G00

(14) Remove the bolt [56] and nut [57] that attach the right VSV actuator [43] to the bell crank assembly [55].

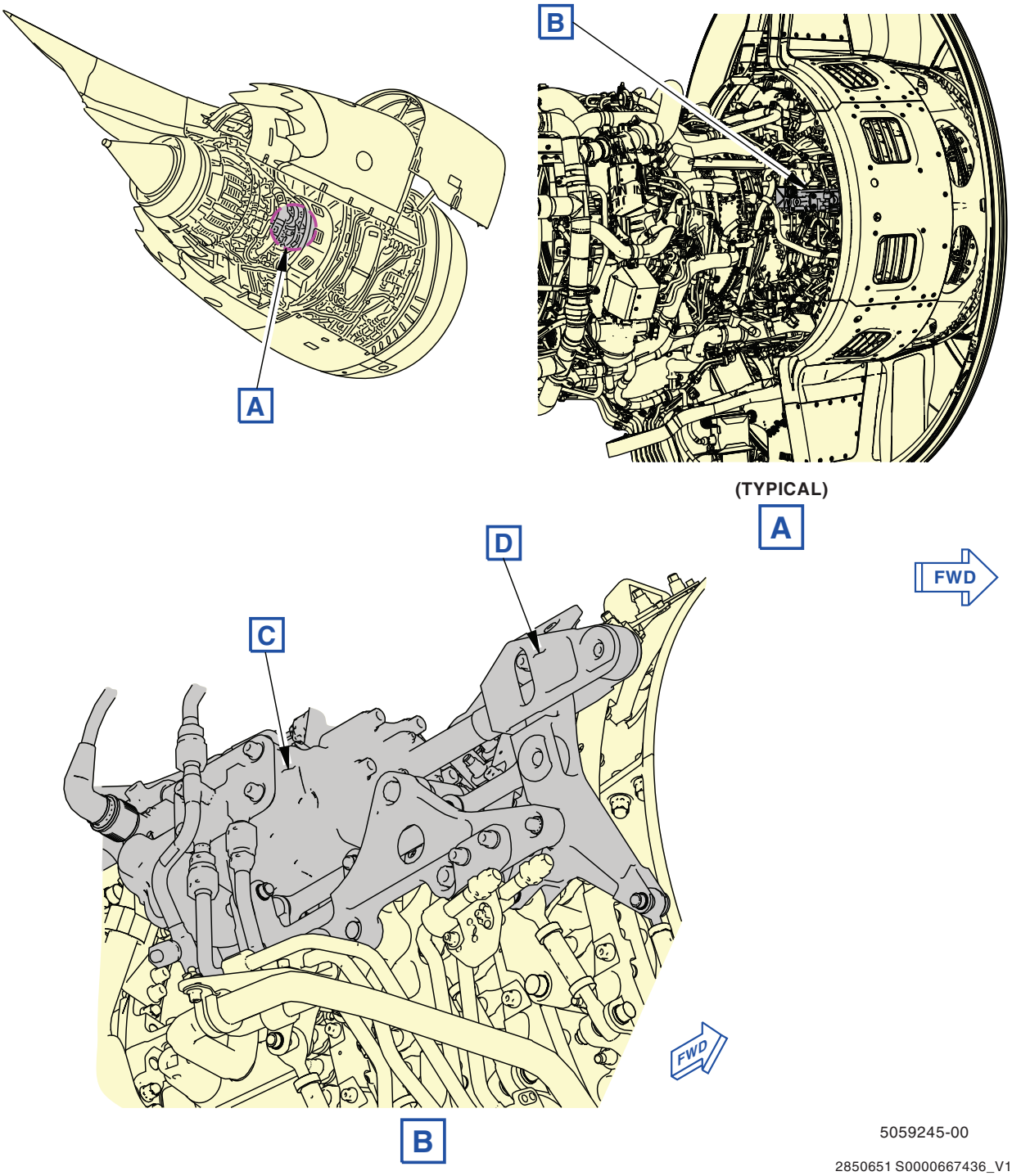
(a) Make sure that the bushing [64] stays in the right VSV actuator [43] housing.

SUBTASK 75-31-02-010-024-G00

(15) Remove the right VSV actuator [43] from the engine.

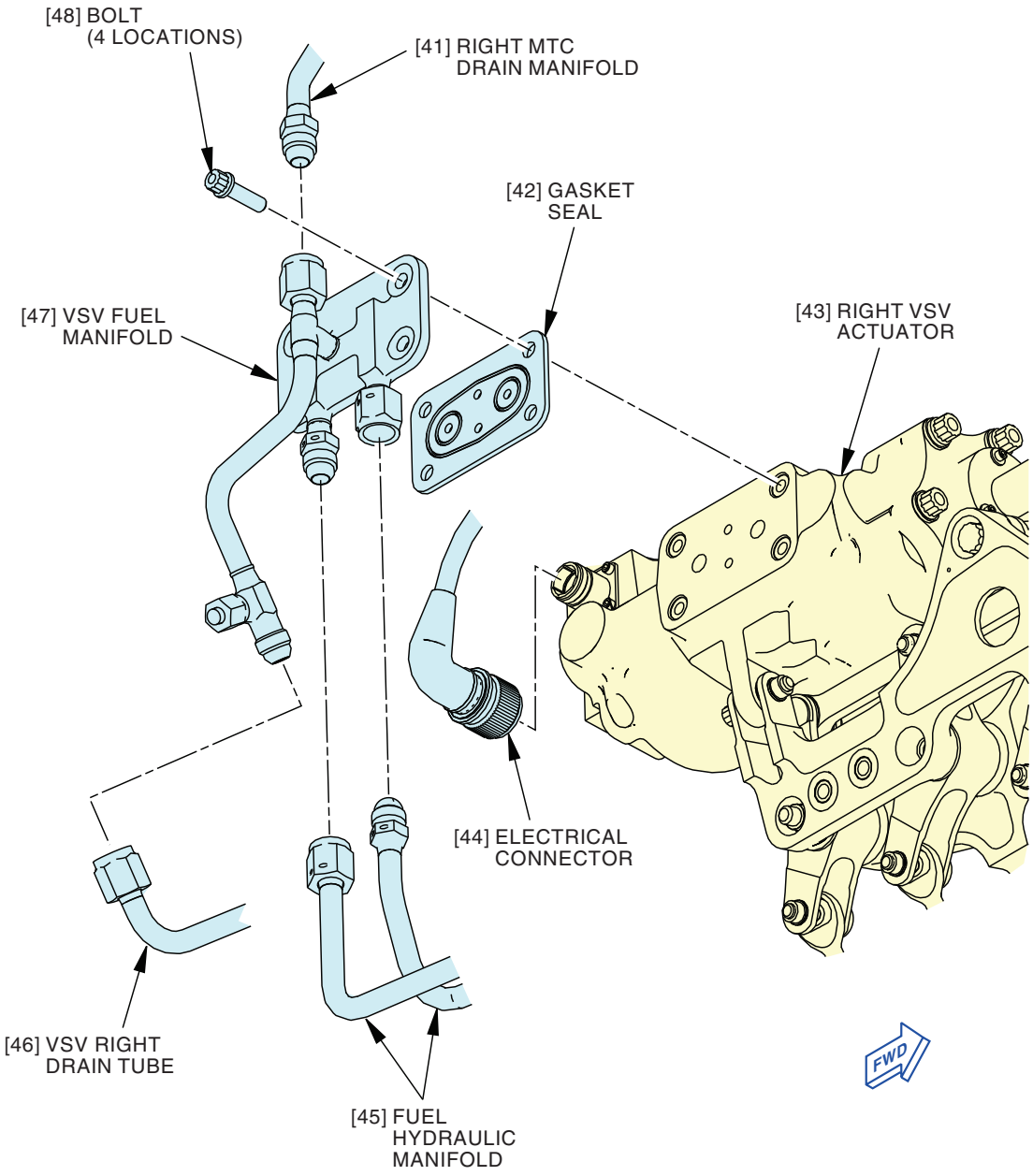
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



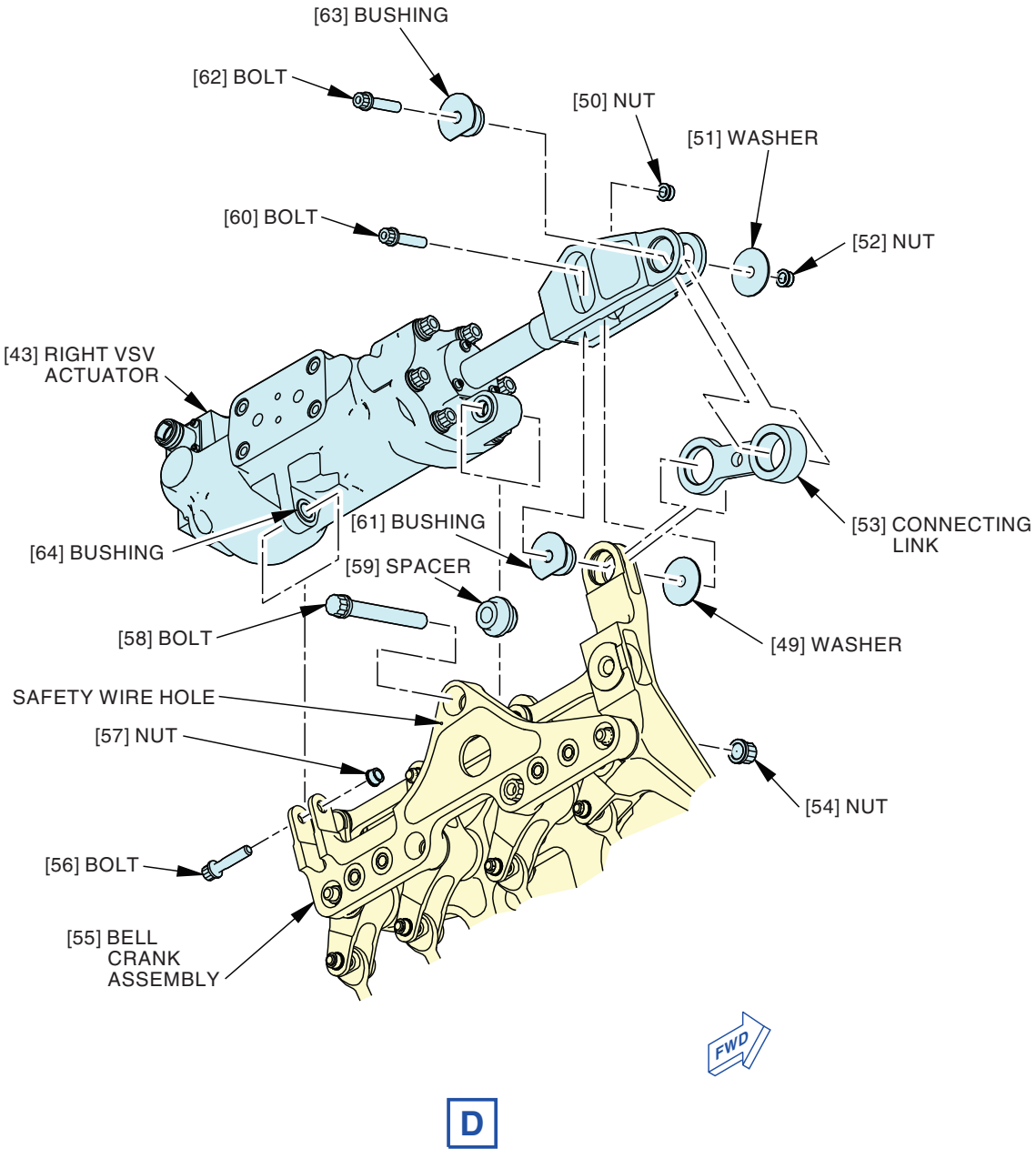
Right VSV Actuator Installation
Figure 402/75-31-02-990-802-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right VSV Actuator Installation
Figure 402/75-31-02-990-802-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

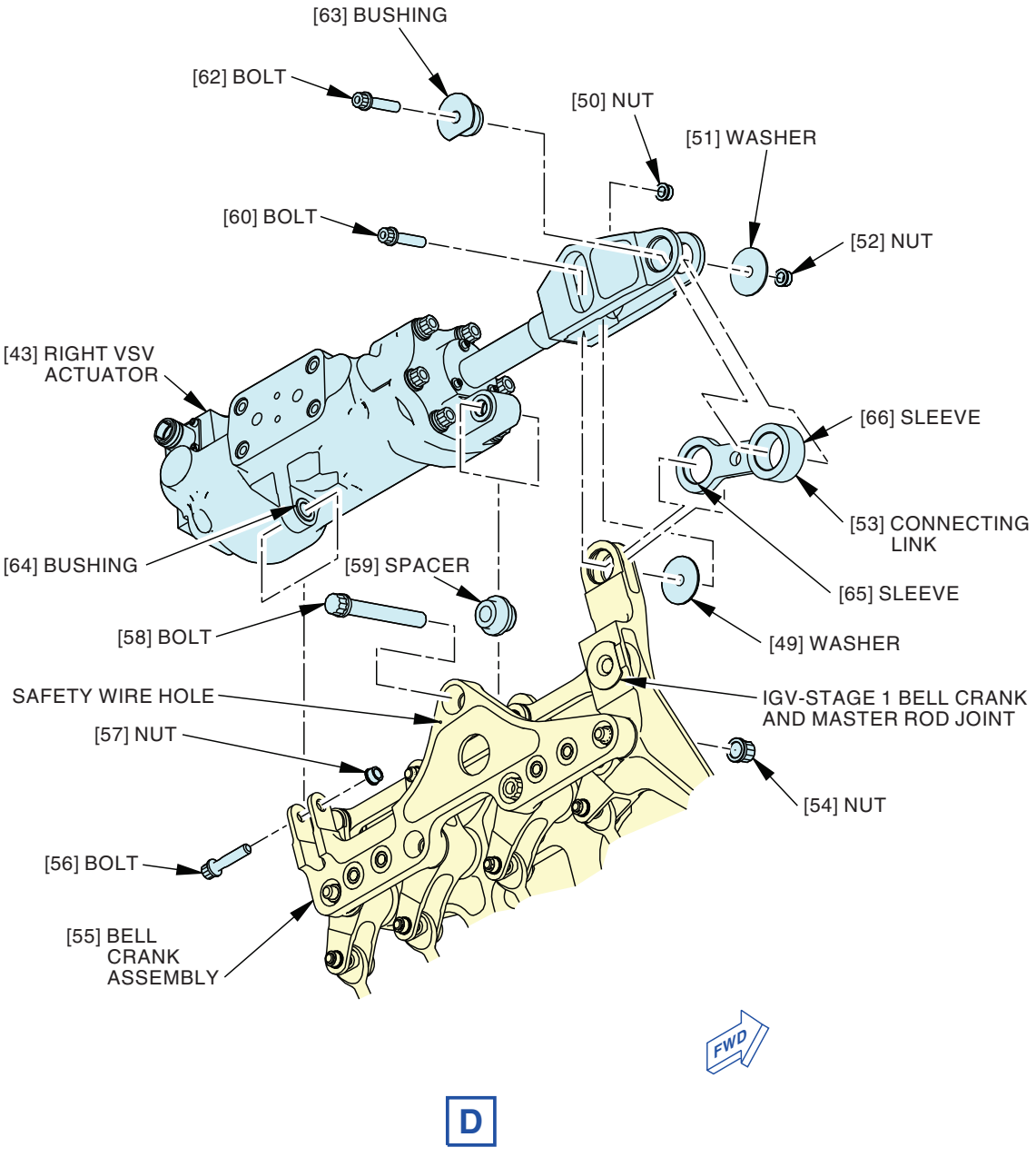


ICN-LEAP-1B-753102-B-07482-00007-A-001-01

2391579 S0000549998_V4

Right VSV Actuator Installation
Figure 402/75-31-02-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-753102-B-07482-00007-A-001-01

3010849 S0000784871_V1

Right VSV Actuator Installation
Figure 402/75-31-02-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-31-02-400-803-G00

5. Right Variable Stator Vane (VSV) Actuator Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the right Variable Stator Vane (VSV) actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
42	Gasket seal	73-11-00-20-410	PTF ALL
43	Right VSV actuator	75-31-02-01-010	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Right Variable Stator Vane (VSV) Actuator Installation

SUBTASK 75-31-02-640-002-G00

(1) Apply engine oil, D00599 [CP2442], on the threads of the nut [50], nut [52], nut [54], and nut [57].

PTF ALL; PRE LEAP-1B SB 72-0343

SUBTASK 75-31-02-420-006-G00

(2) Put the bushing [61], washer [49], and connecting link [53] into position on the right VSV actuator [43].

SUBTASK 75-31-02-420-007-G00

(3) Put the right VSV actuator [43] into position on the bell crank assembly [55].

SUBTASK 75-31-02-420-008-G00

(4) Loosely install the bolt [60] and nut [50] that attach the bushing [61], washer [49], and connecting link [53] to the bell crank assembly [55].

PTF ALL; POST LEAP-1B SB 72-0343

SUBTASK 75-31-02-420-012-G00

(5) Put the washer [49] and connecting link [53] into position on the right VSV actuator [43].

SUBTASK 75-31-02-420-013-G00

(6) Put the right VSV actuator [43] into position on the bell crank assembly [55].

SUBTASK 75-31-02-420-014-G00

(7) Loosely install the bolt [60] and nut [50] that attach the washer [49] and connecting link [53] to the bell crank assembly [55].

PTF ALL

SUBTASK 75-31-02-410-014-G00

(8) Loosely install the bolt [62], bushing [63], washer [51], and nut [52] that attach the right VSV actuator [43] to the connecting link [53].

PTF ALL; POST LEAP-1B SB 72-0343

SUBTASK 75-31-02-210-002-G00

(9) Check that the assembly fit of the connecting link [53] with the bushing [63] and bolt [56] has play on the direction perpendicular to the right VSV actuator [43] extending/retracting direction similar to the fit of IGV-stage 1 bell crank and master rod joint bell crank assembly [55].

(a) If the assembly fit feels loose, make sure that the sleeve [65] and sleeve [66] are correctly installed.

PTF ALL

SUBTASK 75-31-02-420-015-G00

(10) Make sure that the bushing [64] is on the right VSV actuator [43].

SUBTASK 75-31-02-420-016-G00

(11) Loosely install the bolt [56] and nut [57] that attach the right VSV actuator [43] to the bell crank assembly [55].

SUBTASK 75-31-02-410-016-G00

(12) Loosely install the bolt [58], spacer [59], and nut [54] that attach the bell crank assembly [55] to the right VSV actuator [43].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-31-02-410-017-G00

(13) Tighten the nut [50], nut [52], nut [57], and nut [54] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 75-31-02-010-026-G00

(14) Remove the protective covers from the right VSV actuator [43], VSV fuel manifold [47], fuel hydraulic manifolds [45], VSV right drain tube [46], and right MTC drain manifold [41].

SUBTASK 75-31-02-410-018-G00

(15) Connect the VSV fuel manifold [47] to the right VSV actuator [43] as follows:

(a) Install the new gasket seal [42] between the VSV fuel manifold [47] and right VSV actuator [43].

(b) Apply engine oil, D00599 [CP2442], on the threads of the four bolts [48].

(c) Install the four bolts [48] that attach the VSV fuel manifold [47] to the right VSV actuator [43].

(d) Tighten the bolts [48] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 75-31-02-410-019-G00

(16) Connect the fuel hydraulic manifolds [45] to the VSV fuel manifold [47] as follows:

(a) Connect the fuel hydraulic manifold [45] B-nut to the VSV fuel manifold [47].

(b) Connect the VSV fuel manifold [47] B-nut to the fuel hydraulic manifold [45].

(c) Tighten the two B-nuts to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

SUBTASK 75-31-02-410-020-G00

(17) Connect the VSV right drain tube [46] to the VSV fuel manifold [47] as follows:

(a) Connect the VSV right drain tube [46] B-nut to the VSV fuel manifold [47].

(b) Tighten the VSV right drain tube [46] B-nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

SUBTASK 75-31-02-410-021-G00

(18) Connect the right MTC drain manifold [41] to the right VSV actuator [43] as follows:

(a) Connect the VSV fuel manifold [47] B-nut to the right MTC drain manifold [41].

(b) Tighten the VSV fuel manifold [47] B-nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

SUBTASK 75-31-02-420-011-G00

(19) Install safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006], on the B-nuts and fittings that connect the VSV fuel manifold [47] to the fuel hydraulic manifolds [45] (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

SUBTASK 75-31-02-420-009-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(20) Connect the electrical connector [44] as follows (TASK 70-00-01-910-803-G00):

(a) Remove the protective covers from the electrical receptacle on the right VSV actuator [43] and electrical connector [44].

(b) Connect the electrical connector [44] to the right VSV actuator [43].

(c) Use a teflon-jawed pliers, STD-664, to tighten the electrical connector [44].

SUBTASK 75-31-02-410-022-G00

(21) Install the segment shroud 2 (TASK 72-23-04-400-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-31-02-860-004-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-31-02-410-023-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. Right Variable Stator Vane Actuator (VSV) Installation Test

SUBTASK 75-31-02-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Left variable bleed valve actuator removal

(2) Left variable bleed valve actuator installation

(3) Right variable bleed valve actuator removal

(4) Right variable bleed valve actuator installation.

TASK 75-32-01-000-801-G00

2. Left Variable Bleed Valve Actuator Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the left Variable Bleed Valve (VBV) actuator.

(2) The left VBV actuator is at the 7:30 o'clock position on the fan hub frame.

(3) You must open the left thrust reverser to get access to the left VBV actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-32-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-32-01-860-005-G00

(2) Make sure that the VBV doors are in the closed position.

SUBTASK 75-32-01-010-001-G00

(3) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. Left Variable Bleed Valve Actuator Removal

SUBTASK 75-32-01-010-002-G00

(1) Remove the segment shroud 4 (TASK 72-23-04-000-801-G00).

SUBTASK 75-32-01-030-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(2) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [9] from the left VBV actuator [8] as follows:

(a) Disconnect the electrical connector [9] from the left VBV actuator [8] (TASK 70-00-01-910-803-G00).

(b) Put the protective covers on the electrical connector [9] and the electrical receptacle of the left VBV actuator [8].

SUBTASK 75-32-01-030-002-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Disconnect the VBV fuel manifold [11] from the left VBV actuator [8] as follows:

(a) Remove the four bolts [12] that attach the VBV fuel manifold [11] to the left VBV actuator [8].

(b) Put the fuel resistant container (5 gal), STD-201 below the left VBV actuator [8] and the VBV fuel manifold [11].

(c) Loosen the VBV fuel manifold [11] to permit the gasket seal [10] to fall out.

NOTE: When the VBV fuel manifold [11] loosens, the fuel will drain into the container.

(d) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

SUBTASK 75-32-01-020-001-G00

(4) Discard the gasket seal [10].

SUBTASK 75-32-01-020-002-G00

(5) Remove the three captive bolts [13] that attach the left VBV actuator [8] to the fan hub frame.

SUBTASK 75-32-01-030-003-G00

(6) Move the left VBV actuator [8] away from the fan hub frame enough so you can access the bolt [14].

SUBTASK 75-32-01-020-003-G00

(7) Remove the bolt [14] that attaches the left VBV actuator [8] rod from the actuator turnbuckle clevis [15].

SUBTASK 75-32-01-020-004-G00

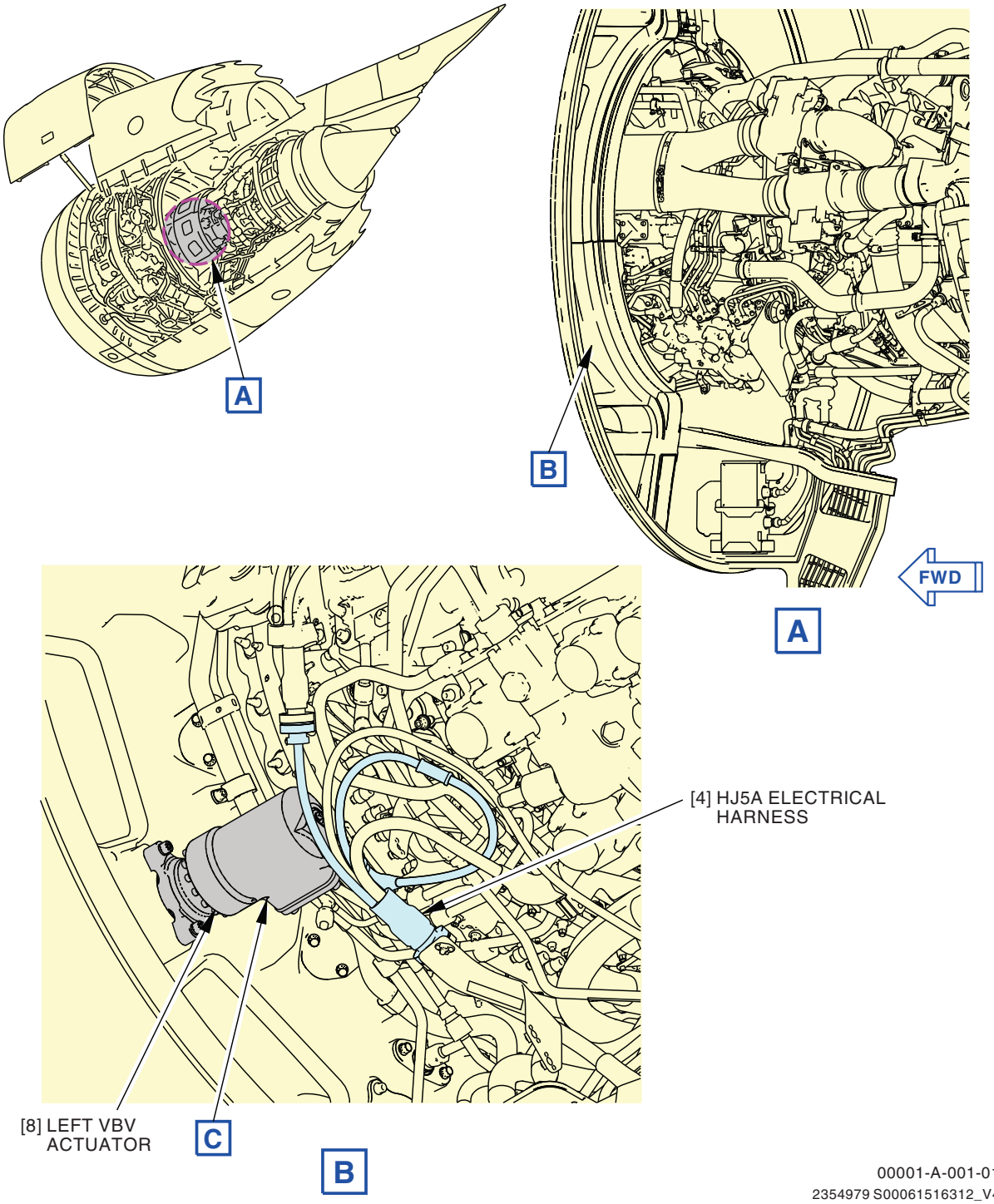
(8) Remove the left VBV actuator [8] from the engine.

SUBTASK 75-32-01-530-001-G00

(9) Install the protective covers on the left VBV actuator [8] and the VBV fuel manifold [11].

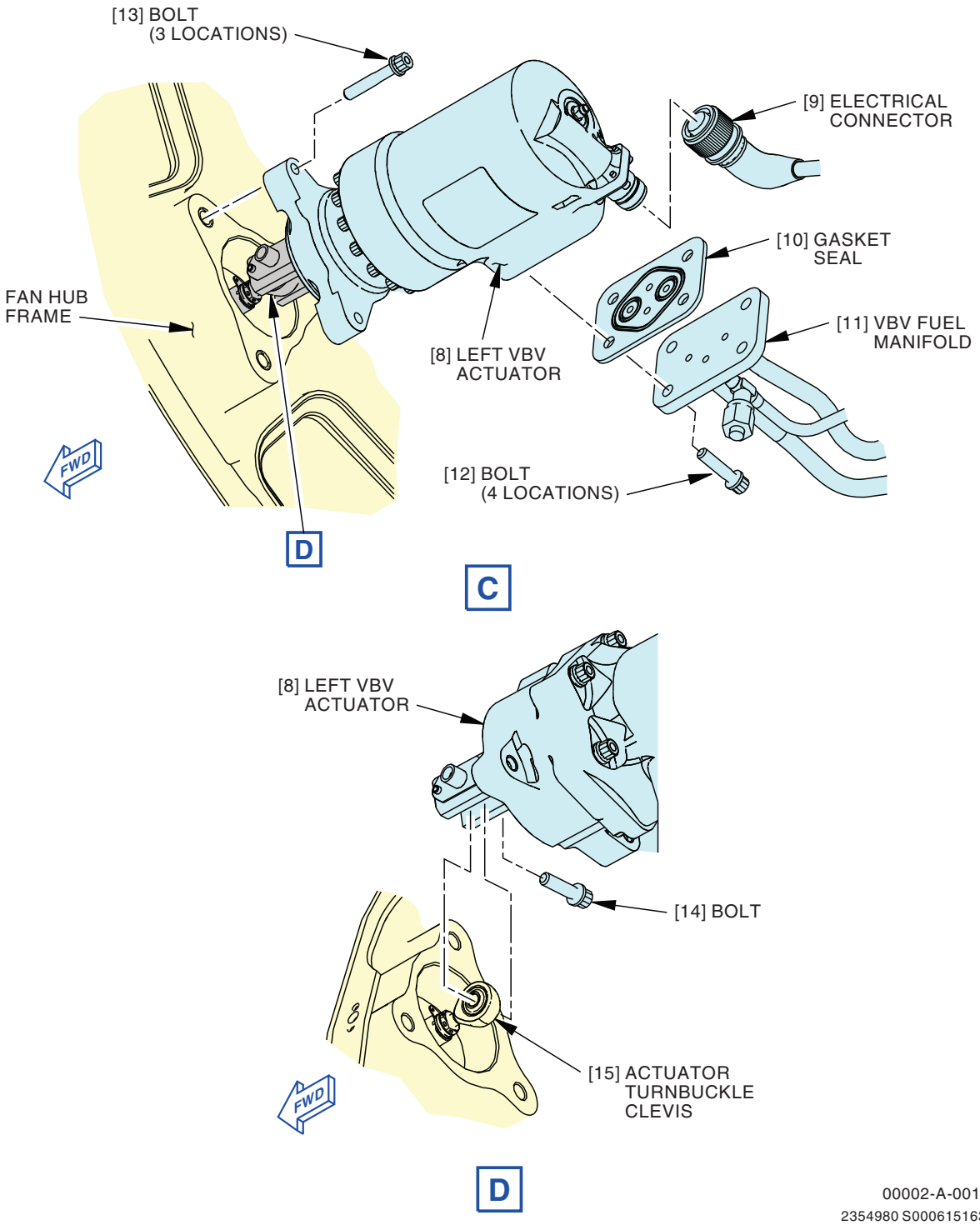
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left VBV Actuator Installation
Figure 401/75-32-01-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left VBV Actuator Installation
Figure 401/75-32-01-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-32-01-400-802-G00

3. Left Variable Bleed Valve Actuator Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the left Variable Bleed Valve (VBV) actuator.

(2) You must do the tests that are listed in the power plant test reference table after you install the left VBV actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Left VBV actuator	75-32-01-01-290	PTF ALL
10	Gasket seal	75-32-01-01-270	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. Left Variable Bleed Valve Actuator Installation

SUBTASK 75-32-01-540-001-G00

(1) Remove the protective covers from the VBV fuel manifold [11] and the left VBV actuator [8].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-32-01-420-001-G00

CAUTION DO NOT MAKE ADJUSTMENTS OR CHANGE THE POSITION OF THE VBV ACTUATOR ROD. IF YOU MAKE THE ADJUSTMENT ON THE VBV ACTUATOR ROD, YOU WILL CHANGE THE VBV SYSTEM RIGGING OR CAN CAUSE ENGINE PROBLEMS OR DAMAGE.

(2) Install the left VBV actuator [8] on the fan hub frame as follows:

(a) Put the left VBV actuator [8] into its position on the fan hub frame.

(b) Put graphite mineral vaseline, D00601 [CP2101] on the threads of the bolt [14].

(c) Pull the left VBV actuator [8] away from the fan hub frame enough to install the bolt [14].

(d) Install the bolt [14] that attaches the left VBV actuator [8] rod to the actuator turnbuckle clevis [15].

1) Tighten the bolt [14] to 65.0 in-lb (7.34 N·m) - 77.0 in-lb (8.70 N·m).

(e) Install the three captive bolts [13] that attach the left VBV actuator [8] to the fan hub frame.

NOTE: Do not tighten the bolts at this time.

SUBTASK 75-32-01-430-001-G00

(3) Connect the VBV fuel manifold [11] to the left VBV actuator [8] as follows:

(a) Put a new new gasket seal [10] between the VBV fuel manifold [11] and the left VBV actuator [8].

NOTE: If necessary, tighten the captive bolts [13] to help with the alignment of the gasket seal [10], the VBV fuel manifold [11], and the left VBV actuator [8].

(b) Apply engine oil, D00599 [CP2442] on the threads of the four bolts [12].

(c) Install the four bolts [12] that attach the VBV fuel manifold [11] to the left VBV actuator [8].

(d) Tighten the bolts [12] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(e) Tighten the bolts [13] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 75-32-01-430-002-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(4) Use teflon-jawed pliers, STD-664 to connect the electrical connector [9] to the left VBV actuator [8] as follows:

(a) Remove the protective covers from the electrical connector [9] and the electrical receptacle of the left VBV actuator [8].

(b) Connect the electrical connector [9] to the left VBV actuator [8] (TASK 70-00-01-910-803-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-32-01-410-005-G00

(5) Install the segment shroud 4 (TASK 72-23-04-400-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-32-01-860-002-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-32-01-410-006-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

J. Left Variable Bleed Valve Actuator Installation Test

SUBTASK 75-32-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 75-32-01-000-802-G00

4. Right Variable Bleed Valve Actuator Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the right Variable Bleed Valve (VBV) actuator.

(2) The right VBV actuator is at the 3:00 o'clock position on the fan hub frame.

(3) You must open the right thrust reverser to get access to the right VBV actuator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-32-01-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-32-01-860-006-G00

(2) Make sure that the VBV doors are in the closed position.

SUBTASK 75-32-01-010-006-G00

(3) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

G. Right Variable Bleed Valve Actuator Removal

SUBTASK 75-32-01-010-007-G00

(1) Remove segment shroud 2 (TASK 72-23-04-000-801-G00).

SUBTASK 75-32-01-010-009-G00

(2) Remove the bolt [35] that attaches the loop clamp [34] and HJ5B electrical harness [33] to the support bracket [36].

SUBTASK 75-32-01-010-010-G00

(3) Remove the W200 electrical harness [38], HJ8A electrical harness [37], HJ8B electrical harness [31], and W300 electrical harness [32] from the clips on support bracket [36].

SUBTASK 75-32-01-010-011-G00

(4) Remove the support bracket [36] from the engine.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

SUBTASK 75-32-01-010-012-G00

(5) Remove the bolt [41] that attaches the loop clamp [40] and HJ8A electrical harness [37] to the support bracket [39].

SUBTASK 75-32-01-010-013-G00

(6) Remove the W200 electrical harness [38], HJ8B electrical harness [31], and W300 electrical harness [32] from the clips on the support bracket [39].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 75-32-01-010-015-G00

(7) Remove the HJ8A electrical harness [37], W200 electrical harness [38], HJ8B electrical harness [31], and W300 electrical harness [32] from the clips on the support bracket [39].

PTF ALL

SUBTASK 75-32-01-010-014-G00

(8) Remove the support bracket [39] from the engine.

SUBTASK 75-32-01-030-004-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(9) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [46] from the right VBV actuator [47] as follows:

(a) Disconnect the electrical connector [46] from the right VBV actuator [47] (TASK 70-00-01-910-803-G00).

(b) Put the protective covers on the electrical connector [46] and electrical receptacle on the right VBV actuator [47].

SUBTASK 75-32-01-030-005-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Disconnect the VBV fuel manifold [44] from the right VBV actuator [47] as follows:

WARNING

WHEN YOU DO WORK WITH SAFETY CABLES, MAKE SURE YOU USE SAFETY GOGGLES. IF PART OF THE SAFETY CABLE BREAKS, INJURY TO PERSONS CAN OCCUR.

(a) Cut and remove the safety cable or safety wire from the VBV fuel manifold [44] B-nuts at fuel manifold head end line [28] and fuel manifold rod end line [30].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Put the fuel resistant container (5 gal), STD-201, below the right VBV actuator [47].

(c) Disconnect the VBV fuel manifold [44] B-nuts from the fuel manifold head end line [28], fuel manifold rod end line [30], and fuel manifold drain line [29].

(d) Remove the four bolts [45] that attach the VBV fuel manifold [44] to the right VBV actuator [47].

(e) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

SUBTASK 75-32-01-020-006-G00

(11) Remove the three bolts [42] that attach the right VBV actuator [47] to the fan hub frame.

SUBTASK 75-32-01-020-005-G00

(12) Remove and discard the gasket seal [43] from between the VBV fuel manifold [44] and right VBV actuator [47].

SUBTASK 75-32-01-030-006-G00

(13) Move the right VBV actuator [47] away from the fan hub frame enough so you can access the bolt [49].

SUBTASK 75-32-01-020-007-G00

(14) Remove the bolt [49] that attaches the right VBV actuator [47] rod to the actuator turnbuckle clevis [48].

SUBTASK 75-32-01-020-008-G00

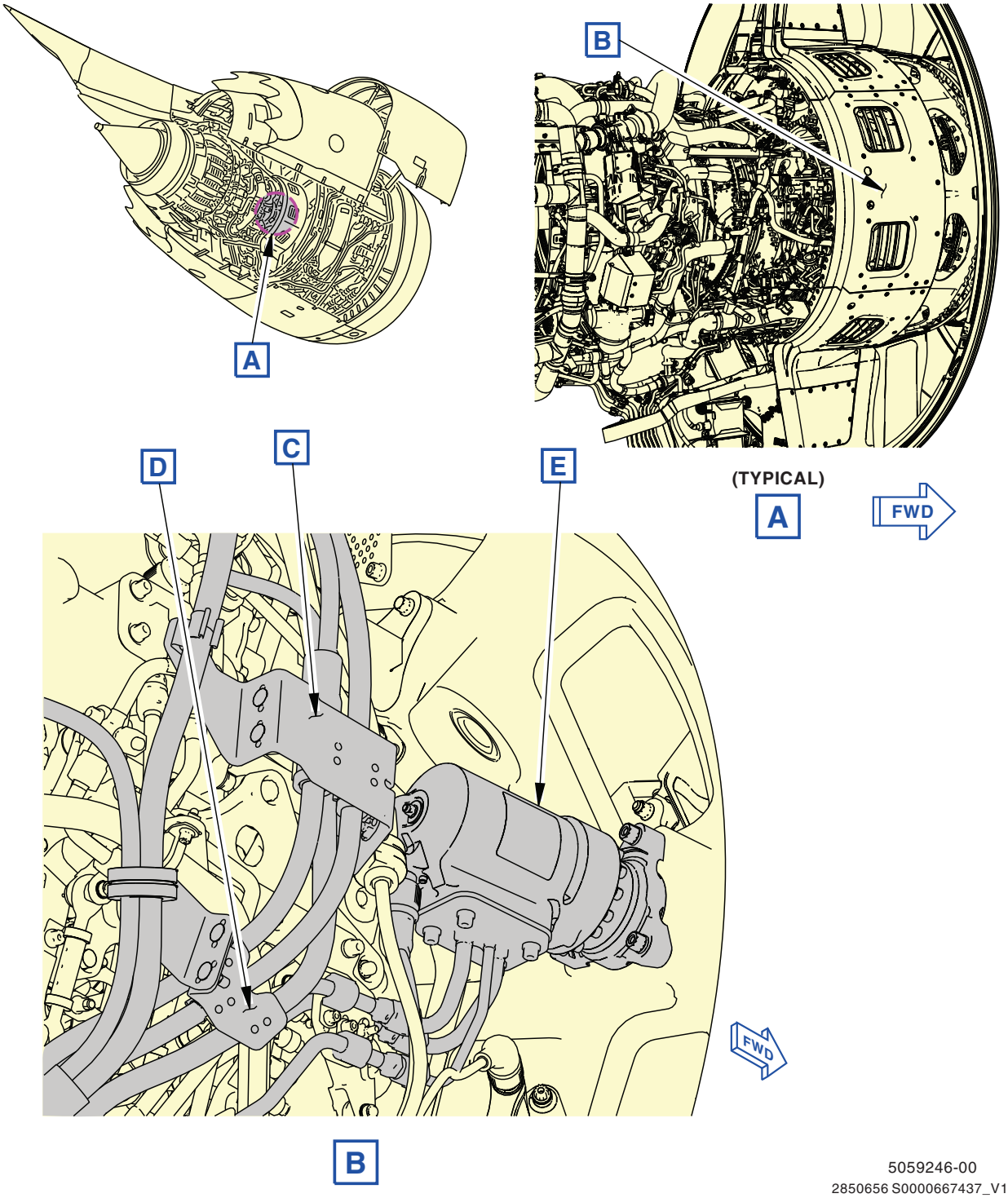
(15) Remove the right VBV actuator [47] from the engine.

SUBTASK 75-32-01-530-002-G00

(16) Install the protective covers on the right VBV actuator [47] and VBV fuel manifold [44].

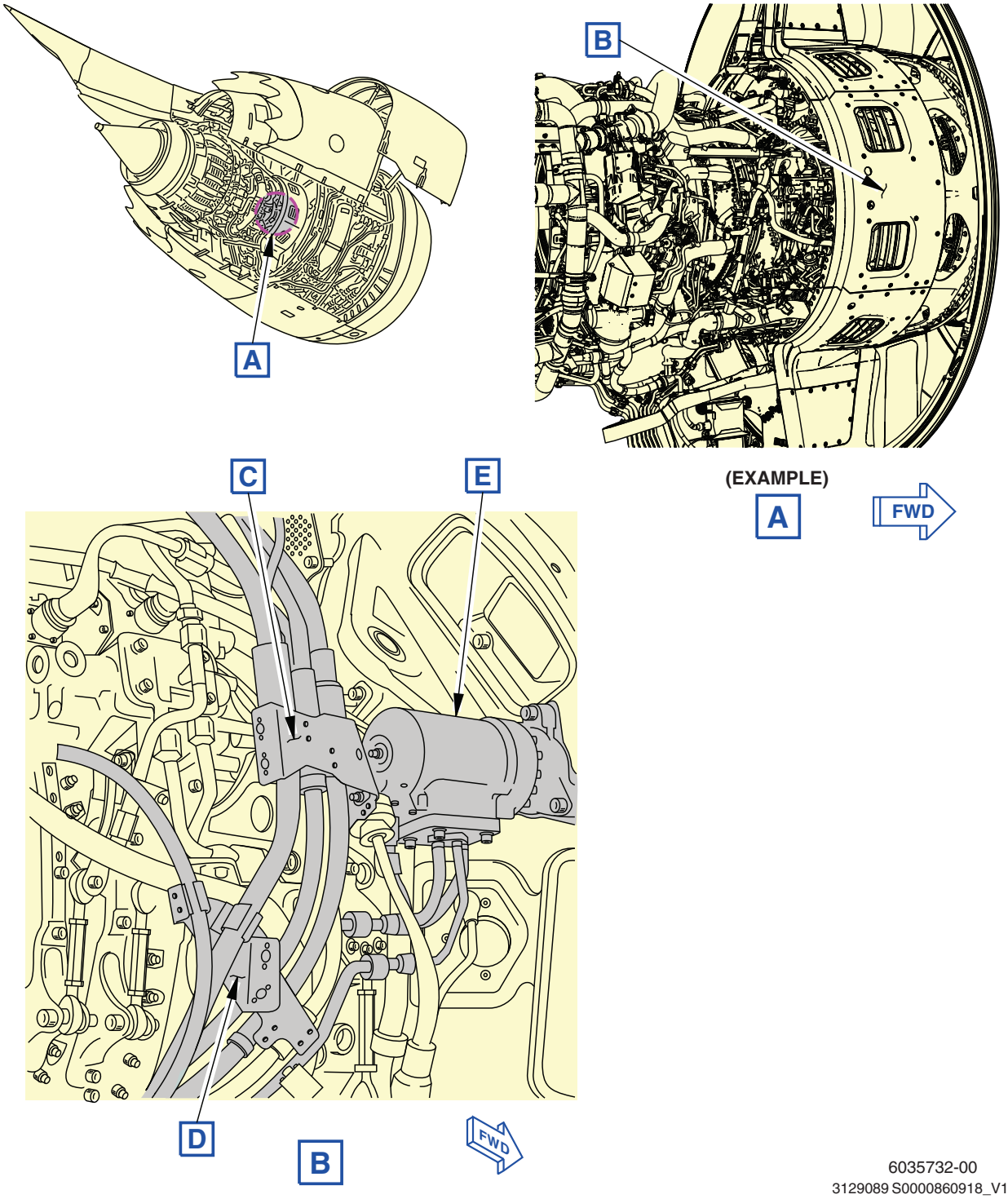
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



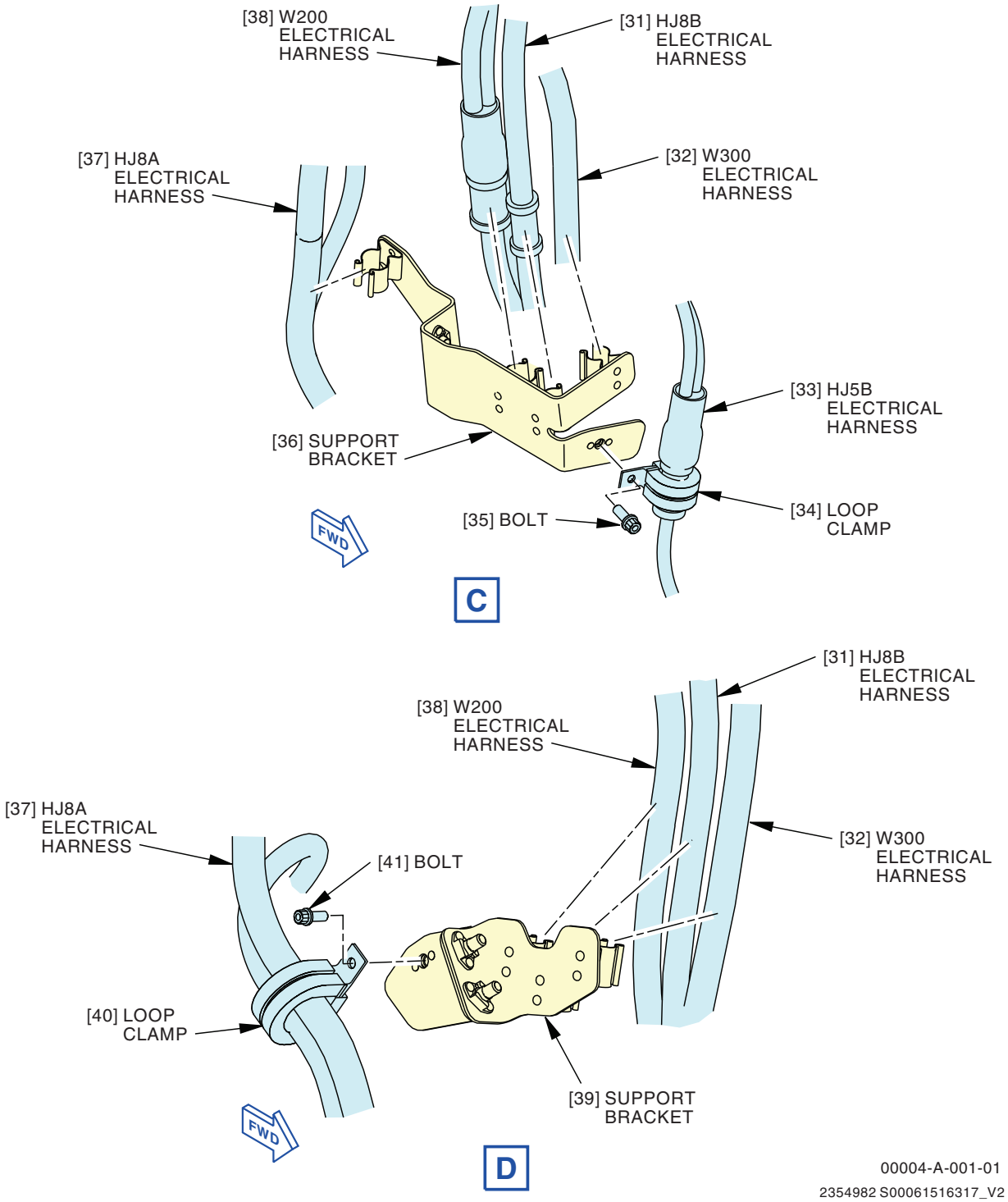
Right VBV Actuator Installation
Figure 402/75-32-01-990-802-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



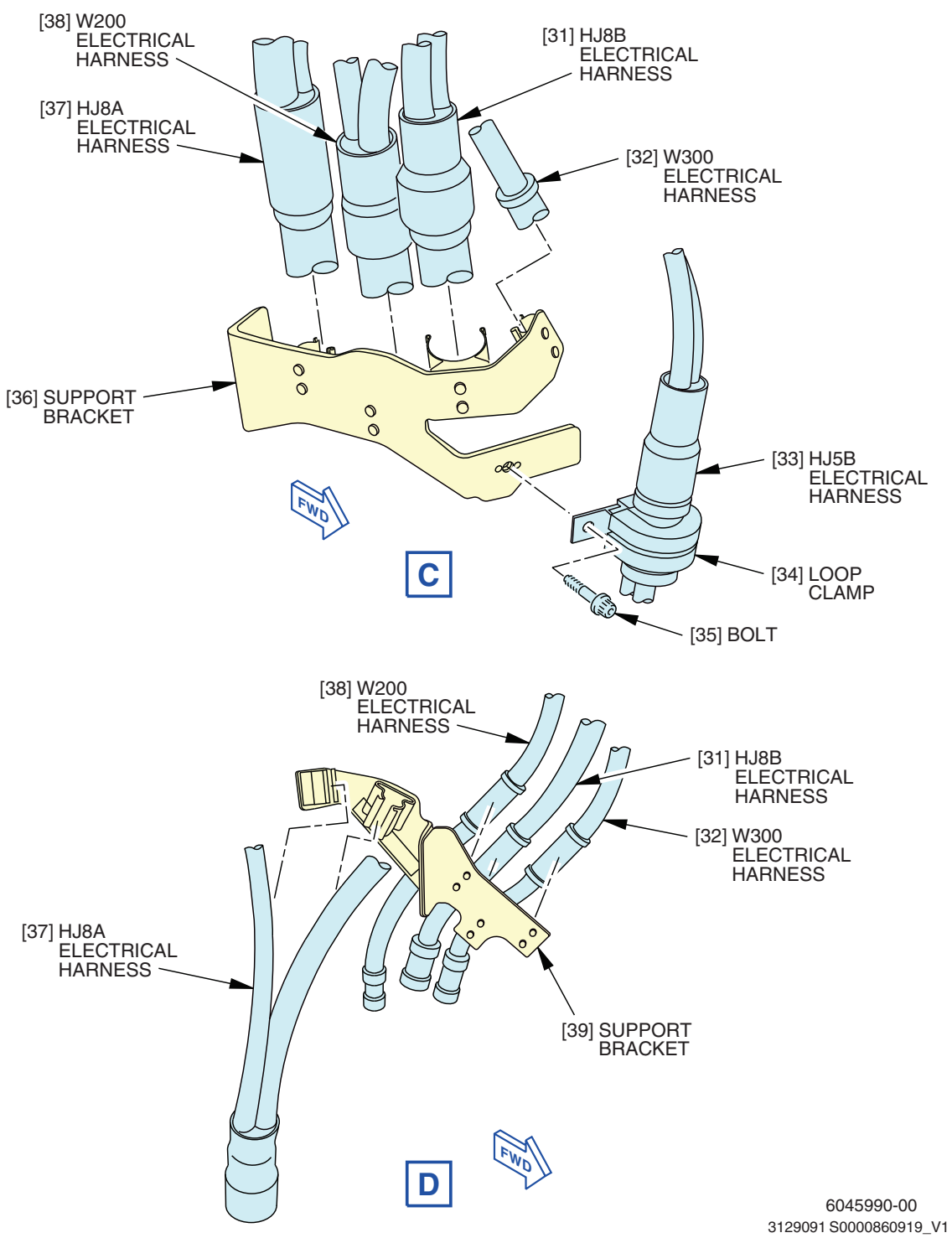
Right VBV Actuator Installation
Figure 402/75-32-01-990-802-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



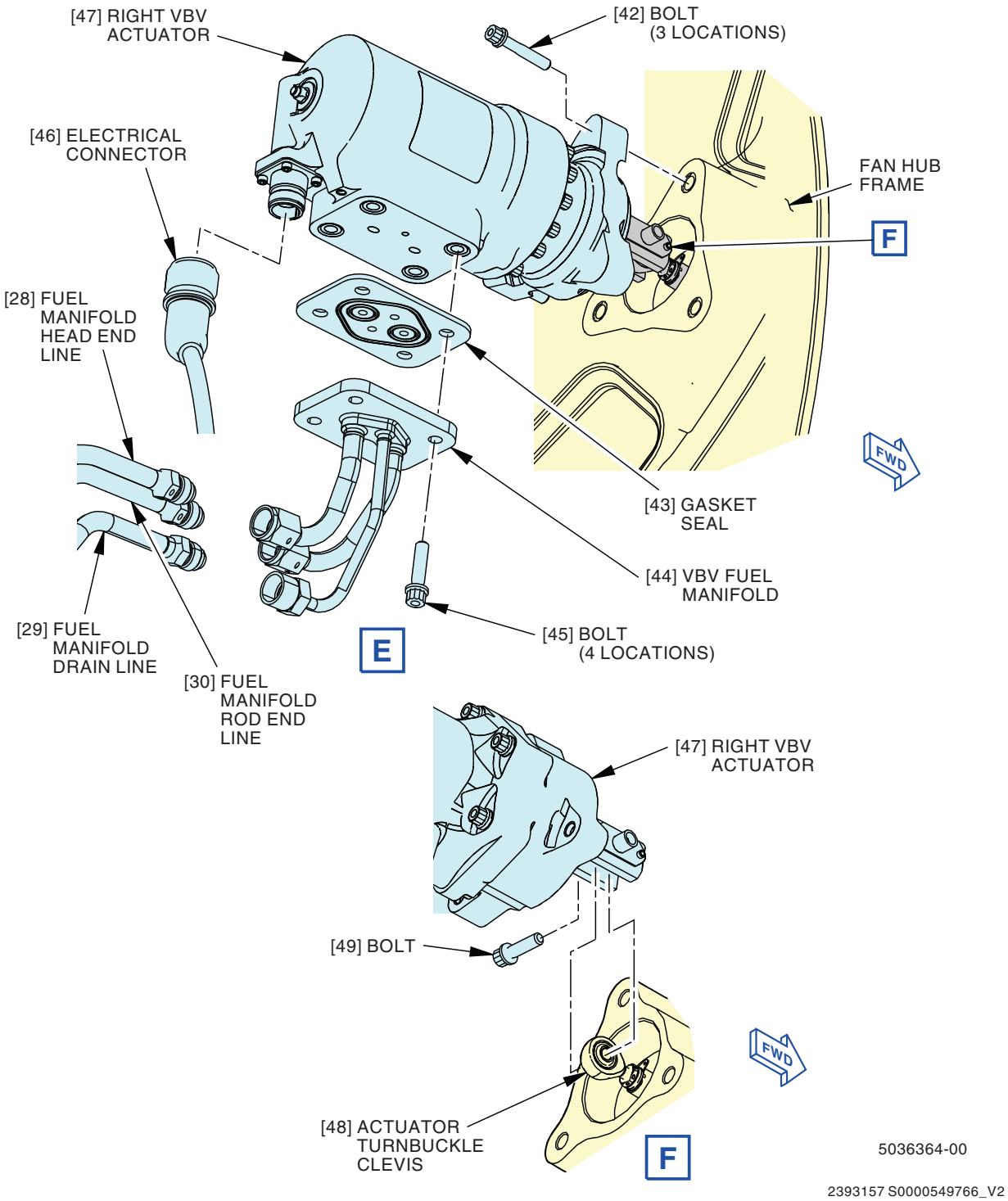
Right VBV Actuator Installation
Figure 402/75-32-01-990-802-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right VBV Actuator Installation
Figure 402/75-32-01-990-802-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right VBV Actuator Installation
Figure 402/75-32-01-990-802-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-32-01-400-803-G00

5. Right Variable Bleed Valve Actuator Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the right Variable Bleed Valve (VBV) actuator.

(2) You must do the tests that are listed in the power plant test reference table after you install the right VBV actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
G02495 [CP8002]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS5689
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
43	Gasket seal	75-32-01-01-090	PTF ALL
47	Right VBV actuator	75-32-01-01-110	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

H. Right Variable Bleed Valve Actuator Installation

SUBTASK 75-32-01-540-002-G00

(1) Remove the protective covers from the VBV fuel manifold [44] and right VBV actuator [47].

SUBTASK 75-32-01-420-002-G00

CAUTION DO NOT MAKE ADJUSTMENTS OR CHANGE THE POSITION OF THE VBV ACTUATOR ROD. IF YOU MAKE THE ADJUSTMENT ON THE VBV ACTUATOR ROD, YOU WILL CHANGE THE VBV SYSTEM RIGGING OR CAN CAUSE ENGINE PROBLEMS OR DAMAGE.

(2) Install the right VBV actuator [47] on the fan hub frame as follows:

(a) Put the right VBV actuator [47] in position on the fan hub frame.

(b) Put graphite mineral vaseline, D00601 [CP2101], on the threads of the bolt [49].

(c) Pull the right VBV actuator [47] away from the fan hub frame enough to install the bolt [49].

(d) Install the bolt [49] that attaches the right VBV actuator [47] rod to the actuator turnbuckle clevis [48].

1) Tighten the bolt [49] to 65.0 in-lb (7.34 N·m) - 77.0 in-lb (8.70 N·m).

(e) Install the three bolts [42] that attach the right VBV actuator [47] to the fan hub frame.

NOTE: Do not tighten the bolts at this time.

SUBTASK 75-32-01-430-003-G00

(3) Connect the VBV fuel manifold [44] to the right VBV actuator [47] as follows:

(a) Put a new new gasket seal [43] between the VBV fuel manifold [44] and right VBV actuator [47].

(b) Apply engine oil, D00599 [CP2442], on the threads of the four bolts [45].

(c) Install the four bolts [45] that attach the VBV fuel manifold [44] to the right VBV actuator [47].

(d) Connect the VBV fuel manifold [44] B-nuts to the fuel manifold head end line [28], fuel manifold rod end line [30], and fuel manifold drain line [29].

(e) Tighten the bolts [42] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(f) Tighten the bolts [45] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(g) Tighten the fuel manifold B-nuts to 285 in-lb (32.2 N·m).

1) Apply the triple torque procedure.

WARNING WHEN YOU DO WORK WITH SAFETY CABLES, MAKE SURE YOU USE SAFETY GOGGLES. IF PART OF THE SAFETY CABLE BREAKS, INJURY TO PERSONS CAN OCCUR.

(h) Tightly attach the VBV fuel manifold [44] B-nuts at the fuel manifold head end line [28] and fuel manifold rod end line [30] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00 or TASK 70-41-00-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-32-01-430-004-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(4) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [46] to the right VBV actuator [47] as follows:

(a) Remove the protective covers from the electrical connector [46] and electrical receptacle on the right VBV actuator [47].

(b) Connect the electrical connector [46] to the right VBV actuator [47] (TASK 70-00-01-910-803-G00).

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

SUBTASK 75-32-01-640-002-G00

(5) Apply engine oil, D00599 [CP2442], on the threads of the bolt [35] and bolt [41].

SUBTASK 75-32-01-410-007-G00

(6) Install the bolt [41] that attaches the loop clamp [40] and HJ8A electrical harness [37] to the support bracket [39].

(a) Tighten the bolt [41] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

SUBTASK 75-32-01-410-008-G00

(7) Install the W200 electrical harness [38], HJ8B electrical harness [31], and W300 electrical harness [32] to the clips on the support bracket [39].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 75-32-01-410-014-G00

(8) Install the HJ8A electrical harness [37], W200 electrical harness [38], HJ8B electrical harness [31], and W300 electrical harness [32] to the clips on the support bracket [39].

PTF ALL

SUBTASK 75-32-01-410-009-G00

(9) Install the bolt [35] that attaches the loop clamp [34] and HJ5B electrical harness [33] to the support bracket [36].

(a) Tighten the bolt [35] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

SUBTASK 75-32-01-410-010-G00

(10) Install the W200 electrical harness [38], HJ8A electrical harness [37], HJ8B electrical harness [31], and W300 electrical harness [32] to the clips on the support bracket [36].

SUBTASK 75-32-01-410-012-G00

(11) Install the segment shroud 2 (TASK 72-23-04-400-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-32-01-860-004-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-32-01-410-013-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

J. Right Variable Bleed Valve Actuator Installation Test

SUBTASK 75-32-01-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRANSIENT BLEED VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Transient bleed valve removal

(2) Transient bleed valve installation.

TASK 75-33-01-000-801-G00

2. Transient Bleed Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the transient bleed valve.

(2) The transient bleed valve is referred to as the TBV.

(3) The transient bleed valve is installed at the 10:00 o'clock position on the engine core.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-33-01-010-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-33-01-865-001-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-33-01-010-002-G00

(3) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Transient Bleed Valve Removal

SUBTASK 75-33-01-030-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [4] from the transient bleed valve [1] as follows:

(a) Disconnect the electrical connector [4] from the transient bleed valve [1] (TASK 70-00-01-910-803-G00).

(b) Put the protective covers on the electrical connector [4] and electrical receptacles on the transient bleed valve [1].

SUBTASK 75-33-01-010-003-G00

(2) Disconnect the start and TBV manifolds [2] from the transient bleed valve [1] as follows:

(a) Cut and remove the safety wire or safety cable from the B-nuts on the start and TBV manifolds [2].

(b) Disconnect the start and TBV manifolds [2] from the transient bleed valve [1].

SUBTASK 75-33-01-010-004-G00

(3) Disconnect the BAI SBV drain tube [3] from the transient bleed valve [1].

SUBTASK 75-33-01-020-001-G00

(4) Loosen the coupling [22] that attaches the transient bleed valve [1] to the air tube [7].

SUBTASK 75-33-01-020-002-G00

(5) Hold the transient bleed valve [1] and loosen the coupling [10] that attaches the transient bleed valve [1] to the transient bleed air tube [9].

SUBTASK 75-33-01-020-003-G00

(6) Loosen the two bolts [20] and two nuts [21] that attach the bracket [18] to the air tube [7].

SUBTASK 75-33-01-020-004-G00

(7) Loosen the two bolts [24] that attach the bracket [17] to the bracket [18].

SUBTASK 75-33-01-020-005-G00

(8) Remove the bolts [25] that attach the bracket [18] to the adapter cap [19].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-33-01-020-006-G00

(9) Loosen the coupling [15] that connects the air tube [14] to the air tube [7].

PTF ALL

SUBTASK 75-33-01-020-007-G00

(10) Remove the two bolts [13] that attach the saddle clamp [12] and air tube [7] to the bracket [17].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-33-01-010-005-G00

(11) Remove the four bolts [6] that attach the air tube [7] to the turbine rear vane [5].

SUBTASK 75-33-01-010-006-G00

(12) Remove the four bolts [8] that attach the air tube [7] to the turbine rear vane [5].

SUBTASK 75-33-01-010-007-G00

(13) Remove the air tube [7] from the engine.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-33-01-020-009-G00

(14) Remove the coupling [10], coupling [22], and coupling [15].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-33-01-020-010-G00

(15) Remove the coupling [10] and coupling [22].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-33-01-020-008-G00

(16) Remove the seal [16] between the air tube [14] and air tube [7].

(a) Discard the seal [16].

PTF ALL

SUBTASK 75-33-01-010-010-G00

(17) Remove the transient bleed valve [1] from the engine.

SUBTASK 75-33-01-010-011-G00

(18) Remove the seal [11] and seal [23] from the transient bleed valve [1].

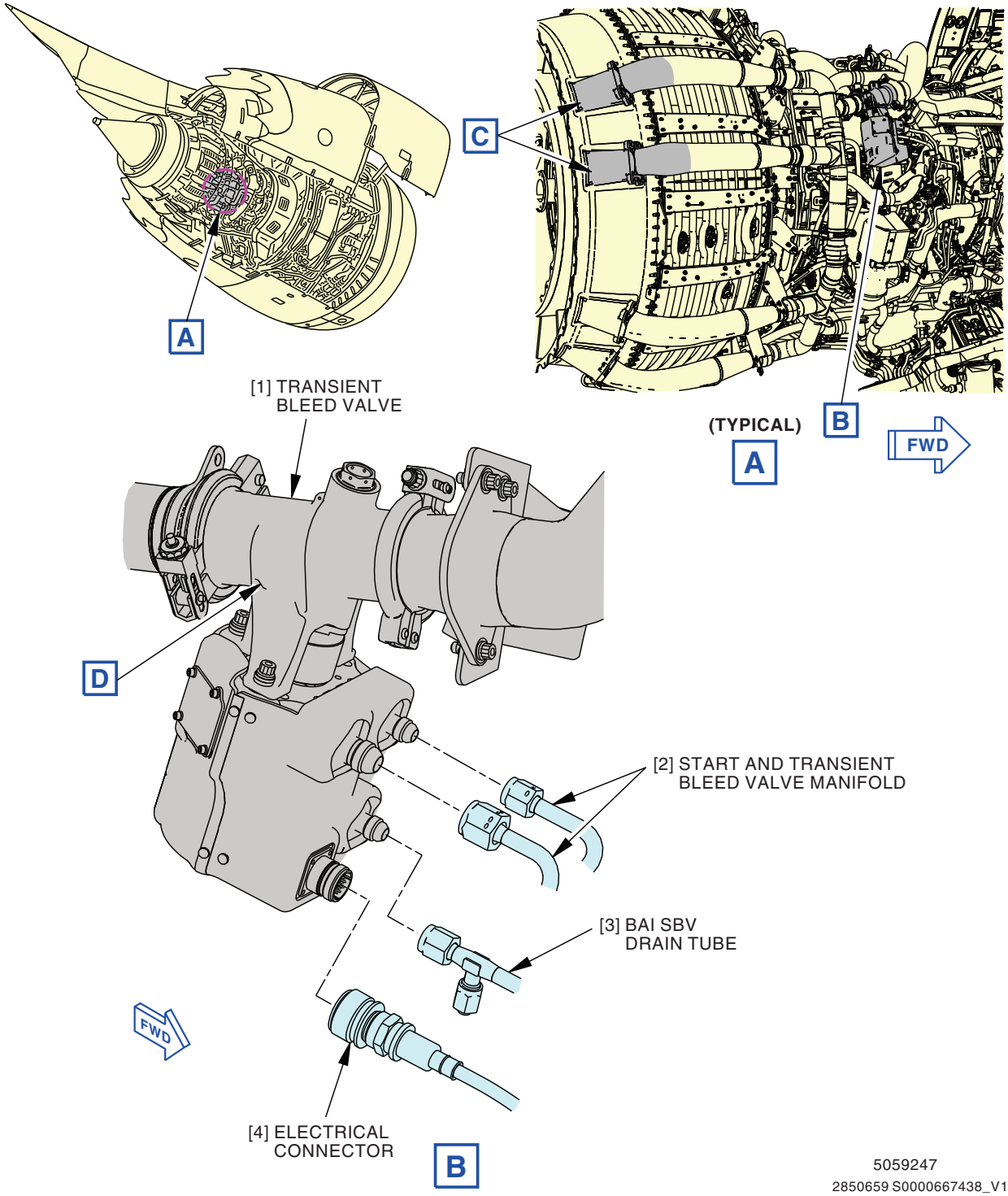
(a) Discard the seal [11] and seal [23].

SUBTASK 75-33-01-860-001-G00

(19) Install the protective covers on all tubes and fittings and the transient bleed valve [1].

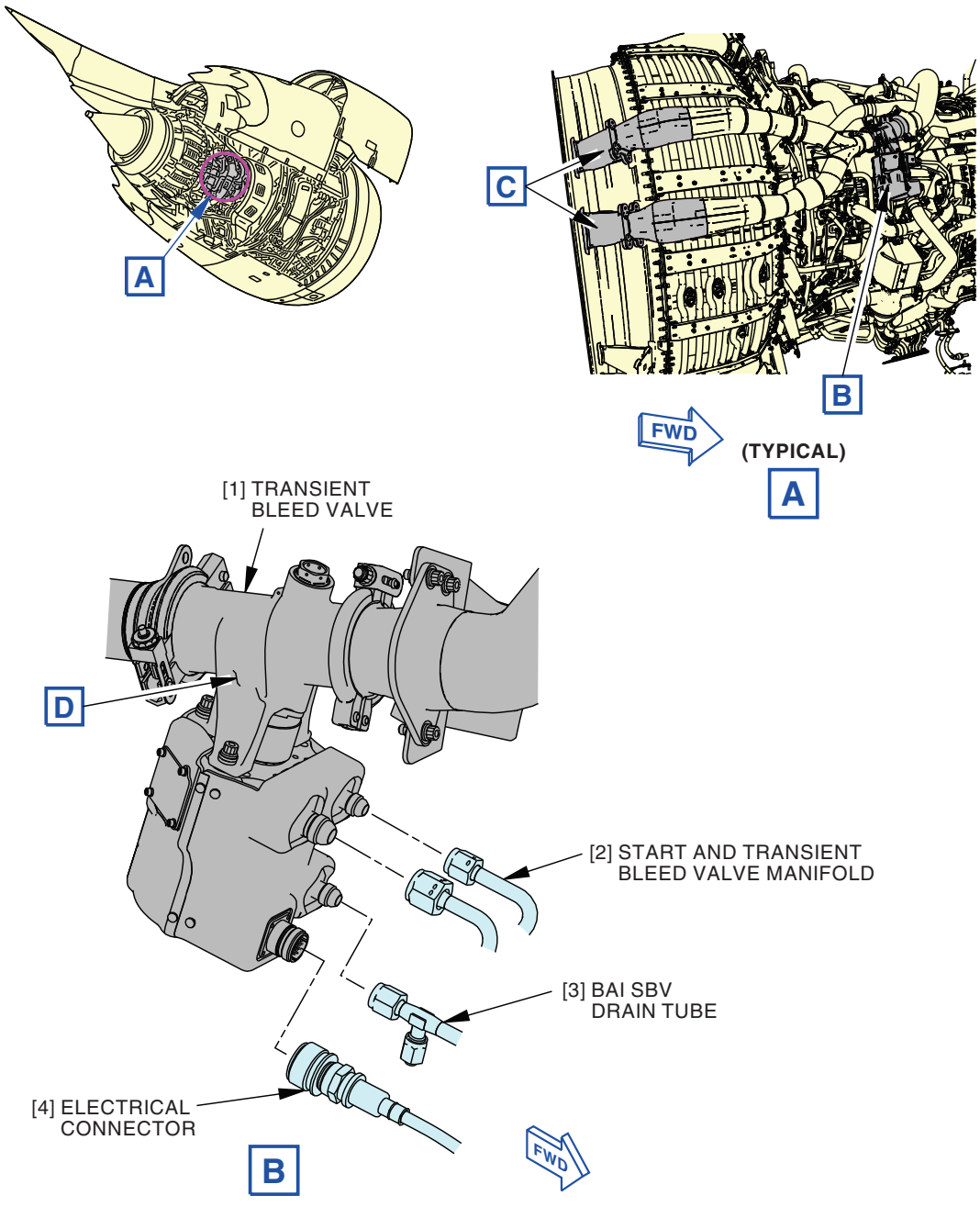
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transient Bleed Valve Installation
Figure 401/75-33-01-990-801-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

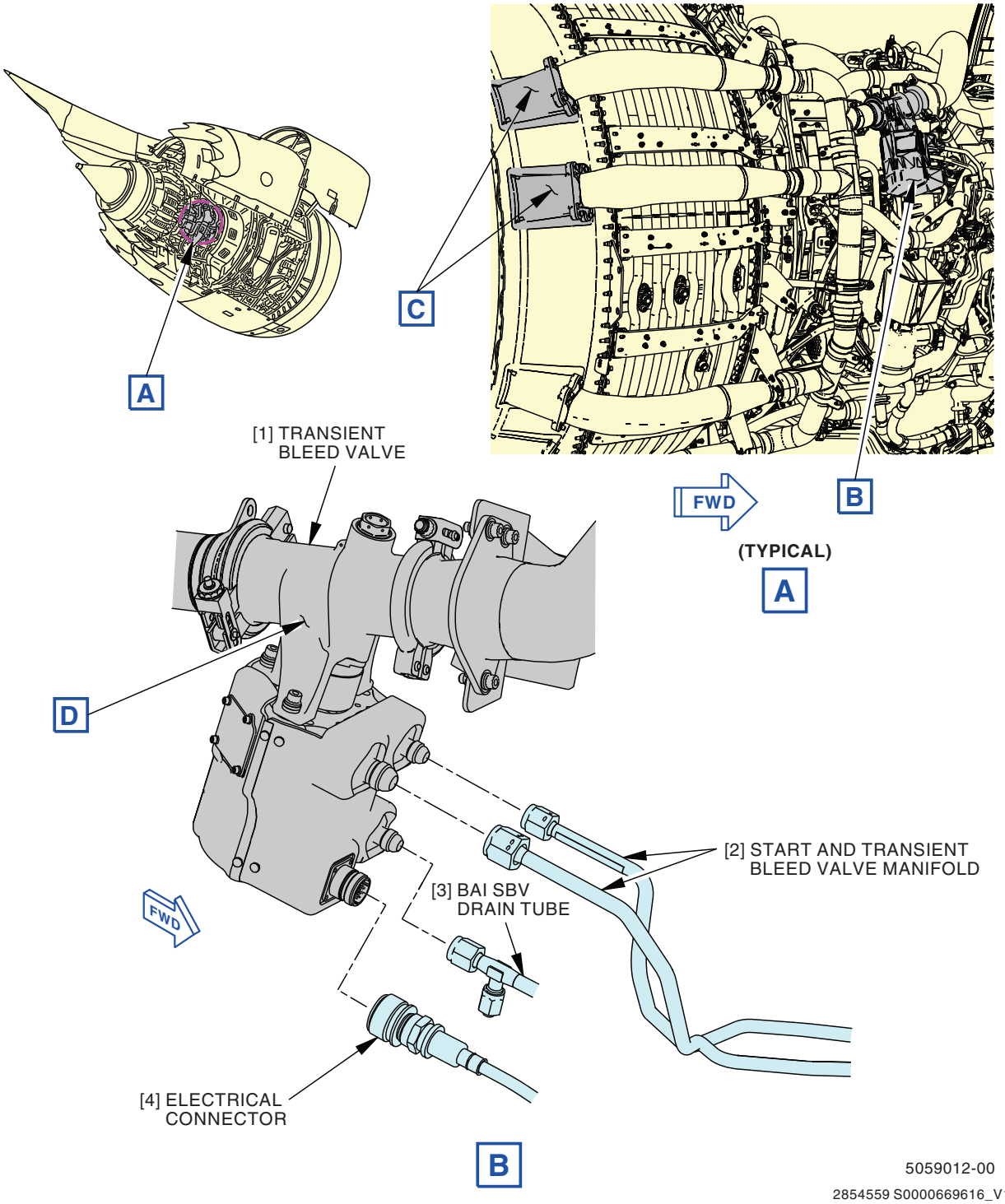


5055202-00

2834394 S0000659450_V2

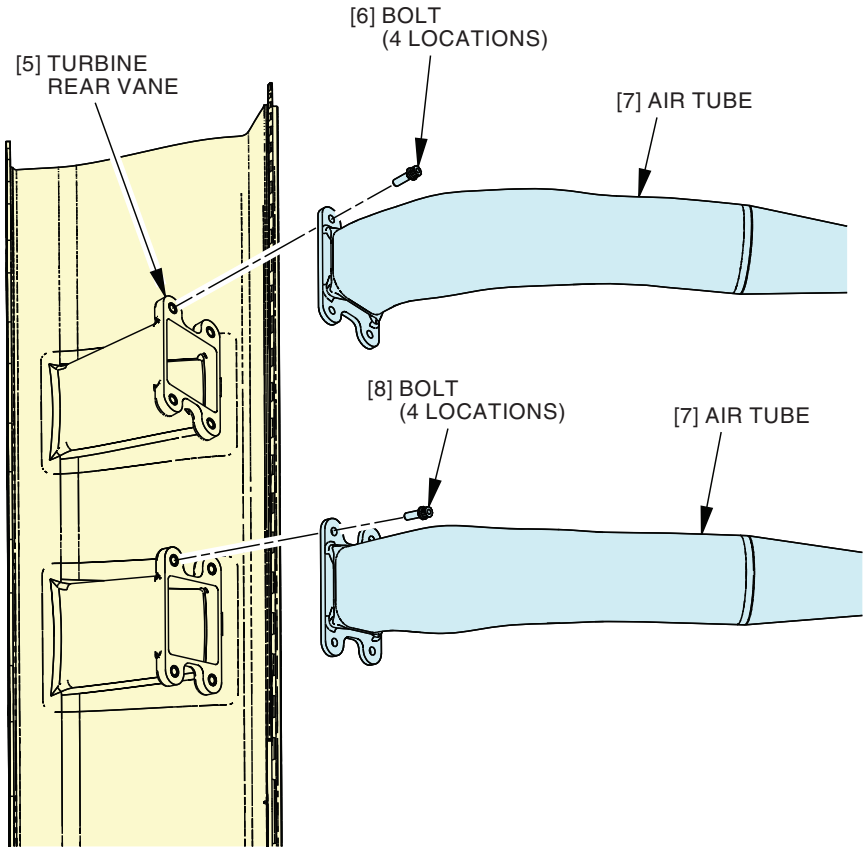
Transient Bleed Valve Installation
Figure 401/75-33-01-990-801-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transient Bleed Valve Installation
Figure 401/75-33-01-990-801-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

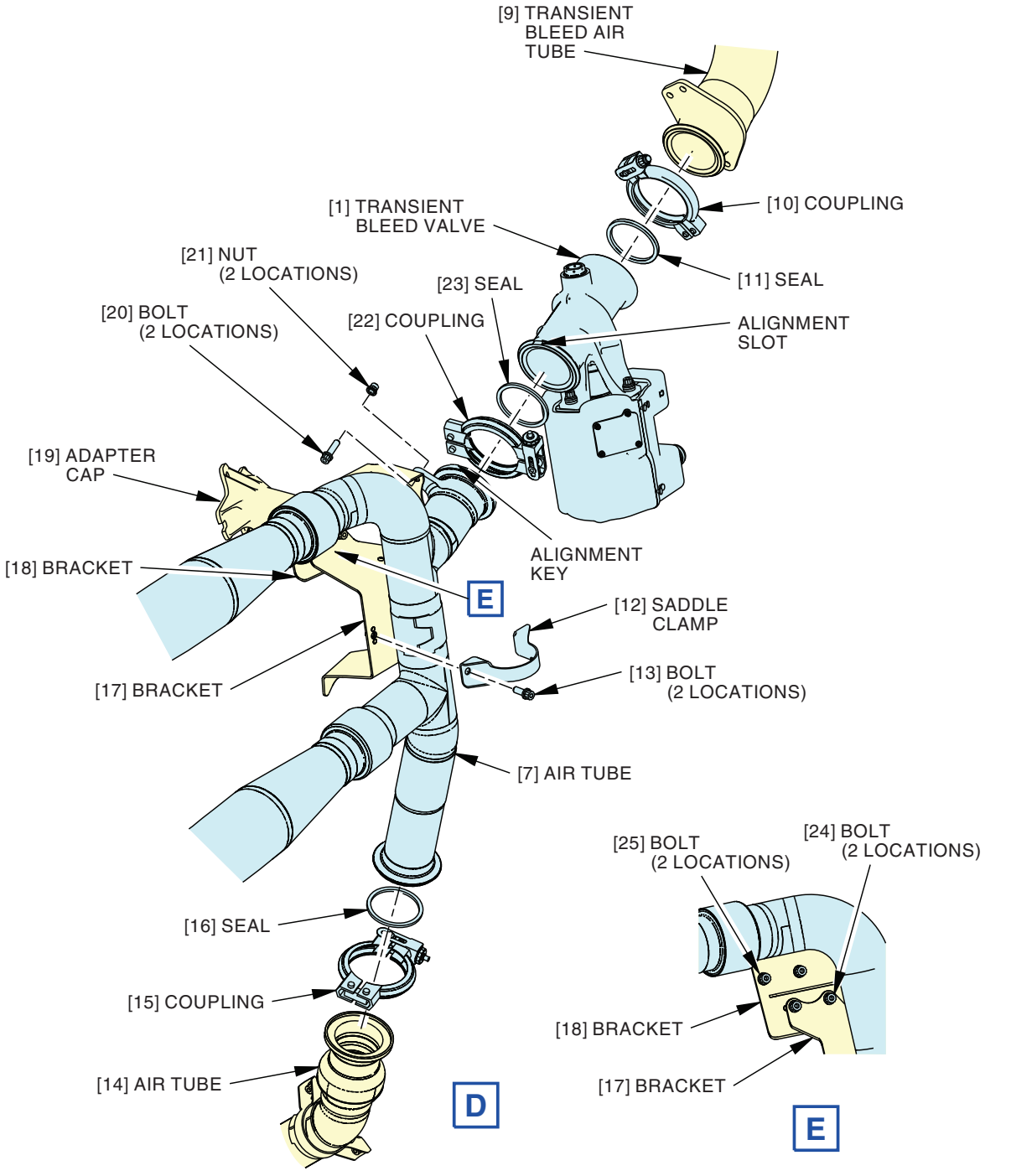


5035146-00

2354984 S00061516324_V3

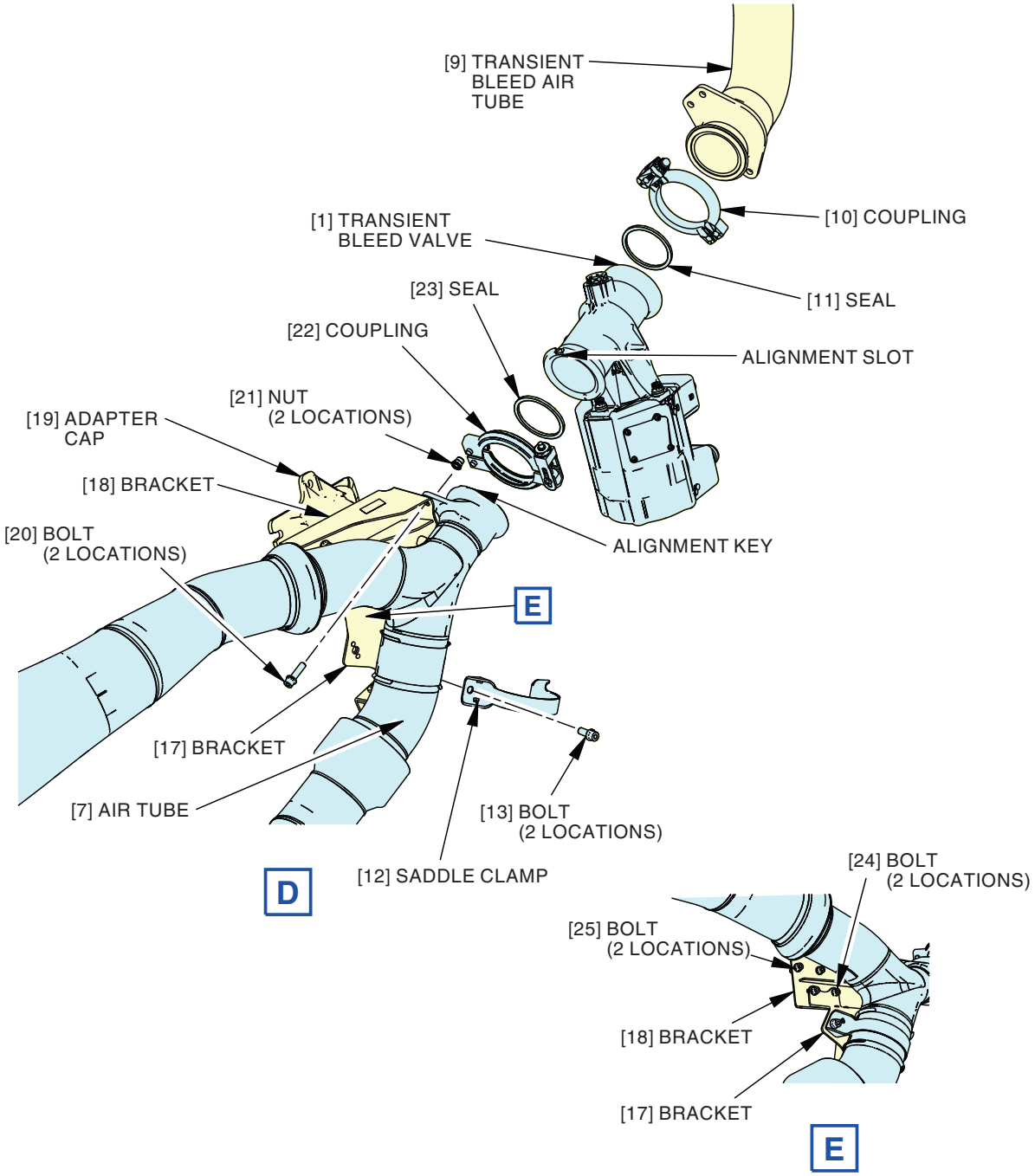
Transient Bleed Valve Installation
Figure 401/75-33-01-990-801-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transient Bleed Valve Installation
Figure 401/75-33-01-990-801-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transient Bleed Valve Installation
Figure 401/75-33-01-990-801-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-33-01-400-801-G00

3. Transient Bleed Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the transient bleed valve.

(2) The transient bleed valve is referred to as the TBV.

(3) You must do the tests that are listed in the power plant test reference table after you install the transient bleed valve.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Seal	72-50-00-02-180	PTF ALL
		75-23-00-02-310	PTF ALL
16	Seal	Not Specified	
23	Seal	72-00-00-15A-580	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

H. Transient Bleed Valve Installation

SUBTASK 75-33-01-540-001-G00

(1) Remove the protective covers from all tubes and fittings and the transient bleed valve [1].

SUBTASK 75-33-01-430-001-G00

(2) Put the new seal [11] and new seal [23] in position on the transient bleed valve [1].

SUBTASK 75-33-01-430-002-G00

(3) Put the transient bleed valve [1] in position on the transient bleed air tube [9].

SUBTASK 75-33-01-820-001-G00

(4) Loosely install the coupling [10] that attaches the transient bleed valve [1] to the transient bleed air tube [9].

(a) Make sure that the coupling [10] is in the correct position as shown in Figure 401.

SUBTASK 75-33-01-560-001-G00

(5) Put the air tube [7] in position on the transient bleed valve [1] and turbine rear vane [5].

(a) Adjust the alignment slot on the transient bleed valve [1] with the alignment key on the air tube [7].

SUBTASK 75-33-01-430-003-G00

(6) Loosely install the coupling [22] that attaches the transient bleed valve [1] to the air tube [7].

SUBTASK 75-33-01-640-002-G00

(7) Apply graphite mineral vaseline, D00601 [CP2101], to the threads of the four bolts [8] and four bolts [6].

SUBTASK 75-33-01-430-005-G00

(8) Install the four bolts [8] that attach the air tube [7] to the turbine rear vane [5].

SUBTASK 75-33-01-430-006-G00

(9) Install the four bolts [6] that attach the air tube [7] to the turbine rear vane [5].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-33-01-560-002-G00

(10) Put the new seal [16] between the air tube [14] and air tube [7].

SUBTASK 75-33-01-420-001-G00

(11) Loosely install the coupling [15] that attaches the air tube [14] to the air tube [7].

(a) Make sure that the coupling [15] is in the correct position as shown in Figure 401.

PTF ALL

SUBTASK 75-33-01-640-003-G00

(12) Lightly lubricate the threads of the two bolts [13] with graphite mineral vaseline, D00601 [CP2101].

SUBTASK 75-33-01-420-002-G00

(13) Install the two bolts [13] that attach the saddle clamp [12] and air tube [7] to the bracket [17].

(a) Tighten the bolts [13] to 66.0 in-lb (7.5 N·m) - 77.0 in-lb (8.7 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-33-01-640-005-G00

(14) Lubricate the threads of the two bolts [25] with graphite mineral vaseline, D00601 [CP2101].

SUBTASK 75-33-01-420-007-G00

(15) Install the two bolts [25] that attach the bracket [18] to the adapter cap [19].

SUBTASK 75-33-01-420-008-G00

(16) Tighten the four bolts [8] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

SUBTASK 75-33-01-420-003-G00

(17) Tighten the four bolts [6] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

SUBTASK 75-33-01-420-004-G00

(18) Tighten the two bolts [25] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

SUBTASK 75-33-01-640-004-G00

(19) Tighten the two bolts [24] that attach the bracket [17] to the bracket [18] to 66.0 in-lb (7.5 N·m) - 77.0 in-lb (8.7 N·m).

SUBTASK 75-33-01-420-005-G00

(20) Tighten the two nuts [21] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-33-01-420-006-G00

(21) Tighten the coupling [10], coupling [22], and coupling [15] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-33-01-420-009-G00

(22) Tighten the coupling [10] and coupling [22] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

PTF ALL

SUBTASK 75-33-01-410-005-G00

(23) Connect the BAI SBV drain tube [3] to the transient bleed valve [1].

(a) Tighten the BAI SBV drain tube [3] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

SUBTASK 75-33-01-410-006-G00

(24) Connect the start and TBV manifolds [2] to the transient bleed valve [1].

(a) Tighten the small B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

(b) Tighten the large B-nut to 460.0 in-lb (52.0 N·m) - 540.0 in-lb (61.0 N·m).

(c) Install safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006], to the two B-nuts (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

SUBTASK 75-33-01-430-007-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(25) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [4] to the transient bleed valve [1] as follows:

(a) Remove the protective covers from the electrical receptacles on the transient bleed valve [1] and electrical connector [4].

(b) Connect the electrical connector [4] to the transient bleed valve [1] (TASK 70-00-01-910-803-G00).

1) Tighten the electrical connector [4].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-33-01-410-007-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-33-01-865-002-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-33-01-860-002-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Transient Bleed Valve Installation Test

SUBTASK 75-33-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

REVERSE BLEED SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) A visual inspection of the reverse bleed system.

TASK 75-34-00-200-801-G00

2. Reverse Bleed System Visual Inspection

(Figure 601)

A. General

(1) This task includes the steps to do visual inspection of the engine reverse bleed system.

(2) This task includes the steps for visual inspection of these components:

(a) The reverse bleed check valve assembly

(b) The reverse bleed blower

(c) The Reverse Bleed Valve (RBV).

B. References

Reference	Title
70-00-01-300-801-G00	Correction of Leaks (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-00-05-200-801-G00	Tube and Hose Inspection Criteria and Damage Limits (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
73-22-01-000-808-G00	Electronic Engine Control (EEC) Electrical Harness (HJ8A) Removal (P/B 401)
73-22-01-400-808-G00	Electronic Engine Control (EEC) Electrical Harness (HJ8A) Installation (P/B 401)
75-34-01-000-801-G00	Reverse Bleed Check Valve Assembly Removal (P/B 401)
75-34-01-400-801-G00	Reverse Bleed Check Valve Assembly Installation (P/B 401)
75-34-02-000-801-G00	Reverse Bleed Blower Removal (P/B 401)
75-34-02-400-801-G00	Reverse Bleed Blower Installation (P/B 401)
75-34-03-000-801-G00	Reverse Bleed Valve Removal (P/B 401)
75-34-03-400-801-G00	Reverse Bleed Valve Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 75-34-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-34-00-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Reverse Bleed System Visual Inspection

SUBTASK 75-34-00-210-001-G00

(1) Examine the reverse bleed check valve assembly (Figure 601).

(a) Remove the reverse bleed check valve assembly, do this task: Reverse Bleed Check Valve Assembly Removal, TASK 75-34-01-000-801-G00.

(b) Look for cracks on the flappers and housing.

1) If a crack is present, replace the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

(c) Look for signs of wear on the hinge pin or flap.

1) If the wear is present on the hinge pin or flap, replace the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

a) Gaps are not permitted between the sealing surface of the flap and valve body ring.

2) If any amount of wear on the hinge pin is present and valve doors do not overlap, replace the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

(d) Look for signs of erosion on the face.

1) If erosion of more than 0.005 in. (0.13 mm) in depth is present on the face, replace the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

(e) Make sure that the check valve flaps move freely.

NOTE: The check valve flaps should move freely to open or close by gravity only (without any binding or sticking).

1) If the flap does not move freely, replace the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

2) If the carbon build-up or other foreign object restricts the flap movement, replace the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

(f) Look for signs of nicks, scratches, or dents on the sealing surface.

1) If the nicks, scratches, or dents are present, replace the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

(g) Look for signs of nicks or scratches on the seating surface (TASK 70-00-05-200-801-G00).

1) If the findings are more than the maximum serviceable limits, replace the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Install the reverse bleed check valve assembly, do this task: Reverse Bleed Check Valve Assembly Installation, TASK 75-34-01-400-801-G00.

SUBTASK 75-34-00-210-002-G00

(2) Examine the reverse bleed blower.

(a) Remove the segment shroud 1 and segment shroud 2 (TASK 72-23-04-000-801-G00).

(b) Look for cracks, nicks, dents, or scratches.

1) If the crack, nick, dent, or scratch is present, refer to the manufacturer instructions for permitted limits.

a) If the findings are more than the maximum permitted limits, replace the reverse bleed blower (TASK 75-34-02-000-801-G00 and TASK 75-34-02-400-801-G00).

(c) Check for looseness.

1) Looseness is not permitted, tighten the mount locations and mount brackets (TASK 75-34-02-400-801-G00).

(d) Examine the electrical receptacle and electrical connector.

1) If it is loose, tighten the electrical connector (TASK 70-00-01-910-803-G00).

2) If there is more than small damage to the electrical connector that prevents correct installation, replace the HJ8A wiring harness (TASK 73-22-01-000-808-G00 and TASK 73-22-01-400-808-G00).

3) If there is more than small damage to the electrical receptacle that prevents correct installation, refer to the manufacturer instructions for permitted limits.

a) If the findings are more than the maximum permitted limits, replace the reverse bleed blower (TASK 75-34-02-000-801-G00 and TASK 75-34-02-400-801-G00).

(e) Look for signs of corrosion or erosion.

1) If the corrosion or erosion is present, refer to the manufacturer instructions for permitted limits.

a) If the findings are more than the maximum permitted limits, replace the reverse bleed blower (TASK 75-34-02-000-801-G00 and TASK 75-34-02-400-801-G00).

(f) Look for any deposits.

1) If any deposits are present, refer to the manufacturer instructions for permitted limits.

a) If the findings are more than the maximum permitted limits, replace the reverse bleed blower (TASK 75-34-02-000-801-G00 and TASK 75-34-02-400-801-G00).

(g) Install the segment shroud 1 and segment shroud 2 (TASK 72-23-04-400-801-G00).

SUBTASK 75-34-00-210-003-G00

(3) Examine the RBV.

(a) Remove the segment shroud 1 and segment shroud 5 (TASK 72-23-04-000-801-G00).

(b) Look for signs of fuel leaks.

1) It is permitted to have a valve that has a fuel stain, if the drops of fuel are not present.

2) A fuel leak that makes drops of fuel is not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) A fuel leak from the valve housing or shaft is not permitted.

4) If the signs of fuel leak are present, do these steps:

a) Replace the gasket seal between the fuel manifold and RBV (TASK 75-34-03-000-801-G00 and TASK 75-34-03-400-801-G00).

b) If the damage or contamination is present on the fuel fittings, refer to the manufacturer instructions for permitted limits.

<1> If the findings are more than the maximum permitted limits, replace the RBV (TASK 75-34-03-000-801-G00 and TASK 75-34-03-400-801-G00).

c) Make sure that the installation of the fuel hydraulic manifold is correct and tight.

<1> If the installation of the fuel hydraulic manifold is not correct or loose, re-install and tighten the fuel tube B-nuts (TASK 75-34-03-000-801-G00 and TASK 75-34-03-400-801-G00).

d) If the fuel leak continues, refer to the manufacturer instructions for permitted limits.

<1> If the findings are more than the maximum permitted limits, replace the RBV (TASK 75-34-03-000-801-G00 and TASK 75-34-03-400-801-G00).

(c) Look for cracks, nicks, or dents in the RBV.

1) If the crack, nick, or dent is present, refer to the manufacturer instructions for permitted limits.

a) If the findings are more than the maximum permitted limits, replace the RBV (TASK 75-34-03-000-801-G00 and TASK 75-34-03-400-801-G00).

(d) Look for signs of air leaks at the V-retainer couplings on the ends of the RBV.

1) Identify the air leakage (TASK 70-00-01-300-801-G00).

2) Replace the V-band seals and install the V-retainer couplings on the RBV (TASK 75-34-03-000-801-G00 and TASK 75-34-03-400-801-G00).

3) Make sure that the installation of the air tubes connecting to the RBV are correct and tight.

a) If the installation of the air tubes is not correct or loose, re-install and tighten the nuts (TASK 75-34-03-000-801-G00 and TASK 75-34-03-400-801-G00).

(e) Install the segment shroud 1 and segment shroud 5 (TASK 72-23-04-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-34-00-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-34-00-860-002-G00

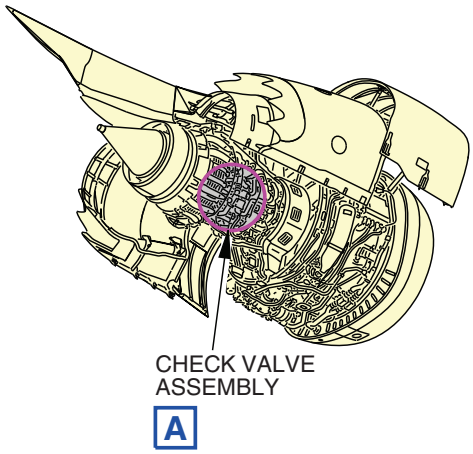
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

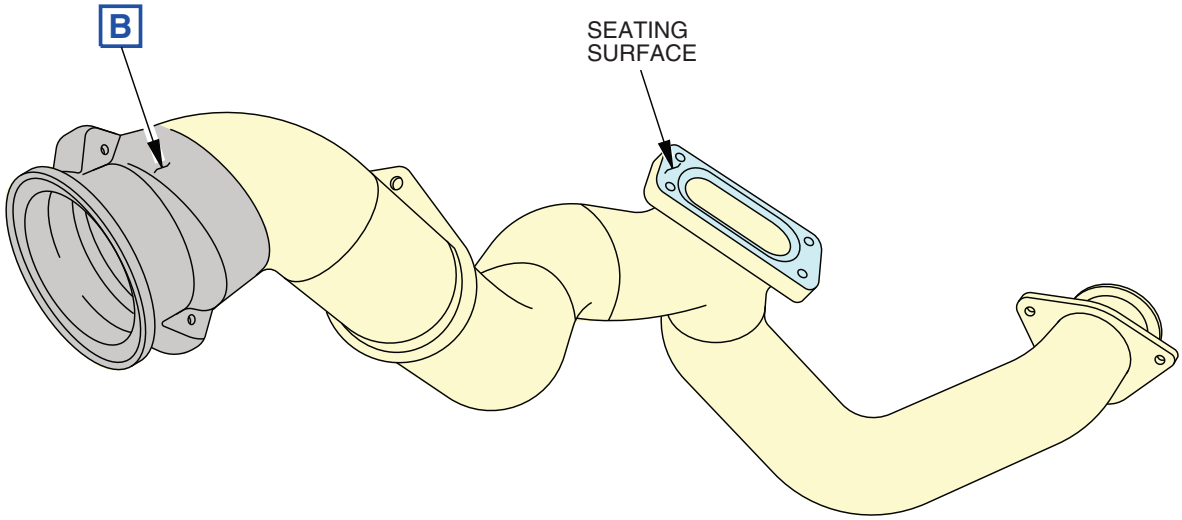
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



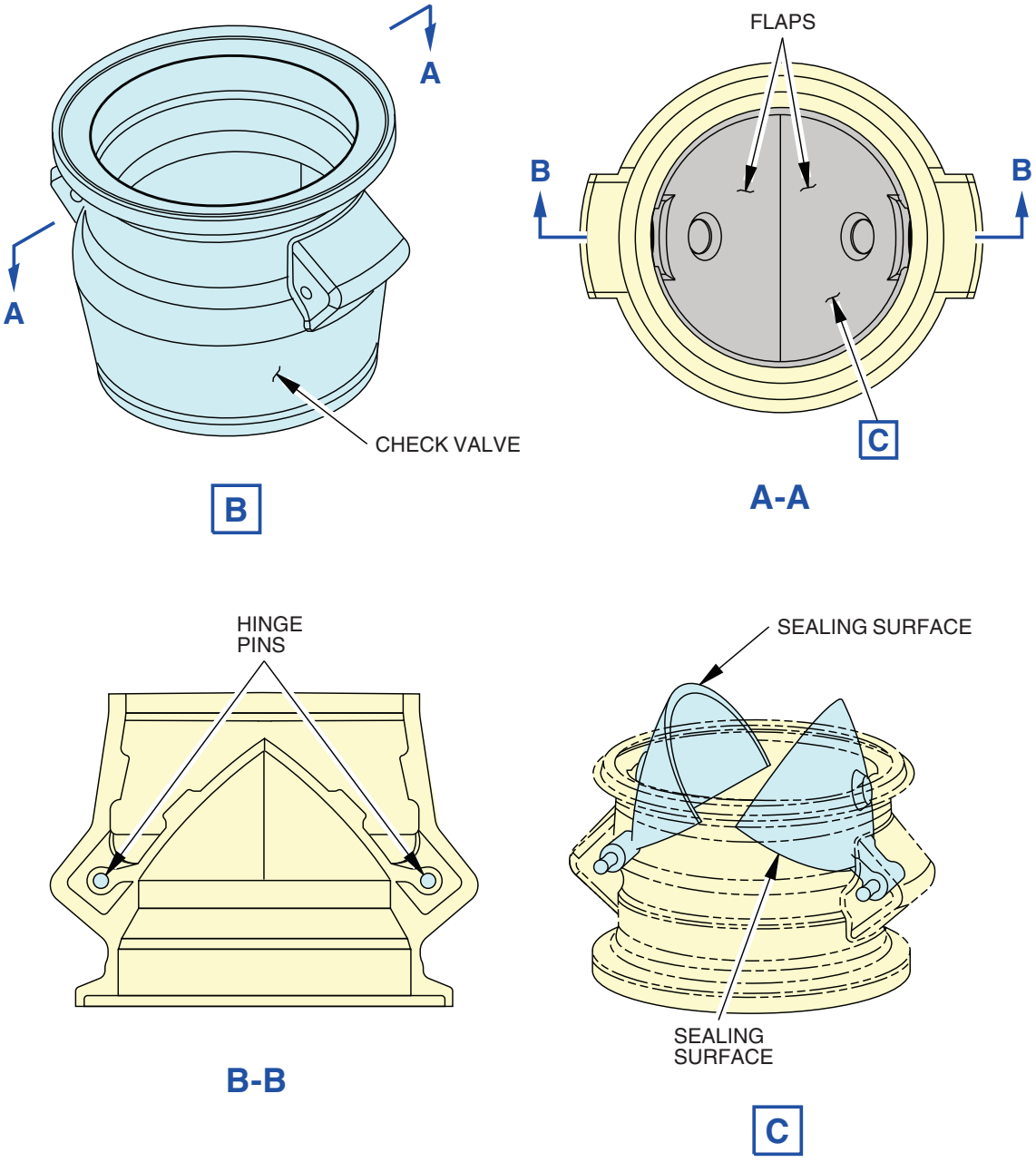


6035879-00

3130382 S0000861258_V1

Reverse Bleed Check Valve Assembly Visual Inspection
Figure 601/75-34-00-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Reverse Bleed Check Valve Assembly Visual Inspection
Figure 601/75-34-00-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

REVERSE BLEED CHECK VALVE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the reverse bleed check valve assembly

(2) An installation of the reverse bleed check valve assembly.

TASK 75-34-01-000-801-G00

2. Reverse Bleed Check Valve Assembly Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the reverse bleed check valve assembly.

NOTE: The reverse bleed check valve assembly is installed between 11:00 and 3:00 o'clock positions on the right of the High Pressure Compressor (HPC) Case and Combustion Case.

B. References

Reference	Title
36-11-01-000-801	Engine Pneumatic Duct Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Removal

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

SUBTASK 75-34-01-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-34-01-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING

DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Reverse Bleed Check Valve Assembly Removal

SUBTASK 75-34-01-020-001-G00

(1) If it is necessary, remove the crossover ECS duct assembly [1] (TASK 36-11-01-000-801).

SUBTASK 75-34-01-020-002-G00

(2) If it is necessary, remove the bai air tube [16], bai air tube [19], bai air manifold [20], and bai air tube [21].

(a) Remove the coupling [13] that connects the bai air manifold [20] to the bai air tube [19].

(b) Remove the coupling [13] that connects the bai air manifold [20] to the bai air tube [21].

(c) Remove the retaining strap [9] and bolts [23] that attach the bai air manifold [20] to the bai tube support bracket [22] on the HPC case.

(d) Remove the retaining strap [9] and bolts [10] that attach the bai air manifold [20] to the bai tube support bracket [24] and bracket [25].

(e) Remove the coupling [28] that connects the bai air manifold [20] to the bai air tube [26].

(f) Remove the bai air manifold [20].

1) Remove and discard the seal ring [12] and seal ring [27].

(g) Remove the retaining strap [9] and bolts [18] that attach the bai air tube [21] to the bai tube support bracket [17].

(h) Remove the coupling [13] that connects the bai air tube [21] to the anti-icing tube.

(i) Remove the bai air tube [21].

1) Remove and discard the seal ring [12].

(j) Remove the retaining strap [9] and bolts [10] that attach the bai air tube [19] to the bai tube support bracket [8] on the HPC case.

(k) Remove the retaining strap [9] and bolts [18] that attach the bai air tube [19] to the bai tube support bracket [11] on the HPC case.

(l) Remove the coupling [13] that connects the bai air tube [19] to the bai air tube [16].

(m) Remove the bai air tube [19].

1) Remove and discard the seal ring [12].

(n) Remove the bolt [15] that attaches the bai air tube [16] to the fuel supply support bracket [14] on the HPC case.

(o) Remove the coupling [13] that connects the bai air tube [16] to the anti-icing tube.

(p) Remove the bai air tube [16].

1) Remove and discard the seal ring [12].

SUBTASK 75-34-01-020-003-G00

(3) Remove the reverse bleed check valve assembly [2] as follows:

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(a) Apply penetrating fluid, D50186 [CP2691], to the bolts [35].

1) Let the lubricant soak into the threads for a one hour.

a) Let penetrating fluid, D50186 [CP2691], to penetrate while you remove the other hardware at the same time before you break the torque of the bolts [35].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the V-retainer coupling [4] that attaches the reverse bleed check valve assembly [2] to the duct assembly [3].

1) Remove and discard the seal [5].

(c) Remove the coupling clamp [32] that attaches the reverse bleed check valve assembly [2] to the transient bleed valve [33].

1) Remove and discard the seal [31].

(d) Remove the bolts [29] that attach the reverse bleed check valve assembly [2] to the support bracket [30].

NOTE: The transient bleed air tube is replaced with reverse bleed check valve assembly in LEAP-1B SERVICE BULLETIN 72-0378.

(e) Remove the bolts [6] that attach the reverse bleed check valve assembly [2] to the support bracket [7].

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(f) Apply penetrating fluid, D50186 [CP2691], to the bolts [35] again.

1) Let the penetrant time to soak into threads.

NOTE: It is necessary to apply penetrating fluid, D50186 [CP2691], to decrease the risk of damage to the threaded inserts.

(g) Remove the bolts [35] that attach the reverse bleed check valve assembly [2] to the combustion case [36].

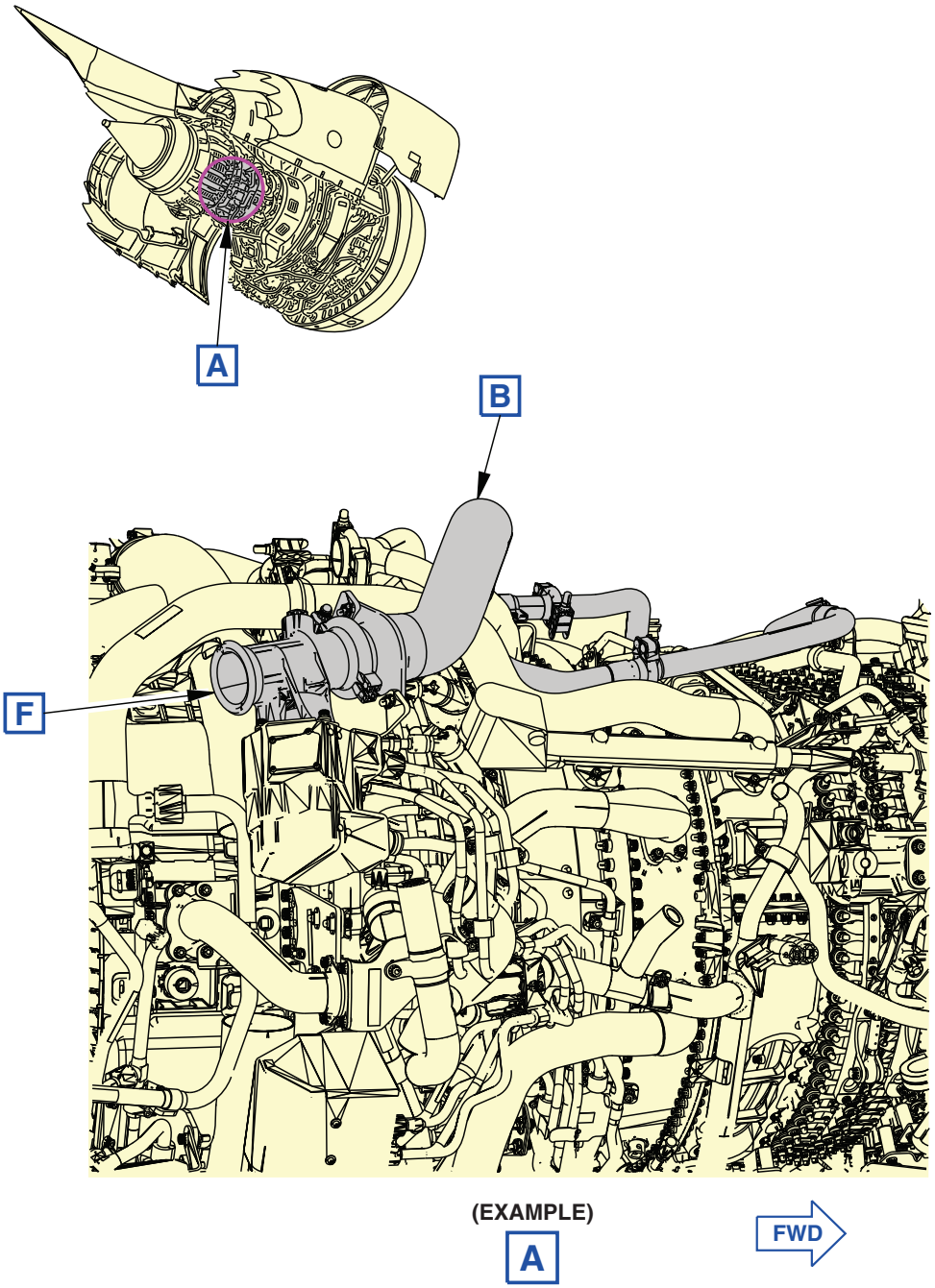
(h) Remove the reverse bleed check valve assembly [2] from the engine.

1) Remove and discard the metal seal ring [34].

(i) Put a protective cover on the open port on the combustion case [36] and on all the tubes and fittings.

END OF TASK

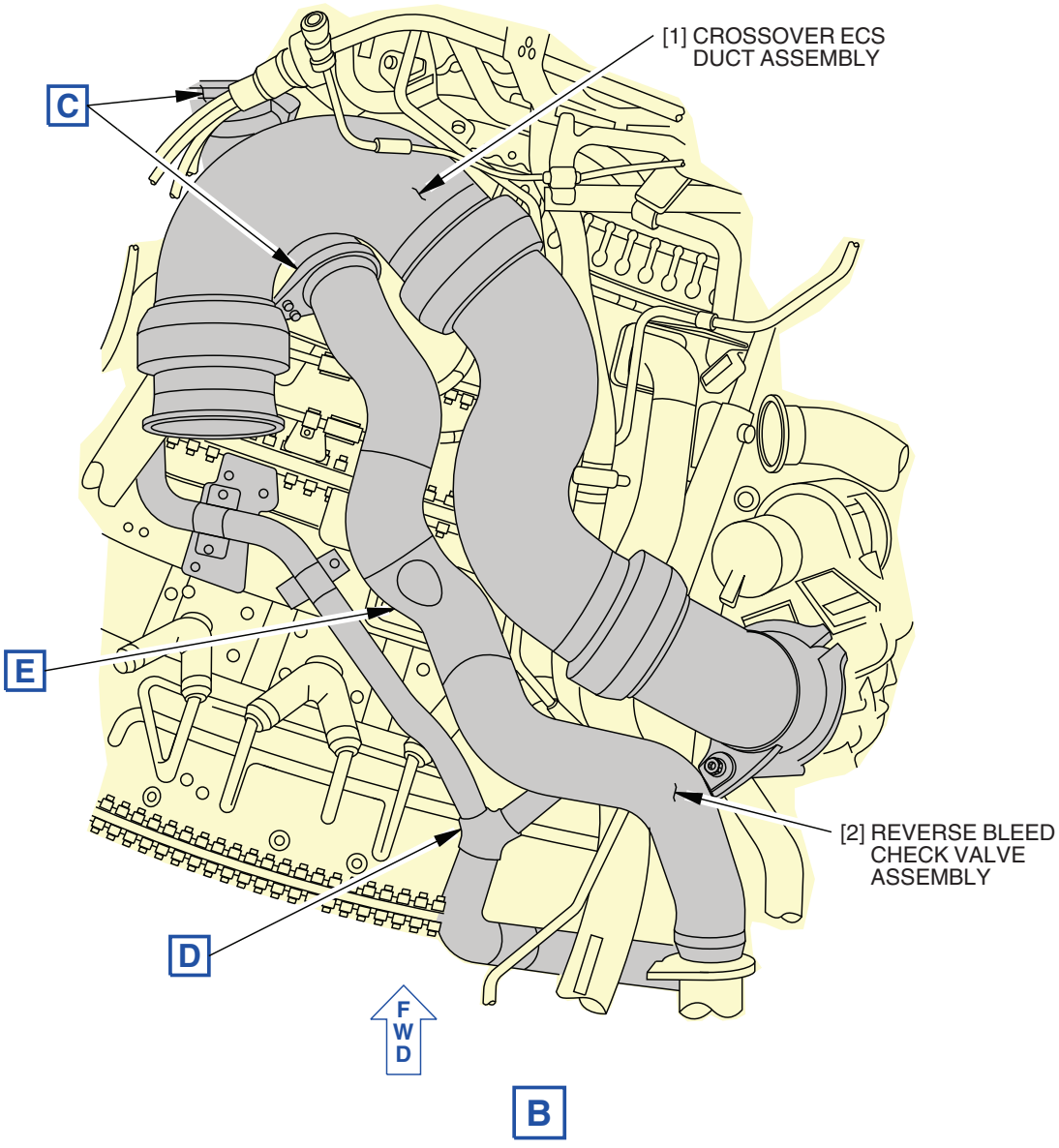
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6036227-00
3128200 S0000860588_V1

Reverse Bleed Check Valve Assembly Installation
Figure 401/75-34-01-990-801-G00 (Sheet 1 of 6)

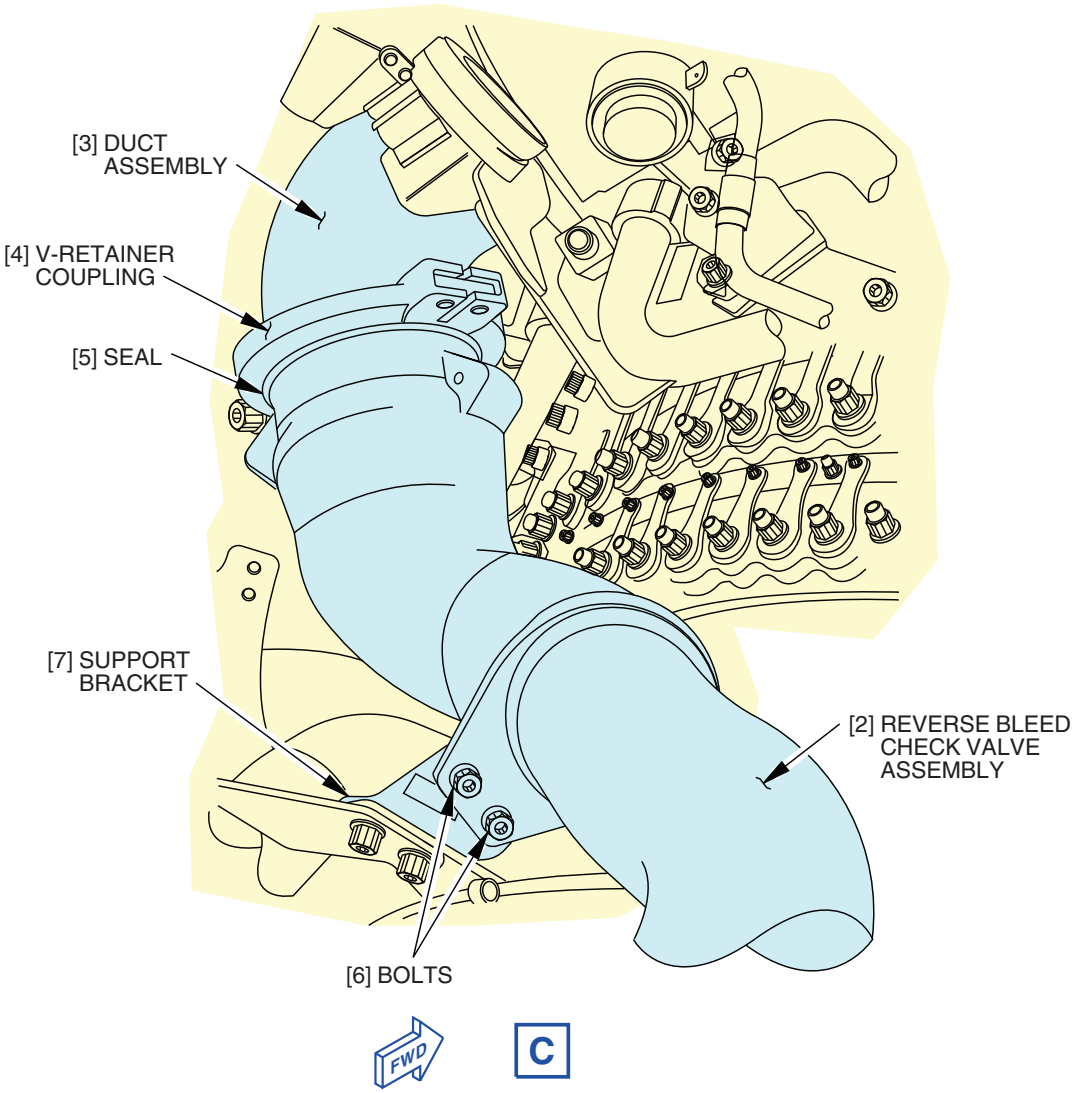
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6036198-00
3128210 S0000860589_V1

Reverse Bleed Check Valve Assembly Installation
Figure 401/75-34-01-990-801-G00 (Sheet 2 of 6)

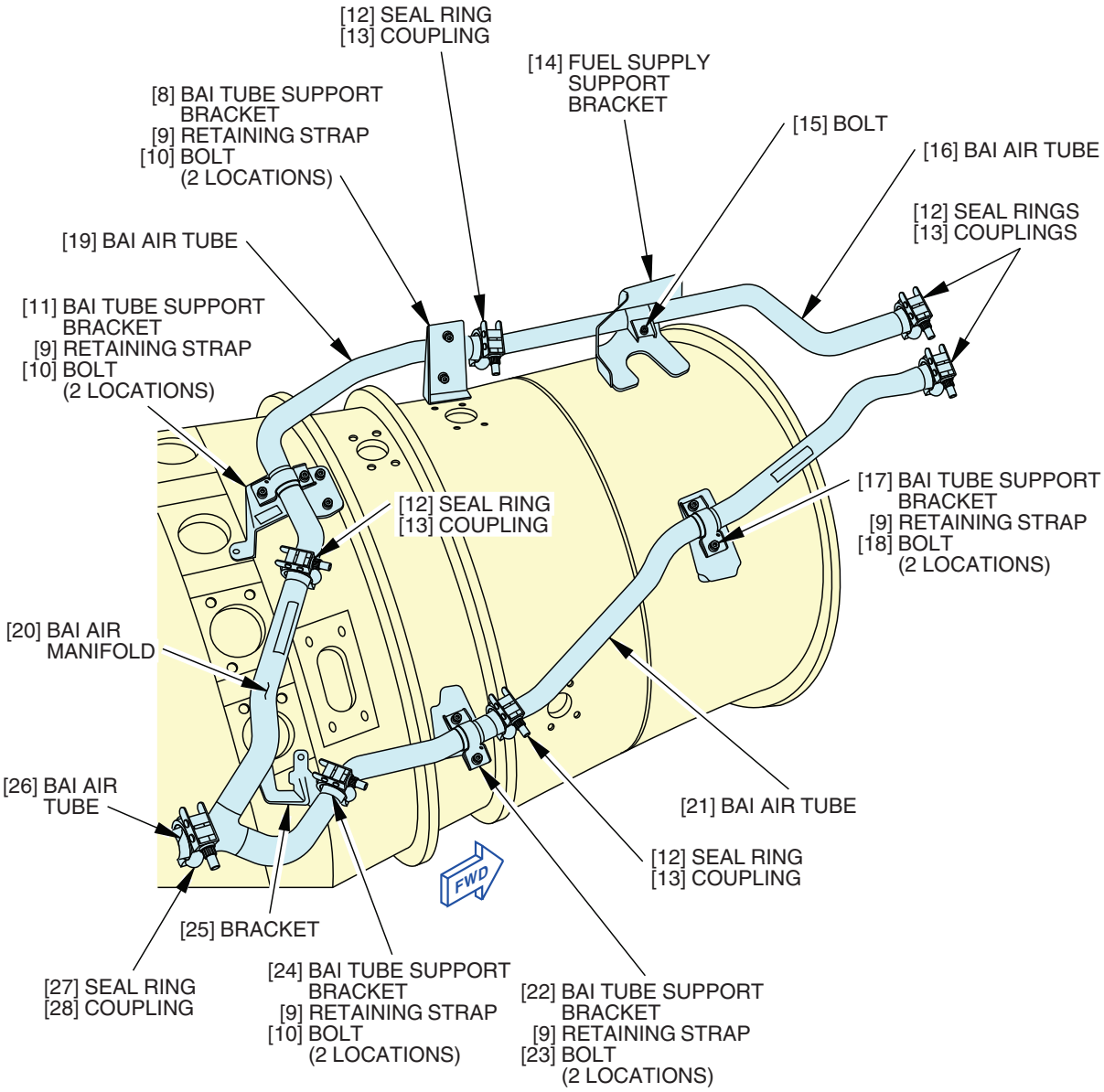
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6036199-00
3128216 S0000860590_V1

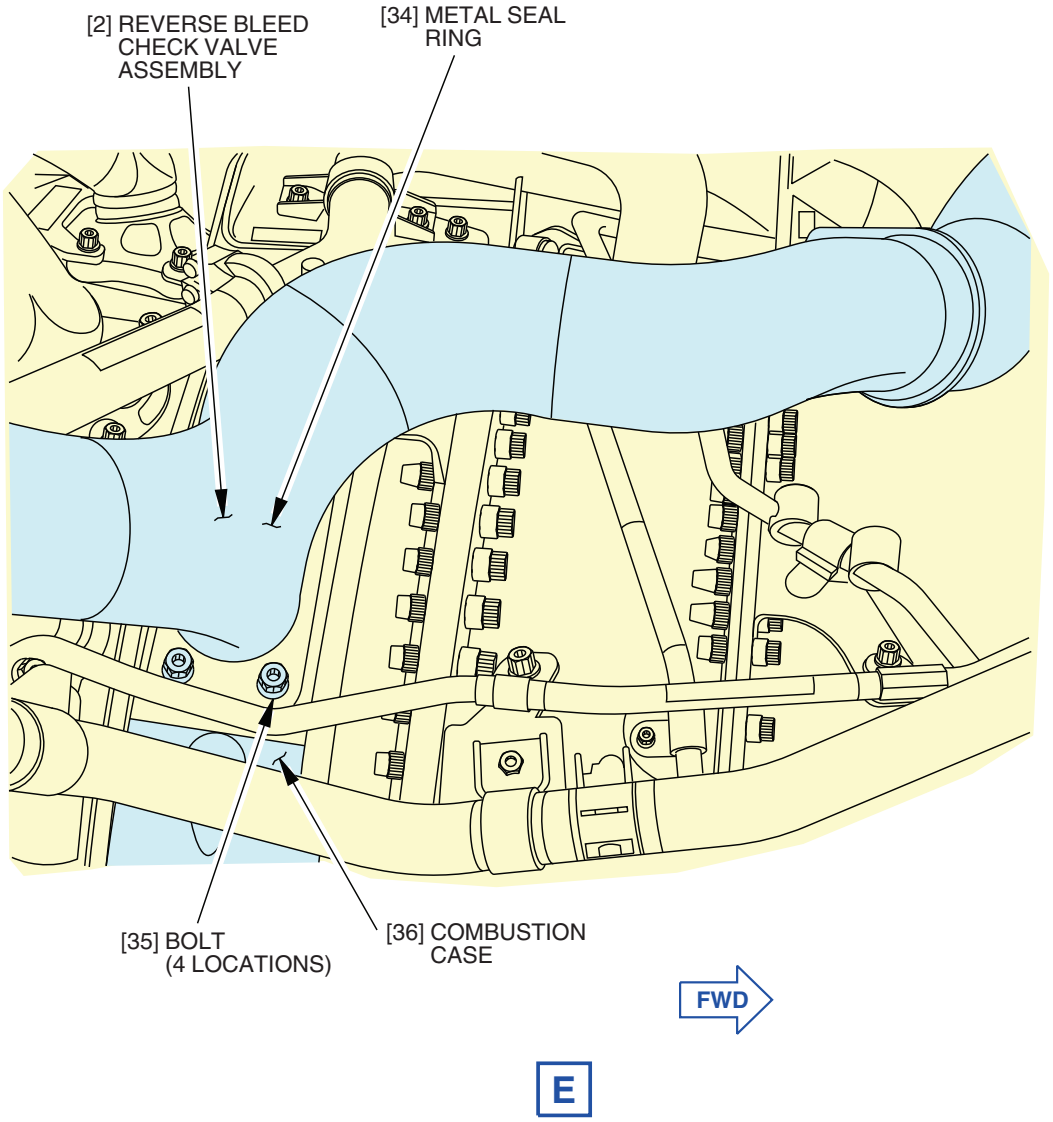
Reverse Bleed Check Valve Assembly Installation
Figure 401/75-34-01-990-801-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Reverse Bleed Check Valve Assembly Installation
Figure 401/75-34-01-990-801-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

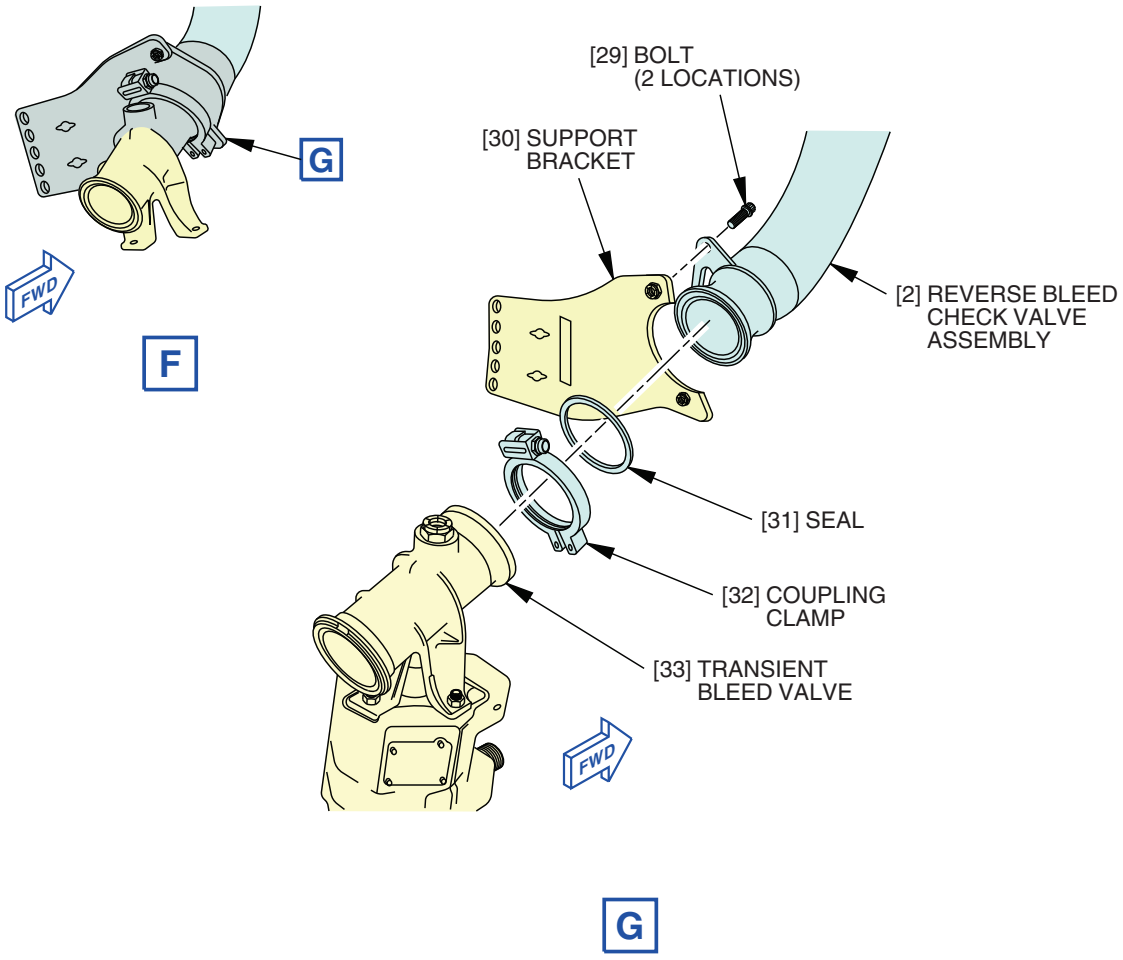


6036222-00

3128293 S0000860593_V1

Reverse Bleed Check Valve Assembly Installation
Figure 401/75-34-01-990-801-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



6036192-00

3128287 S0000860592_V1

Reverse Bleed Check Valve Assembly Installation
Figure 401/75-34-01-990-801-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-34-01-400-801-G00

3. Reverse Bleed Check Valve Assembly Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the reverse bleed check valve assembly.

B. References

Reference	Title
36-11-01-400-801	Engine Pneumatic Duct Installation (P/B 401)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2
G51435 [CP2635]	Wax - Stick	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Installation

SUBTASK 75-34-01-860-001-G00

(1) Remove the protective covers from the open port on the combustion case [36] and from all the tubes and fittings.

H. Reverse Bleed Check Valve Assembly Installation

SUBTASK 75-34-01-420-001-G00

(1) Install the reverse bleed check valve assembly [2] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(a) Install the new metal seal ring [34] as follows:

1) Put the new metal seal ring [34] on the open port of the combustion case [36].

2) Apply a layer of stick wax, G51435 [CP2635], to the metal seal ring [34] to hold it during installation.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(b) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of bolts [35].

(c) Put the reverse bleed check valve assembly [2] in position on the engine.

(d) Install the bolts [35] that attach the reverse bleed check valve assembly [2] to the combustion case [36].

1) Tighten the bolts [35] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the bolts [6] and bolts [29].

(f) Install the bolts [6] that attach the reverse bleed check valve assembly [2] to the support bracket [7].

1) Do not tighten the bolts [6] at this time.

(g) Install the bolts [29] that attach the reverse bleed check valve assembly [2] to the support bracket [30].

1) Do not tighten the bolts [29] at this time.

(h) Put a new seal [31] between the duct assembly [3] and reverse bleed check valve assembly [2].

(i) Loosely install the coupling clamp [32] that attaches the transient bleed valve [33] to the reverse bleed check valve assembly [2].

1) Make sure that the coupling clamp [32] is in the correct position.

(j) Put a new seal [5] between the duct assembly [3] and reverse bleed check valve assembly [2].

(k) Loosely install the V-retainer coupling [4] that attaches the reverse bleed check valve assembly [2] to the duct assembly [3].

1) Make sure that the V-retainer coupling [4] is in the correct position.

(l) Tighten the bolts [6] and bolts [29] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(m) Tighten the coupling clamp [32] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(n) Tighten the V-retainer coupling [4] to 60.0 in-lb (6.8 N·m) - 70 in-lb (7.9 N·m).

SUBTASK 75-34-01-410-001-G00

(2) If removed, install the bai air tube [16], bai air tube [19], bai air manifold [20], and bai air tube [21] as follows:

(a) Apply a layer of oil, D00599 [CP2442], to the threads of the bolt [15], bolts [18], bolts [10], and bolts [23].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Connect the bai air tube [16] to the anti-icing tube with the coupling [13] and new seal ring [12].

(c) Attach the bai air tube [16] to the fuel supply support bracket [14] on the High Pressure Compressor (HPC) case with the bolt [15].

(d) Connect the bai air tube [21] to the anti-icing tube with the coupling [13] and new seal ring [12].

(e) Attach the bai air tube [21] to the bai tube support bracket [17] on the HPC case with the retaining strap [9] and bolts [18].

(f) Connect the bai air tube [19] to the bai air tube [16] with the coupling [13] and new seal ring [12].

(g) Attach the bai air tube [19] to the bai tube support bracket [8] with the retaining strap [9] and bolts [10].

(h) Attach the bai air tube [19] to the bai tube support bracket [11] with the retaining strap [9] and bolts [18].

(i) Attach the bai air manifold [20] to the bai tube support bracket [24] and bracket [25] with the retaining strap [9] and bolts [10].

(j) Attach the bai air manifold [20] to the bai tube support bracket [22] on the HPC case with the retaining strap [9] and bolts [23].

(k) Connect the bai air manifold [20] to the bai air tube [19] with the coupling [13] and new seal ring [12].

(l) Connect the bai air manifold [20] to the bai air tube [21] with the coupling [13] and new seal ring [12].

(m) Connect the bai air manifold [20] to the bai air tube [26] with the coupling [28] and new seal ring [27].

(n) Tighten the couplings [13] to 60 in-lb (6.8 N·m) - 70 in-lb (7.9 N·m).

(o) Tighten the bolt [15], bolts [18], bolts [10], and bolts [23] to 106 in-lb (12.0 N·m) - 124 in-lb (14.0 N·m).

SUBTASK 75-34-01-410-002-G00

(3) If removed, install the crossover ECS duct assembly [1] on the right side of the engine (TASK 36-11-01-400-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-34-01-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-34-01-860-006-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-34-01-860-007-G00

(3) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

REVERSE BLEED BLOWER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the reverse bleed blower

(2) An installation of the reverse bleed blower.

TASK 75-34-02-000-801-G00

2. Reverse Bleed Blower Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the reverse bleed blower.

(2) The reverse bleed blower is installed between 1:00 and 3:00 o'clock positions on the right of the High Pressure Compressor (HPC) case.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-34-02-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-34-02-010-004-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 75-34-02-010-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(3) Remove the segment shroud 1 (TASK 72-23-04-000-801-G00).

SUBTASK 75-34-02-010-003-G00

(4) If it is necessary, remove the segment shroud 2 (TASK 72-23-04-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Reverse Bleed Blower Removal

SUBTASK 75-34-02-020-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS OR RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [2] from the reverse bleed blower [1] as follows:

(a) Disconnect the electrical connector [2] from the reverse bleed blower [1] (TASK 70-00-01-910-803-G00).

(b) Install the protective covers on the electrical connector [2] and electrical receptacle of the reverse bleed blower [1].

SUBTASK 75-34-02-020-002-G00

(2) Loosen the hose clamp [3] that attaches the air rubber duct [4] to the reverse bleed blower [1].

SUBTASK 75-34-02-020-003-G00

(3) Remove the two bolts [5] that attach the reverse bleed blower [1], bonding jumper [10], and bonding jumper [11] to the blower suppurt bracket [6].

SUBTASK 75-34-02-020-007-G00

(4) Remove the two bolts [9] that attach the bonding jumper [10] and bonding jumper [11] to the blower support bracket [8].

(a) Remove and discard the bonding jumper [10] and bonding jumper [11].

SUBTASK 75-34-02-020-004-G00

(5) Remove the two bolts [5] that attach the reverse bleed blower [1] to the blower support bracket [7].

SUBTASK 75-34-02-020-005-G00

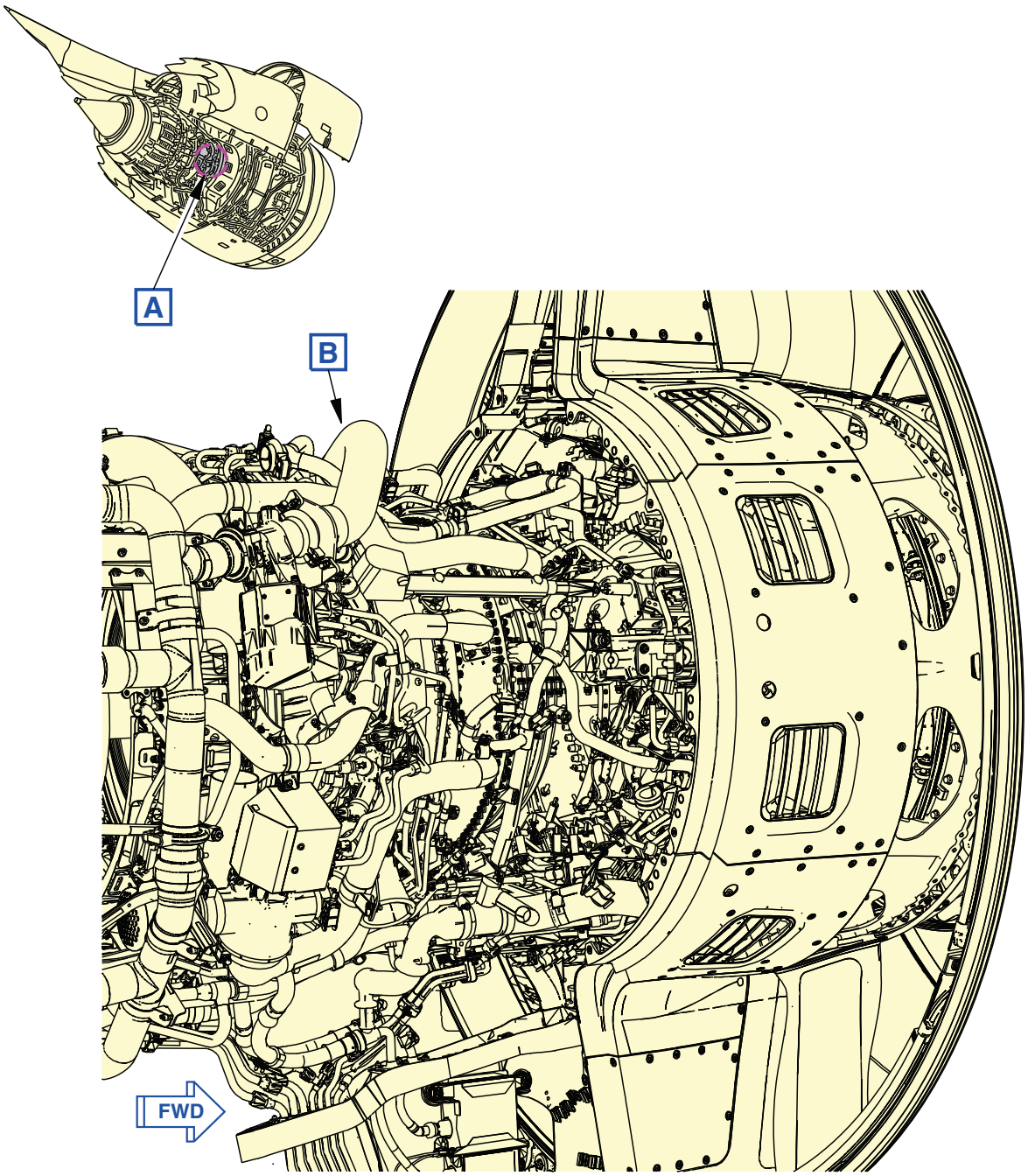
(6) Remove the reverse bleed blower [1] from the engine.

SUBTASK 75-34-02-420-001-G00

(7) Install the protective covers on the air rubber duct [4] and reverse bleed blower [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

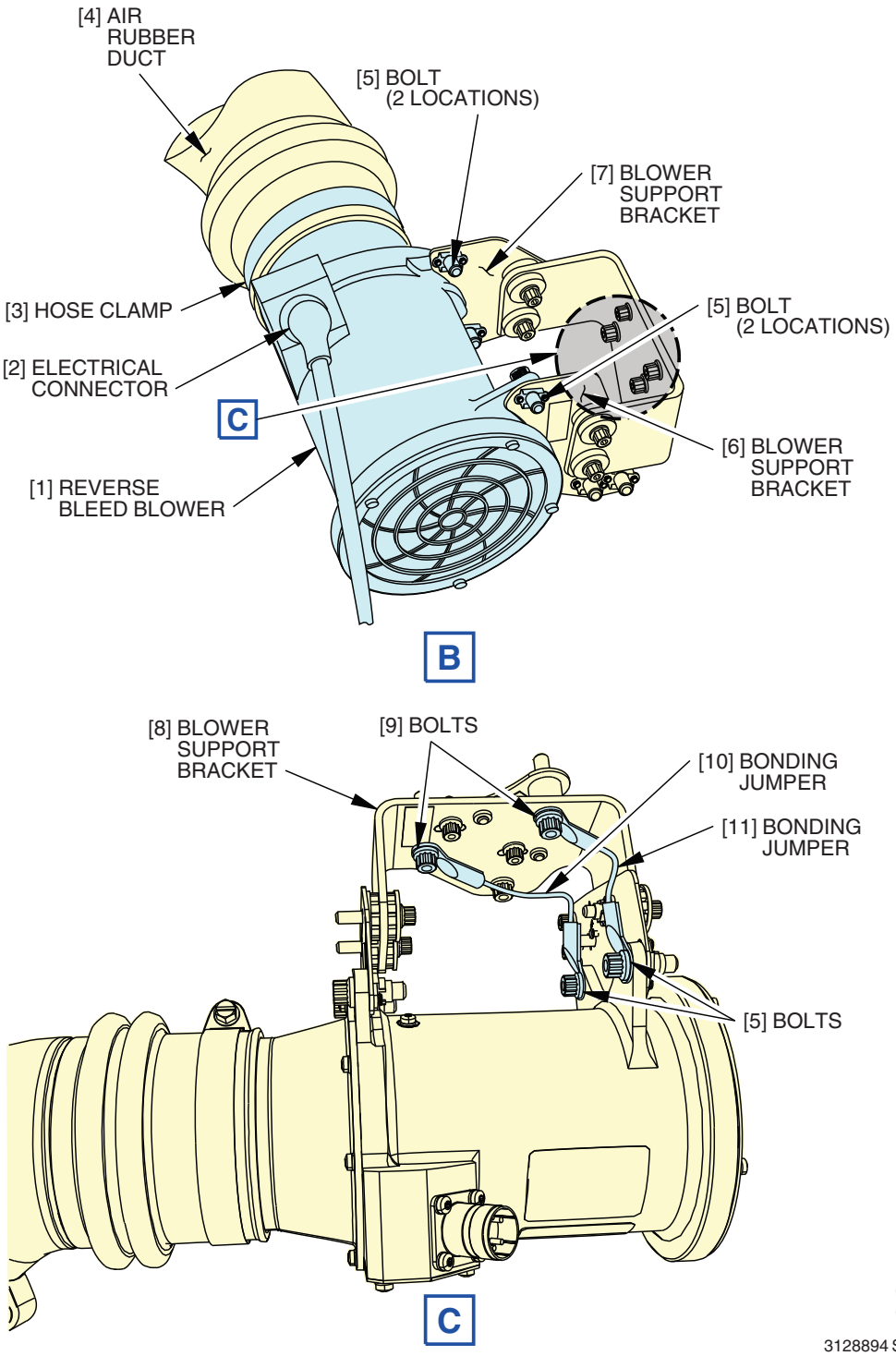
A

6057954-00

3128892 S0000860823_V1

Reverse Bleed Blower Installation
Figure 401/75-34-02-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Reverse Bleed Blower Installation
Figure 401/75-34-02-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-34-02-400-801-G00

3. Reverse Bleed Blower Installation

(Figure 401)

A. General

(1) This task includes the steps to install the reverse bleed blower.

B. References

Reference	Title
70-00-01-400-801-G00	Electrical Bonding Straps Installation (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
412	Engine 1 - Nose Inlet Cowl

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Installation

SUBTASK 75-34-02-020-006-G00

(1) Remove the protective covers from the air rubber duct [4] and reverse bleed blower [1].

H. Reverse Bleed Blower Installation

SUBTASK 75-34-02-420-002-G00

(1) Put and hold the reverse bleed blower [1] in position on the engine.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-34-02-640-001-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

(2) Apply oil, D00599 [CP2442], on the threads of the bolts [5] and bolts [9].

SUBTASK 75-34-02-420-008-G00

(3) Install the two bolts [9] to attach the new bonding jumper [10] and new bonding jumper [11] to the blower support bracket [8] (TASK 70-00-01-400-801-G00).

SUBTASK 75-34-02-420-003-G00

(4) Install the two bolts [5] that attach the reverse bleed blower [1] to the blower support bracket [7].

(a) Do not tighten the bolts [5] at this time.

SUBTASK 75-34-02-420-004-G00

(5) Install the two bolts [5] that attach the reverse bleed blower [1] to the blower suppurt bracket [6].

(a) Install the lugs of the bonding jumper [10] and bonding jumper [11] under the heads of the bolts [5] (TASK 70-00-01-400-801-G00).

(b) Do not tighten the bolts [5] at this time.

SUBTASK 75-34-02-420-005-G00

(6) Install the hose clamp [3] that attaches the air rubber duct [4] to the reverse bleed blower [1].

(a) Tighten the hose clamp [3] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

SUBTASK 75-34-02-420-006-G00

(7) Tighten the bolts [5] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

SUBTASK 75-34-02-420-007-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS OR RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(8) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [2] to the reverse bleed blower [1] as follows:

(a) Remove the protective covers from the electrical connector [2] and electrical receptacles on the reverse bleed blower [1].

(b) Connect the electrical connector [2] to the reverse bleed blower [1] (TASK 70-00-01-910-803-G00).

1) Tighten the electrical connector [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-34-02-410-001-G00

(1) If it was removed, install the segment shroud 2 (TASK 72-23-04-400-801-G00).

SUBTASK 75-34-02-410-002-G00

(2) Install the segment shroud 1 (TASK 72-23-04-400-801-G00).

SUBTASK 75-34-02-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-34-02-860-002-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Reverse Bleed Blower Installation Test

SUBTASK 75-34-02-700-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

REVERSE BLEED VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the reverse bleed valve

(2) An installation of the reverse bleed valve.

TASK 75-34-03-000-801-G00

2. Reverse Bleed Valve Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the Reverse Bleed Valve (RBV).

(2) The RBV is installed between 11:00 and 1:00 o'clock positions on the engine core.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-4049	Container - Fuel Resistant, 1 Gallon (4 Liter)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

SUBTASK 75-34-03-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-34-03-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING

DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-34-03-010-002-G00

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(3) Remove the segment shroud 5 (TASK 72-23-04-000-801-G00).

G. Reverse Bleed Valve Removal

SUBTASK 75-34-03-020-001-G00

(1) Remove the HJ8B electrical harness [7] from the clips on the support bracket [42].

SUBTASK 75-34-03-020-002-G00

(2) Remove the HJ8B electrical harness [7] from the clips on the support bracket [43].

SUBTASK 75-34-03-020-003-G00

(3) Remove the W200 electrical harness [25] from the clips on the harness support bracket [23].

SUBTASK 75-34-03-020-004-G00

(4) Remove the W300 electrical harness [26] from the clips on the support bracket [27].

SUBTASK 75-34-03-020-005-G00

(5) Remove the air cooling tube [17].

(a) Remove the bolt [20] and clamp [22] that attach the air cooling tube [17] and harness support bracket [23] to the bracket that is part of the air tube [19].

(b) Disconnect the air cooling tube [17] from the nipple on the RBV [1].

(c) Remove the air cooling tube [17] from the engine.

SUBTASK 75-34-03-020-006-G00

(6) Remove the air exhaust tube [18].

(a) Disconnect the air exhaust tube [18] from the TBV exhaust air tube [24].

(b) Disconnect the air exhaust tube [18] from the nipple on the RBV [1].

SUBTASK 75-34-03-020-007-G00

(7) If it is necessary, remove the support bracket [42].

NOTE: The support bracket can be removed together with harness support bracket in the next steps.

(a) Remove the two bolts [2] that attach the support bracket [42] to the harness support bracket [9].

(b) Remove the support bracket [42] from the engine.

SUBTASK 75-34-03-020-008-G00

(8) If it is necessary, remove the support bracket [43].

NOTE: The support bracket can be removed together with harness support bracket in the next steps.

(a) Remove the two bolts [3] that attach the support bracket [43] to the harness support bracket [10].

(b) Remove the support bracket [43] from the engine.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-34-03-020-009-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(9) Disconnect the fuel hydraulic drain tube [14] from the fuel hydraulic manifold [8].

(a) Cut and remove the safety wire or safety cable from the fuel hydraulic drain tube [14] B-nut that connects to the fuel hydraulic manifold [8].

(b) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the fuel hydraulic manifold [8].

(c) Disconnect the fuel hydraulic drain tube [14] B-nut from the fuel hydraulic manifold [8].

(d) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

SUBTASK 75-34-03-020-010-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Disconnect the fuel hydraulic rod tube [15] from the fuel hydraulic manifold [8].

(a) Cut and remove the safety wire or safety cable from the fuel hydraulic rod tube [15] B-nut that connects to the fuel hydraulic manifold [8].

(b) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the fuel hydraulic manifold [8].

(c) Disconnect the fuel hydraulic rod tube [15] B-nut from the fuel hydraulic manifold [8].

(d) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

SUBTASK 75-34-03-020-011-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(11) Remove the fuel hydraulic head tube [16].

(a) Cut and remove the safety wire or safety cable from the fuel hydraulic head tube [16] B-nut that connects to the fuel hydraulic manifold [8].

(b) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the fuel hydraulic manifold [8].

(c) Disconnect the fuel hydraulic head tube [16] B-nut from the fuel hydraulic manifold [8].

(d) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

(e) Disconnect the fuel hydraulic head tube [16] B-nut from the fuel manifold [28].

(f) Remove the bolt [29] and clamp [30] that attach the fuel hydraulic head tube [16] to the support bracket [31].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-34-03-020-012-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(12) Remove the fuel hydraulic manifold [8], harness support bracket [9], and harness support bracket [10] from the RBV [1].

(a) Put a 1 gallon (4 l) fuel resistant container, STD-4049, below the fuel hydraulic manifold [8].

(b) Remove the four bolts [4] that attach the fuel hydraulic manifold [8], harness support bracket [9], and harness support bracket [10] to the RBV [1].

(c) Let the fuel drain into the 1 gallon (4 l) fuel resistant container, STD-4049.

(d) Remove the fuel hydraulic manifold [8], harness support bracket [9], and harness support bracket [10] from the RBV [1].

SUBTASK 75-34-03-020-013-G00

(13) Remove and discard the gasket seal [11] from between the fuel hydraulic manifold [8] and RBV [1].

SUBTASK 75-34-03-020-014-G00

(14) Remove the support bracket [27].

(a) Remove the bolt [32] and clamp [33] that attach the HJ8A harness [34] to the support bracket [27].

(b) Remove the bolts [35] that attach the support bracket [27] to the air tube [19].

SUBTASK 75-34-03-020-015-G00

(15) Remove harness support bracket [23].

(a) Remove the bolt [36] that attaches the harness support bracket [23] to the air tube [19].

SUBTASK 75-34-03-020-016-G00

(16) Remove the air tube [19].

(a) Remove the V-retainer coupling [5] that attaches the air tube [19] to the RBV [1].

(b) Remove the two bolts [37] that attach the air tube [19] to the support bracket [38].

(c) Remove the V-retainer coupling [39] that attaches the air tube [19] to the TBV duct assembly [40].

(d) Remove the air tube [19] from the engine.

(e) Remove the V-band seal [12] and V-band seal [41].

1) Do a visual check of the V-band seal [12] and V-band seal [41] for damage.

a) If the V-band seal [12] or V-band seal [41] is damaged, discard the damaged seal.

SUBTASK 75-34-03-020-017-G00

(17) Remove the RBV [1].

(a) Hold the RBV [1] and loosen the V-retainer coupling [6] that attaches the air tube [21] to the RBV [1].

(b) Remove the V-retainer coupling [6] that attaches the air tube [21] to the RBV [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the RBV [1] from the engine.

(d) Remove the V-band seal [13].

1) Do a visual check of the V-band seal [13] for damage.

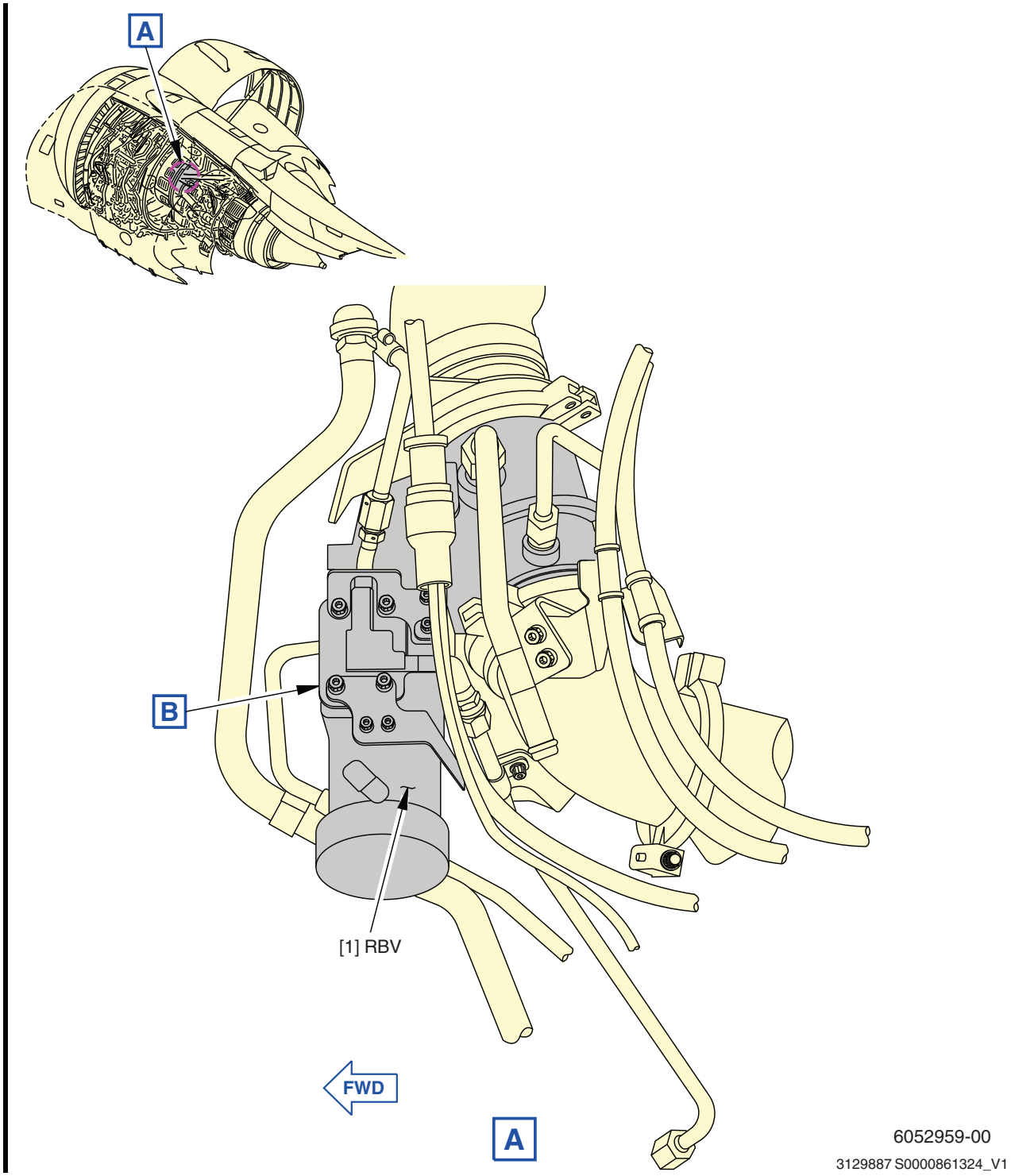
a) If the V-band seal [13] is damaged, discard it.

SUBTASK 75-34-03-420-001-G00

(18) Install protective covers on all the tubes, fittings, and reverse bleed valve.

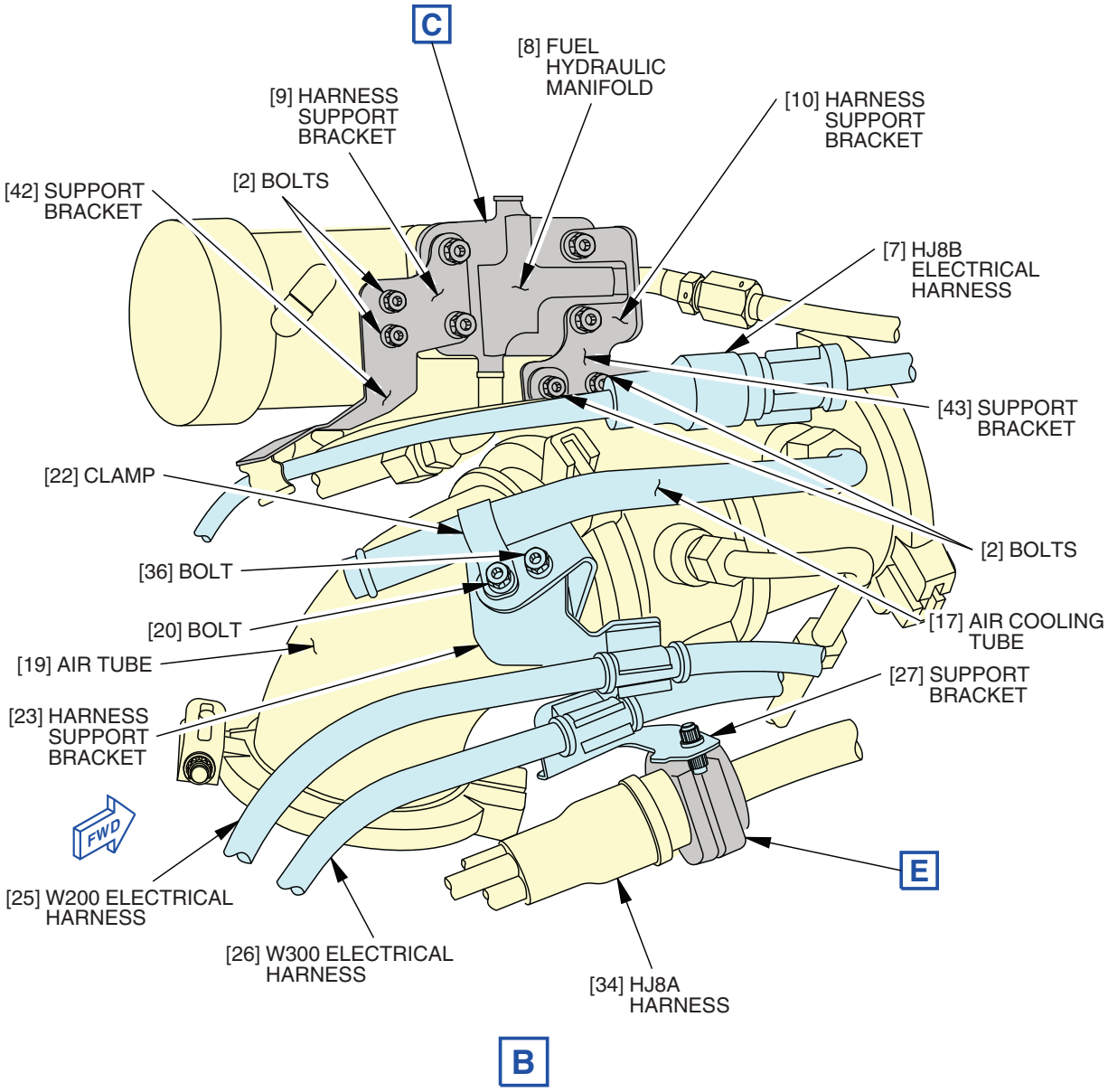
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Reverse Bleed Valve Installation
Figure 401/75-34-03-990-801-G00 (Sheet 1 of 5)

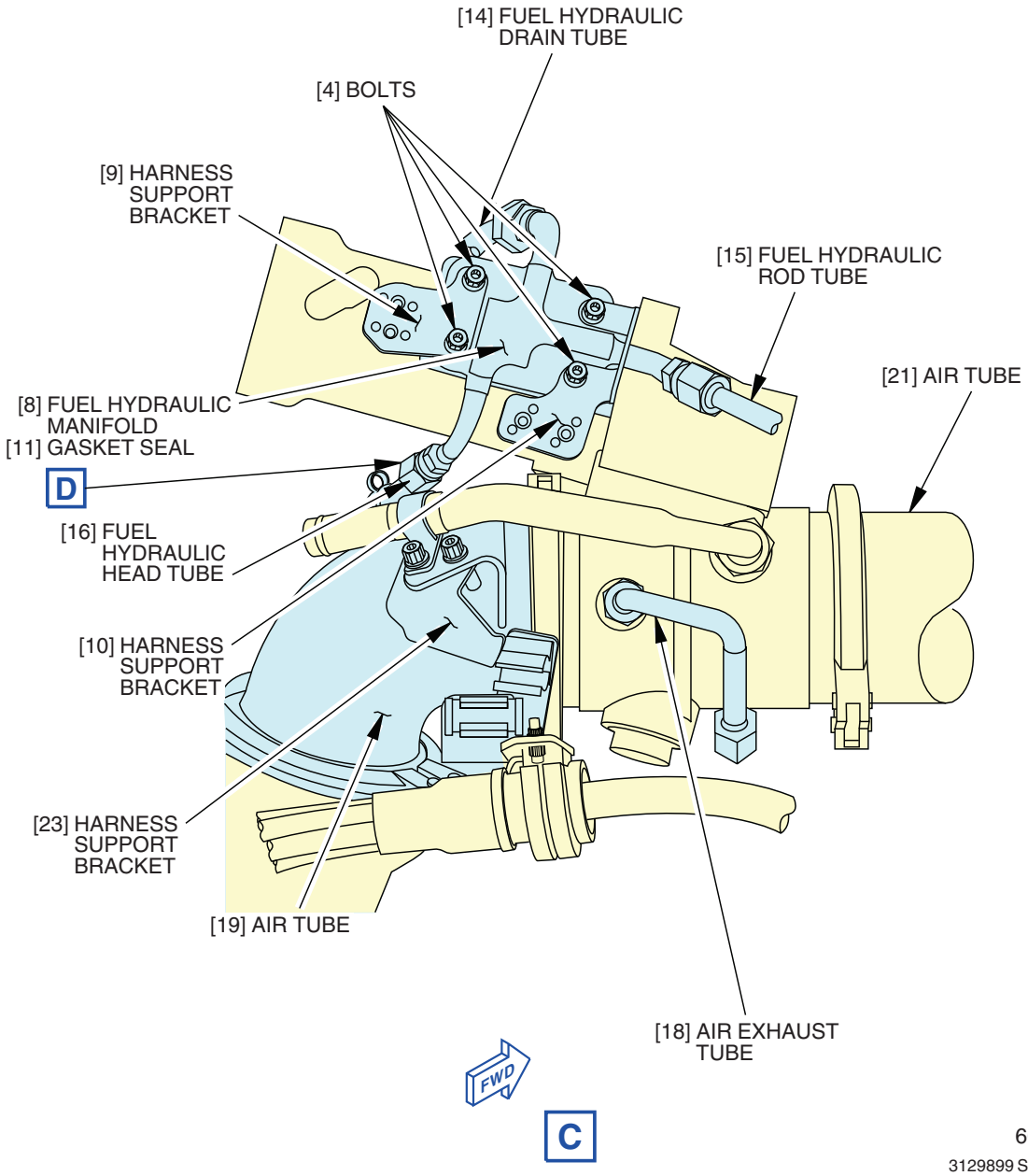
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6052960-00
3129888 S0000861325_V1

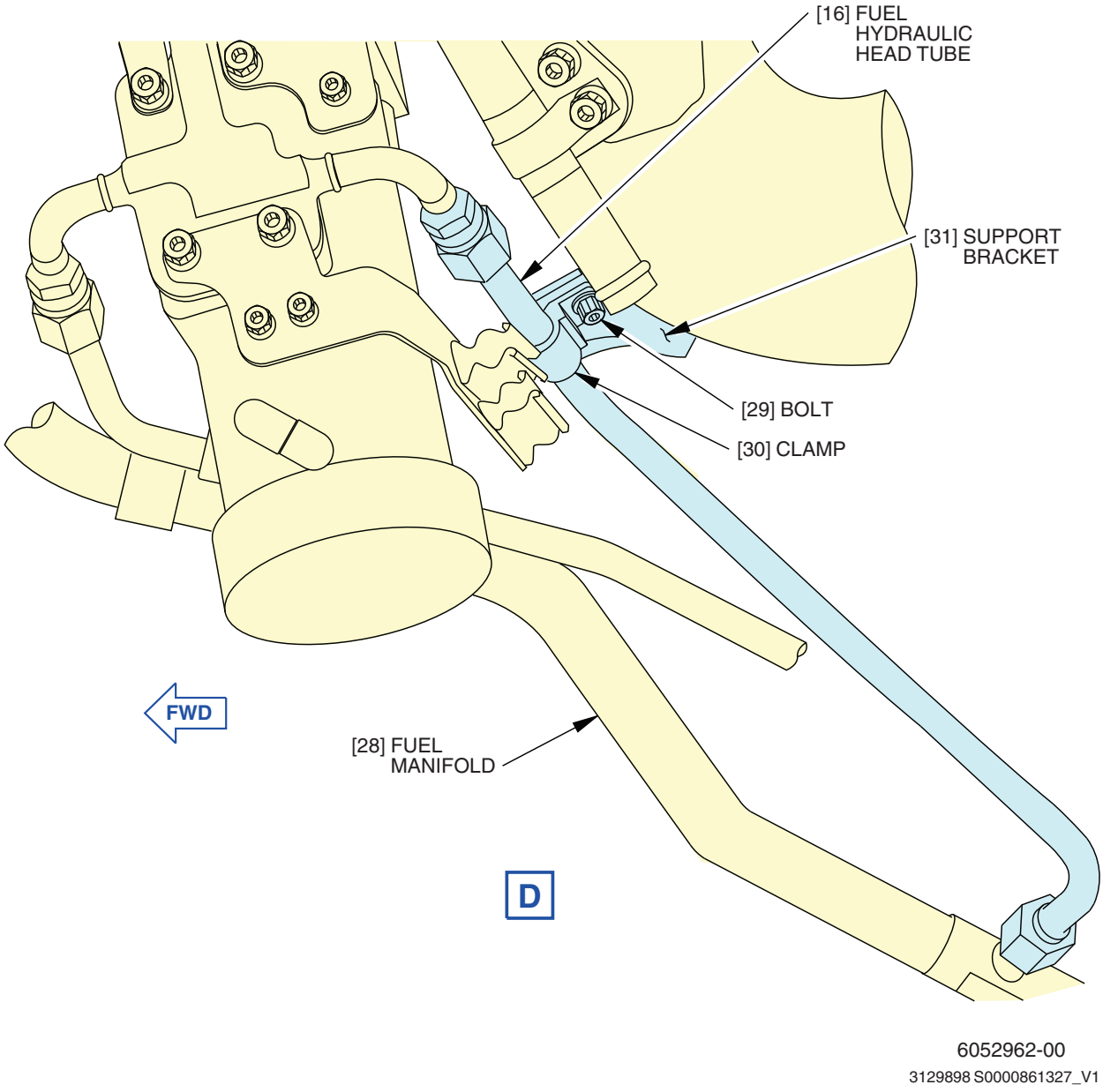
Reverse Bleed Valve Installation
Figure 401/75-34-03-990-801-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



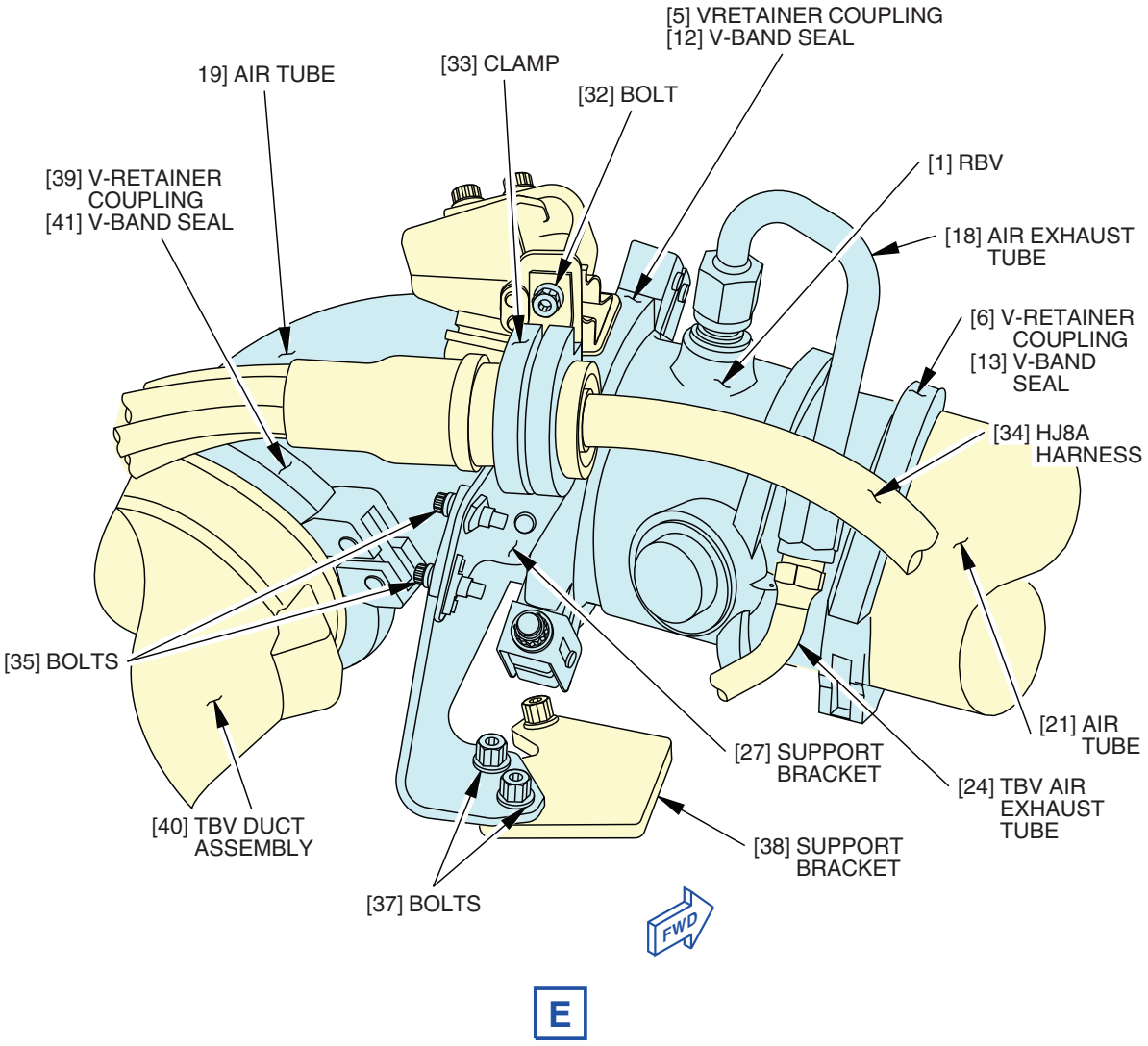
Reverse Bleed Valve Installation
Figure 401/75-34-03-990-801-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Reverse Bleed Valve Installation
Figure 401/75-34-03-990-801-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



6052964-00
3129900 S0000861329_V1

Reverse Bleed Valve Installation
Figure 401/75-34-03-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-34-03-400-801-G00

3. Reverse Bleed Valve Installation

(Figure 401)

A. General

(1) This task includes the steps to install the Reverse Bleed Valve (RBV).

B. References

Reference	Title
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Installation

SUBTASK 75-34-03-020-018-G00

(1) Remove protective covers from all the tubes, fittings, and RBV [1].

H. Reverse Bleed Valve Installation

SUBTASK 75-34-03-420-002-G00

(1) Connect the air tube [19] to the RBV [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the V-band seal [12] and V-band seal [13] into the ports of the RBV [1].

(b) Install the V-band seal [41] between the air tube [19] and TBV duct assembly [40].

(c) Loosely install the V-retainer coupling [5] that attaches the air tube [19] to the RBV [1].

1) Align the V-retainer coupling [5] alignment tab to the alignment tab in the air tube [19] and RBV [1] V-band flanges.

2) Make sure that the V-retainer coupling [5] is in the correct position.

3) Tighten the V-retainer coupling [5] by hand.

SUBTASK 75-34-03-420-003-G00

(2) Install the RBV [1] and air tube [19].

(a) Apply oil, D00599 [CP2442], on the threads of the two bolts [37].

(b) Install the V-band seal [41] between the air tube [19] and TBV duct assembly [40].

(c) Put the RBV [1] and air tube [19] in position between the TBV duct assembly [40] and air tube [21].

(d) Loosely install the V-retainer coupling [6] that attaches the air tube [21] to the RBV [1].

1) Make sure that the V-retainer coupling [6] is in the correct position.

(e) Loosely install the V-retainer coupling [39] that attaches the air tube [19] to the TBV duct assembly [40].

1) Make sure that the V-retainer coupling [39] is in the correct position.

(f) Install the two bolts [37] that attach the air tube [19] to the support bracket [38].

(g) Tighten the V-retainer coupling [6], V-retainer coupling [5], and V-retainer coupling [39] to 111 in-lb (12.5 N·m) - 129 in-lb (14.6 N·m).

(h) Tighten the two bolts [37] to 106 in-lb (12.0 N·m) - 124 in-lb (14.0 N·m).

SUBTASK 75-34-03-420-004-G00

(3) Install the air exhaust tube [18].

(a) Connect the air exhaust tube [18] B-nut to the TBV exhaust air tube [24].

(b) Connect the air exhaust tube [18] B-nut to the RBV [1].

(c) Tighten the B-nuts to 263 in-lb (29.7 N·m) - 307 in-lb (34.7 N·m).

SUBTASK 75-34-03-420-005-G00

(4) Install the fuel hydraulic head tube [16].

(a) Apply oil, D00599 [CP2442] on the threads of the bolt [29].

(b) Connect the fuel hydraulic head tube [16] B-nut to the fuel manifold [28]

(c) Attach the fuel hydraulic head tube [16] to the support bracket [31] with the clamp [30] and bolt [29].

1) Do not tighten the bolt [29] at this time.

(d) Triple tighten the fuel hydraulic head tube [16] B-nut, do these steps:

1) Tighten the B-nut to 263 in-lb (29.7 N·m) - 307 in-lb (34.7 N·m).

2) Loosen the B-nut until it turns freely.

3) Tighten the B-nut to 263 in-lb (29.7 N·m) - 307 in-lb (34.7 N·m).

4) Tighten the B-nut again to 263 in-lb (29.7 N·m) - 307 in-lb (34.7 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Install safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006], on the fuel hydraulic head tube [16] B-nut (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

(f) Tighten the bolt [29] to 33 in-lb (3.7 N·m) - 37 in-lb (4.2 N·m).

SUBTASK 75-34-03-420-006-G00

(5) Connect the support bracket [27] to the air tube [19].

(a) Apply oil, D00599 [CP2442], on the threads of the two bolts [35] and bolt [32].

(b) Attach the support bracket [27] to the air tube [19] with the two bolts [35].

1) Tighten the two bolts [35] to 51 in-lb (5.8 N·m) - 59 in-lb (6.7 N·m).

(c) Attach the HJ8A harness [34] to the support bracket [27] with the clamp [33] and bolt [32].

1) Tighten the bolt [32] to 33 in-lb (3.7 N·m) - 37 in-lb (4.2 N·m).

SUBTASK 75-34-03-420-007-G00

(6) Install the harness support bracket [23].

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [36].

(b) Install the bolt [36] that attaches the harness support bracket [23] to the air tube [19].

1) Do not tighten the bolt [36] at this time.

SUBTASK 75-34-03-420-008-G00

(7) Install the air cooling tube [17].

(a) Connect the air cooling tube [17] B-nut to the RBV [1].

1) Do not tighten the air cooling tube [17] B-nut at this time.

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [20].

(c) Attach the air cooling tube [17] and harness support bracket [23] to the bracket that is part of the air tube [19] with clamp [22] and bolt [20].

1) Tighten the bolt [20] by hand.

(d) Triple tighten the air cooling tube [17] B-nut on the RBV [1], do these steps:

1) Tighten the B-nut to 460 in-lb (52.0 N·m) - 540 in-lb (61.0 N·m).

2) Loosen the B-nut until it turns freely.

3) Tighten the B-nut to 460 in-lb (52.0 N·m) - 540 in-lb (61.0 N·m).

4) Tighten the B-nut again to 460 in-lb (52.0 N·m) - 540 in-lb (61.0 N·m).

(e) Tighten the bolt [20] and bolt [36] to 60 in-lb (6.8 N·m) - 70 in-lb (7.9 N·m).

SUBTASK 75-34-03-420-009-G00

(8) Connect the fuel hydraulic manifold [8], harness support bracket [9], and harness support bracket [10] to the RBV [1].

(a) Put the new gasket seal [11] between the fuel hydraulic manifold [8] and RBV [1].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [4].

(c) Install the fuel hydraulic manifold [8], harness support bracket [9], and harness support bracket [10] in position on the RBV [1].

(d) Install the four bolts [4] that attach the fuel hydraulic manifold [8], harness support bracket [9], and harness support bracket [10] to the RBV [1].

1) Do not tighten the bolts [4] at this time.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-34-03-420-010-G00

(9) Connect the fuel hydraulic head tube [16], fuel hydraulic rod tube [15], and fuel hydraulic drain tube [14] to the fuel hydraulic manifold [8].

(a) Connect the fuel hydraulic head tube [16] B-nut to the fuel hydraulic manifold [8].

(b) Connect the fuel hydraulic rod tube [15] B-nut to the fuel hydraulic manifold [8].

(c) Connect the fuel hydraulic drain tube [14] B-nut to the fuel hydraulic manifold [8].

(d) Triple tighten the B-nuts that connect the fuel hydraulic drain tube [14], fuel hydraulic rod tube [15], and fuel hydraulic head tube [16] to the fuel hydraulic manifold [8], do these steps:

1) Tighten the B-nuts to 263 in-lb (29.7 N·m) - 307 in-lb (34.7 N·m).

2) Loosen the B-nuts until it turns freely.

3) Tighten the B-nuts to 263 in-lb (29.7 N·m) - 307 in-lb (34.7 N·m).

4) Tighten the B-nuts again to 263 in-lb (29.7 N·m) - 307 in-lb (34.7 N·m).

(e) Install safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006], on the fuel hydraulic drain tube [14], fuel hydraulic rod tube [15], and fuel hydraulic head tube [16] B-nuts (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

SUBTASK 75-34-03-420-011-G00

(10) Tighten the bolts [4] to 106 in-lb (12.0 N·m) - 124 in-lb (14.0 N·m).

SUBTASK 75-34-03-420-012-G00

(11) If removed, connect the support bracket [43] to the harness support bracket [10].

(a) Install the support bracket [43] in position on the harness support bracket [10].

(b) Apply oil, D00599 [CP2442], on the threads of the two bolts [2].

(c) Install the two bolts [2] that attach the support bracket [43] to the harness support bracket [10].

1) Tighten the bolts [2] to 51 in-lb (5.8 N·m) - 59 in-lb (6.7 N·m).

SUBTASK 75-34-03-420-013-G00

(12) If removed, connect the support bracket [42] to the harness support bracket [9].

(a) Install the support bracket [42] in position on the harness support bracket [9].

(b) Apply oil, D00599 [CP2442], on the threads of the two bolts [2].

(c) Install the two bolts [2] that attach the support bracket [42] to the harness support bracket [9].

1) Tighten the bolts [2] to 51 in-lb (5.8 N·m) - 59 in-lb (6.7 N·m).

SUBTASK 75-34-03-420-014-G00

(13) Attach the HJ8B electrical harness [7] to the clips on the support bracket [42].

SUBTASK 75-34-03-420-015-G00

(14) Attach the HJ8B electrical harness [7] to the clips on the support bracket [43].

SUBTASK 75-34-03-420-016-G00

(15) Attach the W200 electrical harness [25] to the clips on the harness support bracket [23].

SUBTASK 75-34-03-420-017-G00

(16) Attach the W300 electrical harness [26] to the clips on the support bracket [27].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-34-03-410-001-G00

(1) Install the segment shroud 5 (TASK 72-23-04-400-801-G00).

SUBTASK 75-34-03-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-34-03-860-002-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Reverse Bleed Valve Installation Test

SUBTASK 75-34-03-700-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details